

ELECTRONICS



ΠΑΝΕΠΙΣΤΗΜΙΟ
ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
UNIVERSITY OF WEST ATTICA

DEPARTMENT OF COMPUTER AND INFORMATION
TECHNOLOGY AND COMPUTER ENGINEERING

PROJECT 5
BIPOLAR JUNCTION TRANSISTOR (BJT)

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DELIVERY DATE : 27/1/2023

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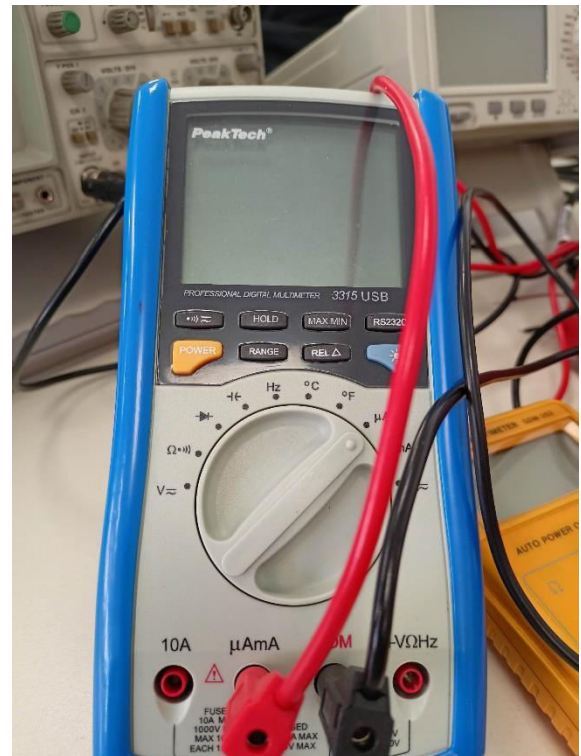
STUDENT PHOTO :



PHOTOGRAPHS OF EQUIPMENT USED IN THE LABORATORY

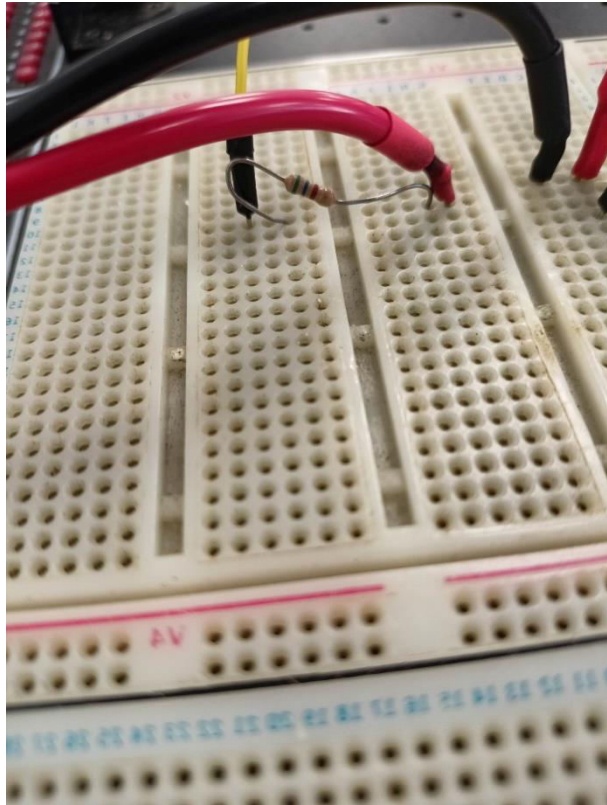


*Digital
Multimeter Bench MT8045*



Digital multimeter

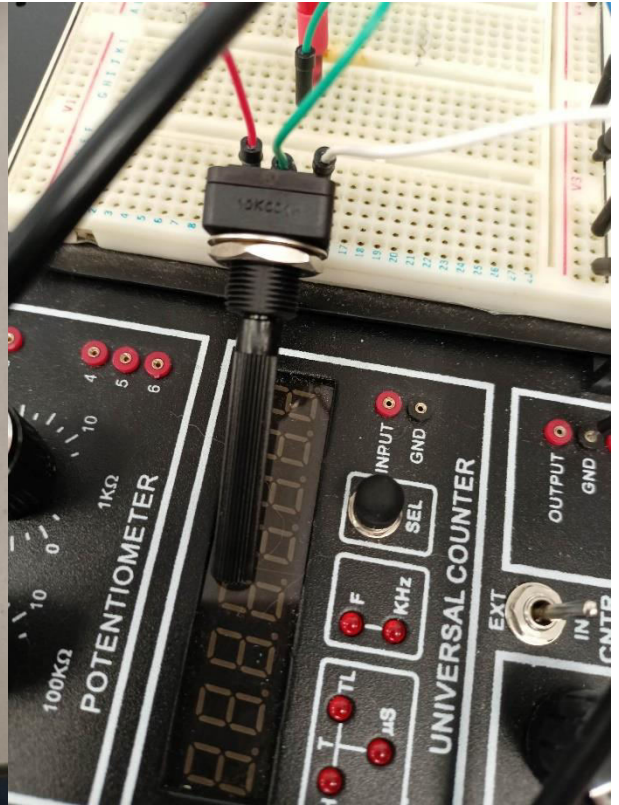
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560 Ohm resistor



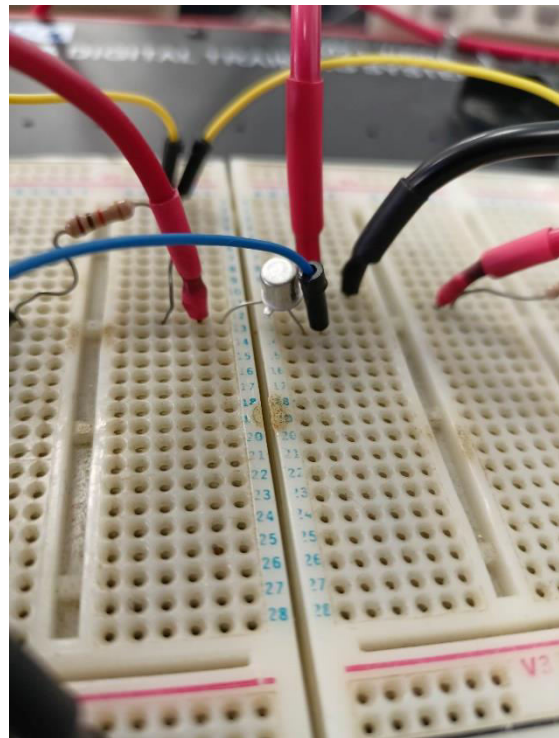
10K Ohm resistor



Potentiometer



Cable



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1. COMMON EMISSION CIRCUIT

1.1 OUTPUT CHARACTERISTICS $I_C = f(V_{CE})$

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 1.1.1.1 shows the schematic of the common emitter circuit.

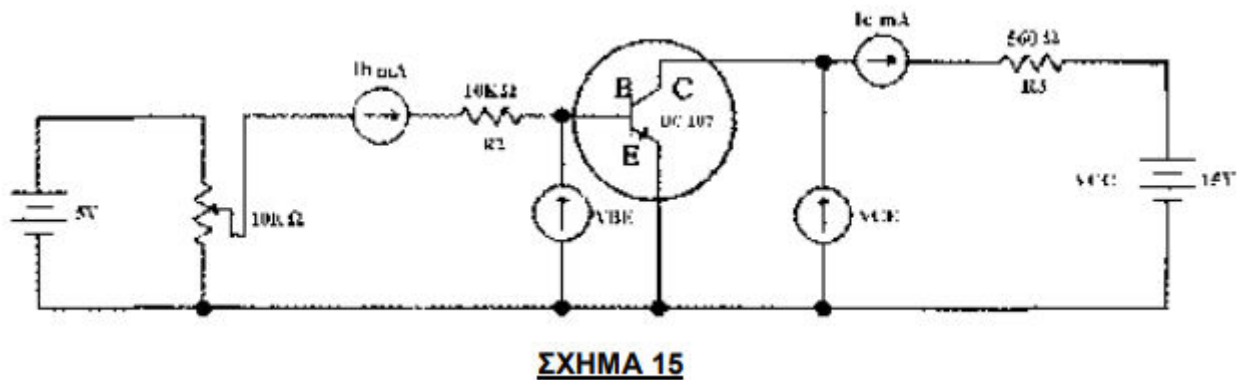


Figure 1.1.1.1 Schematic of the common emitter circuit

Table 1.1.1 shows the measurements made for the current I_C flowing in the collector in mA, for three values of the base current I_b (30 mA, 75 mA, 100 mA) and different potential differences V_{ce} at the collector-emitter ends in Volts.

Table 1.1.1

Vce (Volt)		0.1	0.3	0.6	0.9	1.5	2	3	4	5	7	9
$I_b = 30$ mA	I_c (mA)	2.75	7.21	7.40	7.50	7.71	7.83	8.00	8.33	8.83	9.71	10.38
$I_b = 75$ mA	I_c (mA)	13.2	22.7	23.0	23.1	23.3	23.4	23.7	24.0	24.3	24.8	25.5
$I_b = 100$ μA	I_c (mA)	15.8	27.5	27.6	27.7	27.9	28.1	28.5	28.9	29.2	30.0	30.7

Figure 1.1.2 shows the output characteristics $I_C = f(V_{CE})$ with the load line drawn on it according to the values in Table 1.1.1.1. The load line is described by the equation $V_{CC} - I_C R_C - V_{CE} = 0$. The point Q on the line indicates the operating range of the transistor.

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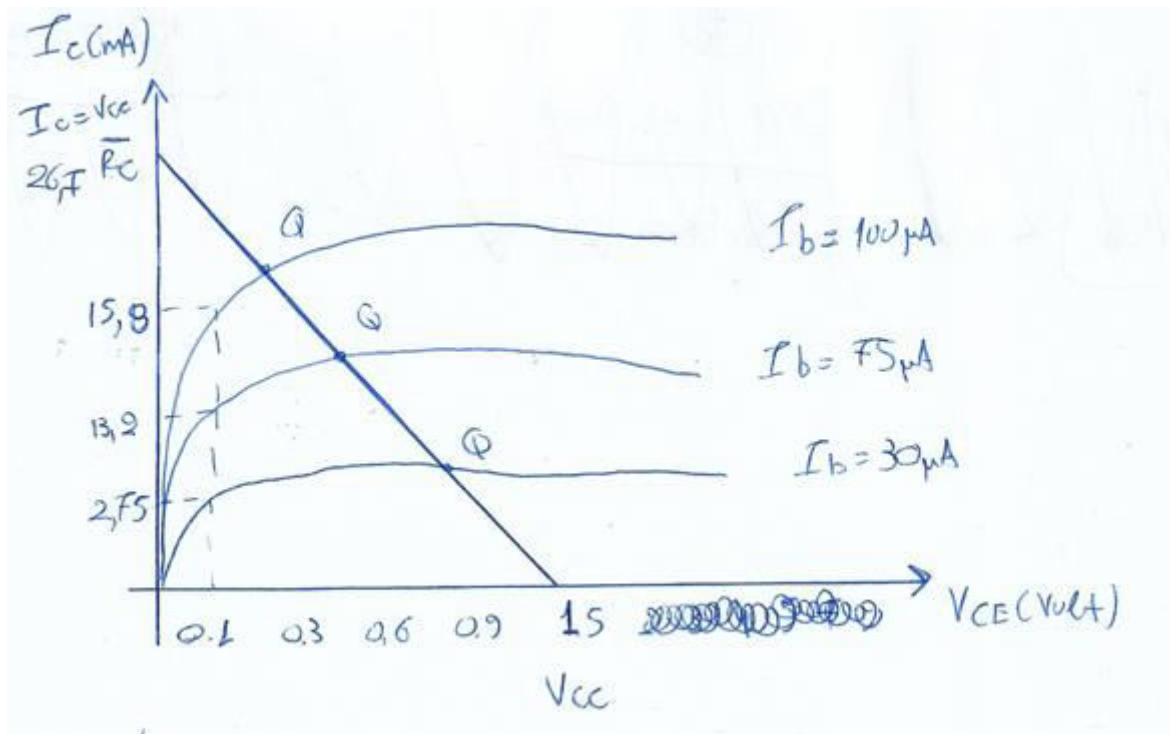


Figure 1.1.2 Output characteristics $I_c = f(V_{ce})$ and load line

Simulative Resolution

Figure 1.1.3 shows the simulated representation of the common emitter circuit in the simulation software "Multisim".

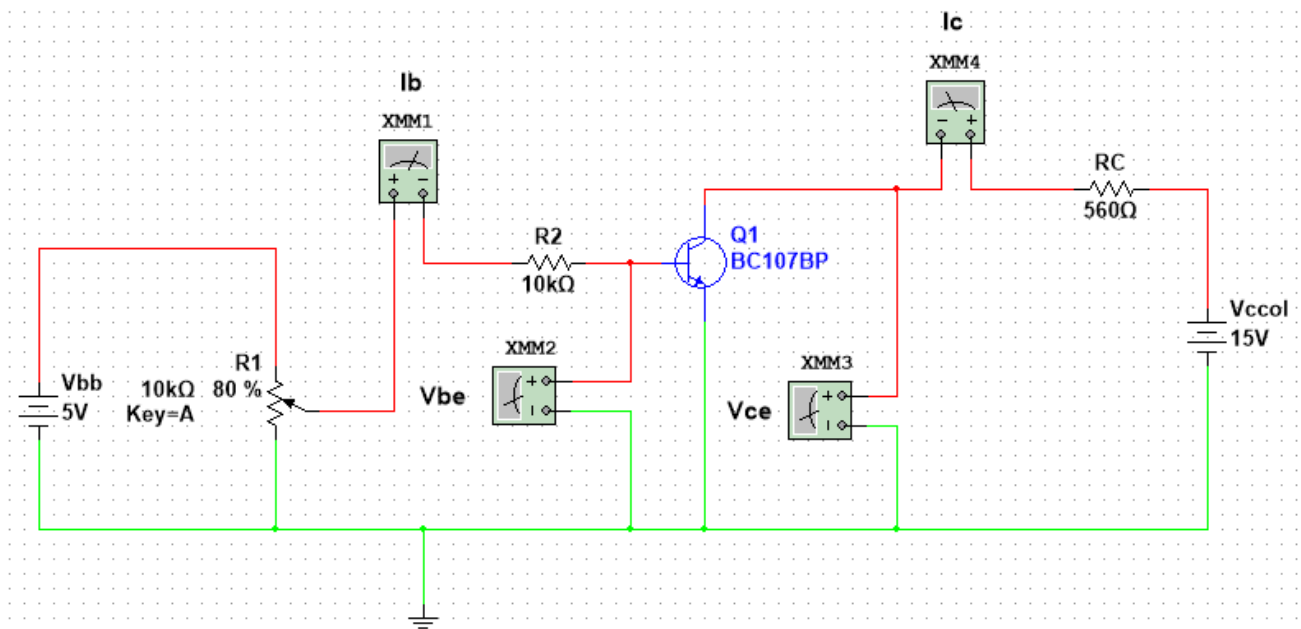


Figure 1.1.1.3 Simulation of the common emitter circuit in "Multisim"

Figure 1.1.4 shows the common emitter circuit with a base current $I_b = 30 \text{ mA}$.

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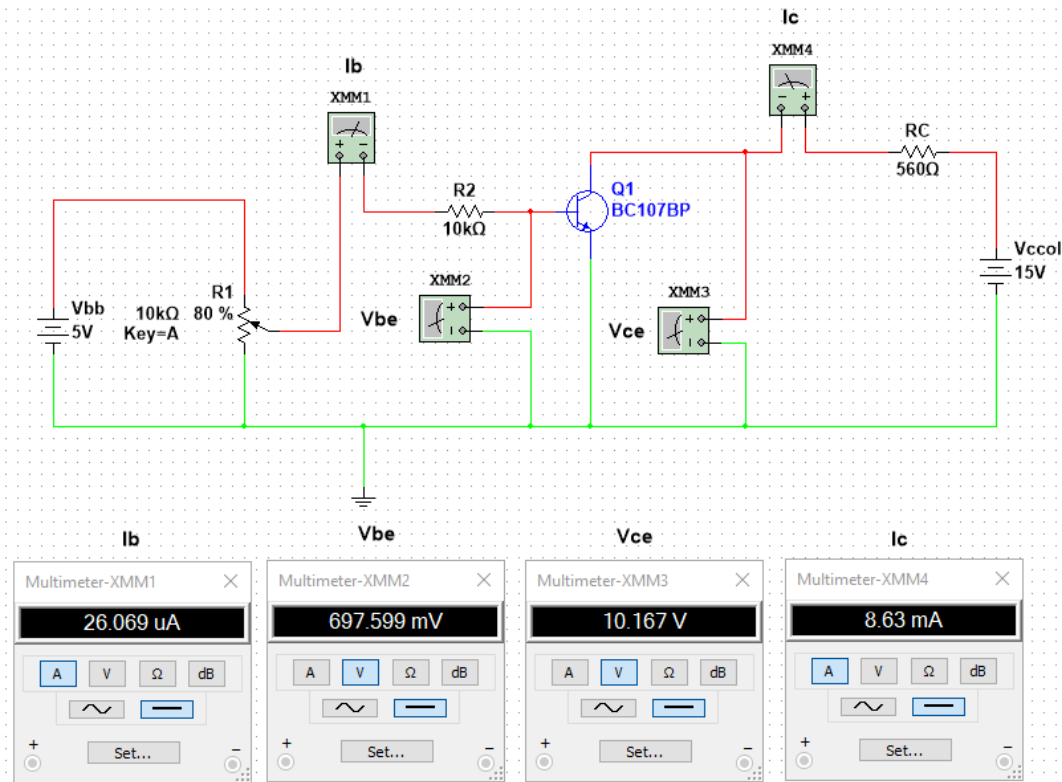


Figure 1.1.4 Simulation of the common emitter circuit with $I_b = 30$ mA in "Multisim"

Figure 1.1.5 shows the common emitter circuit with a base current $I_b = 75$ mA.

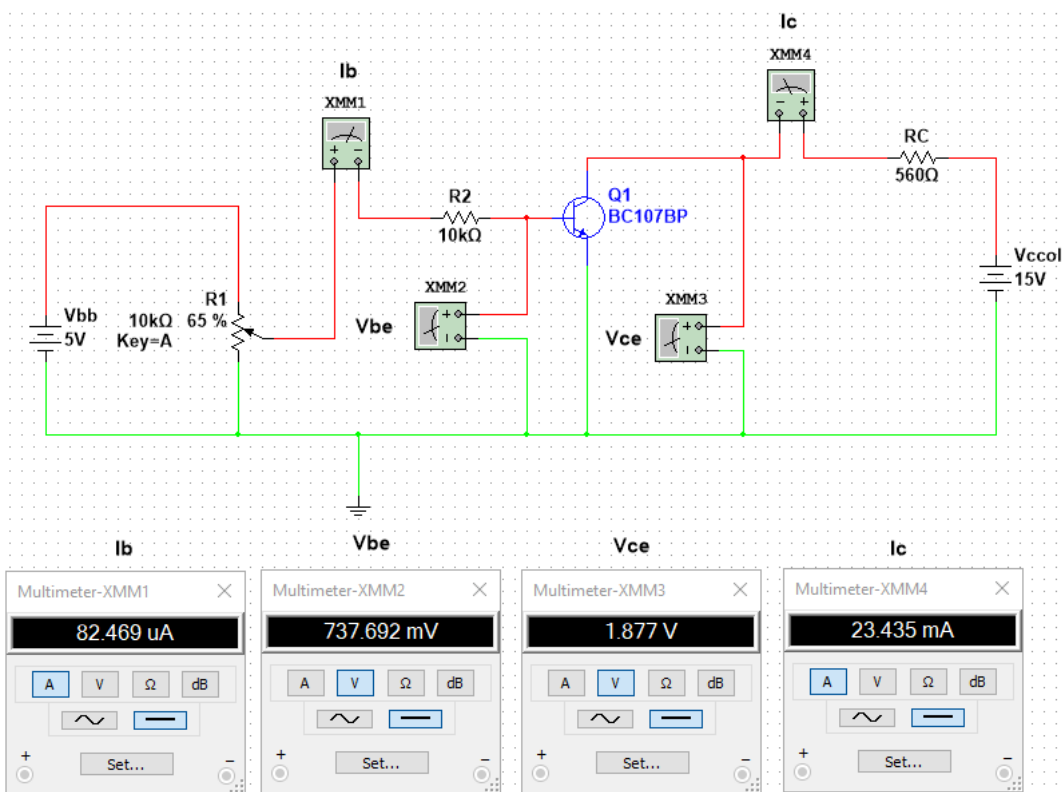


Figure 1.1.1.5 Simulation of the common emitter circuit with $I_b = 75$ mA in "Multisim"

Figures 1.1.6 - 1.1.16 simulatively verify the values recorded in Table 1.1.1 for $I_b = 75$ mA.

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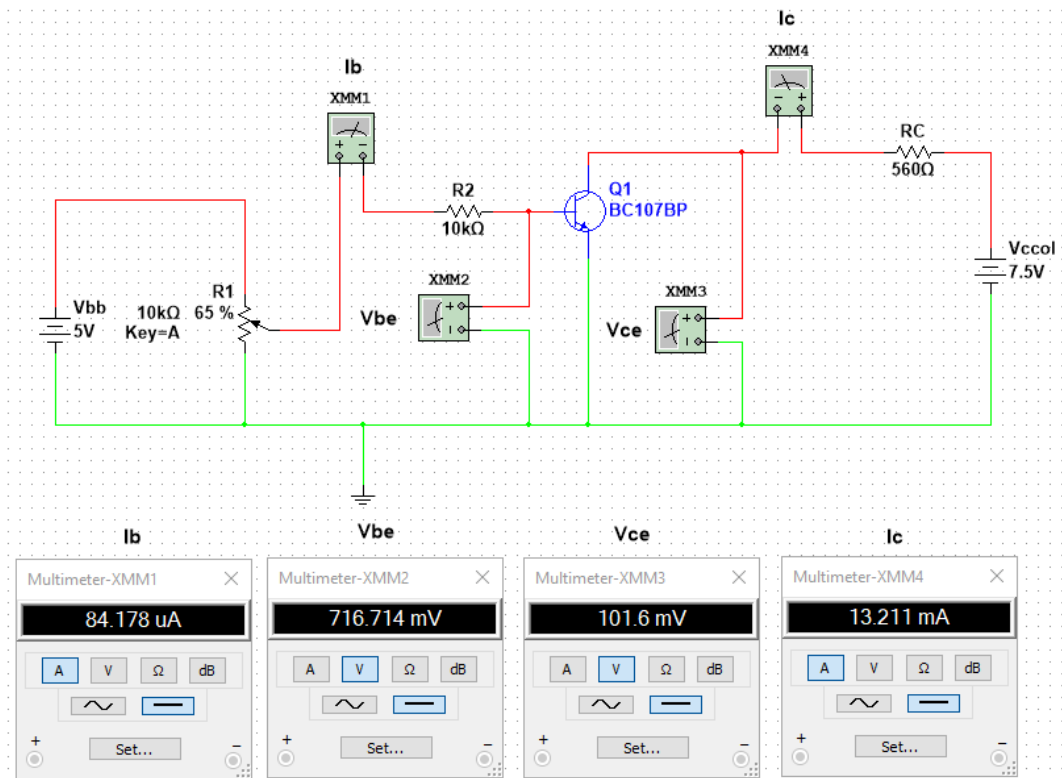


Figure 1.1.6 $I_b = 75 \text{ mA}$, $V_{ce} = 0.1 \text{ Volt}$

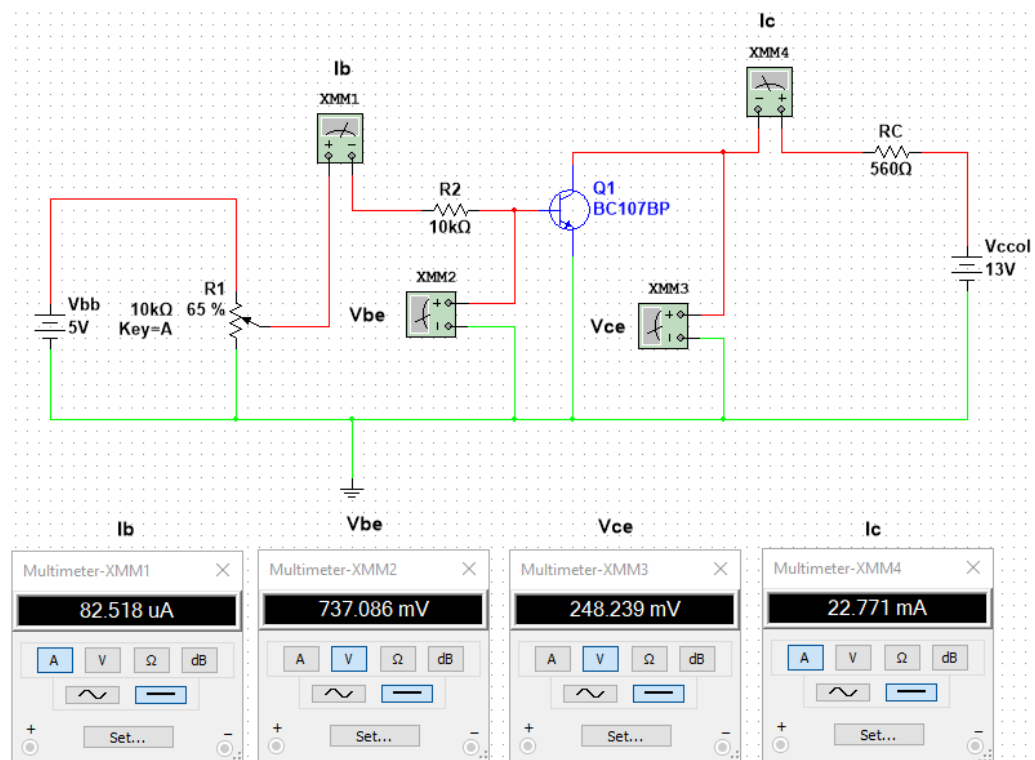


Figure 1.1.1.7 $I_b = 75 \text{ mA}$, $V_{ce} = 0.3 \text{ Volt}$

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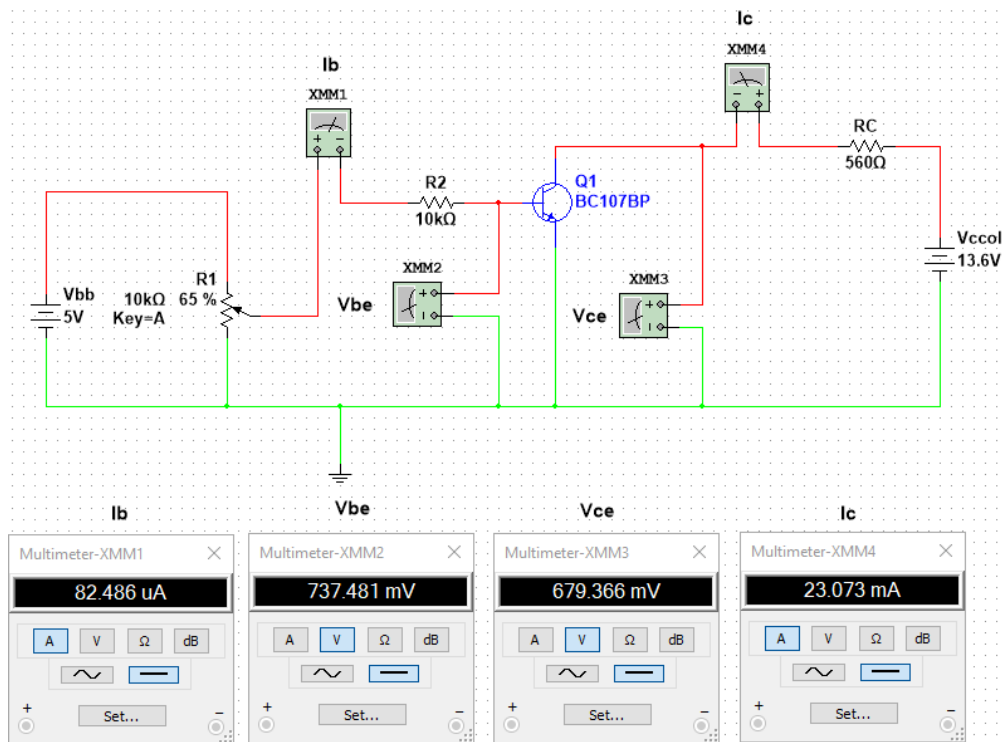


Figure 1.1.8 $I_b = 75 \text{ mA}$, $V_{ce} = 0.6 \text{ Volt}$

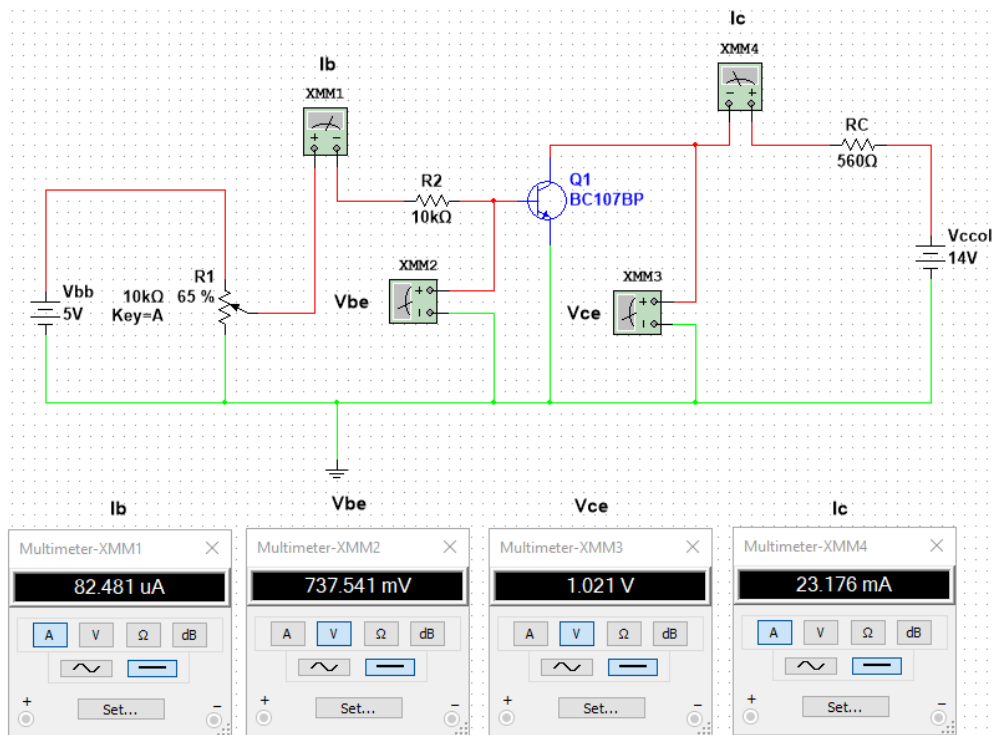


Figure 1.1.9 $I_b = 75 \text{ mA}$, $V_{ce} = 0.9 \text{ Volt}$

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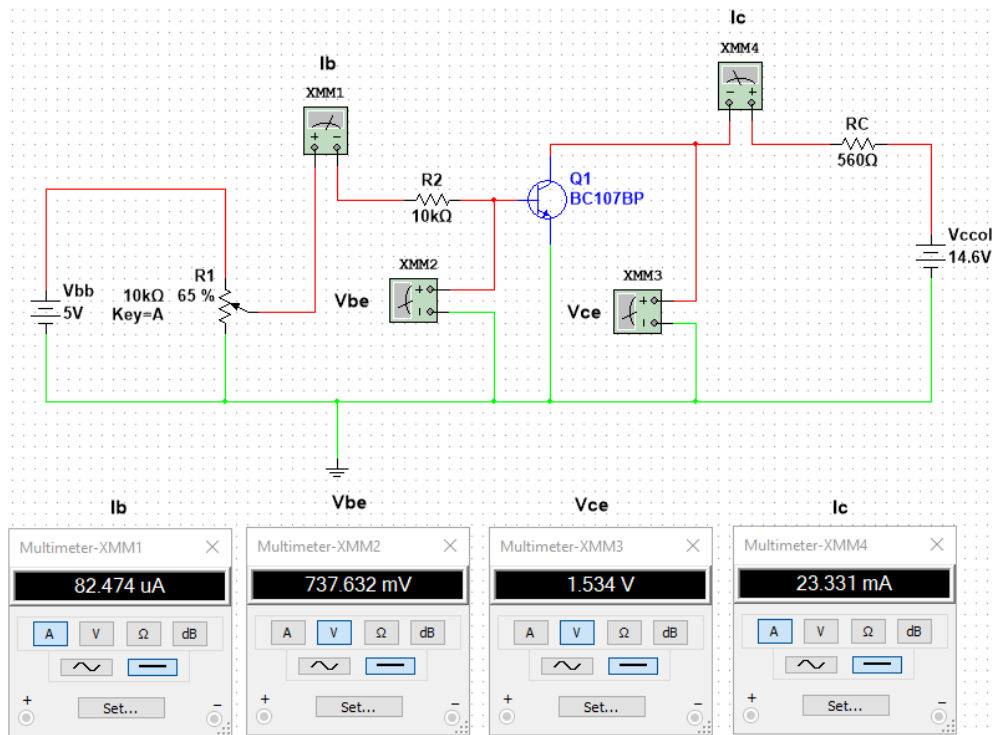


Figure 1.1.10 $I_b = 75 \text{ mA}$, $V_{ce} = 1.5 \text{ Volt}$

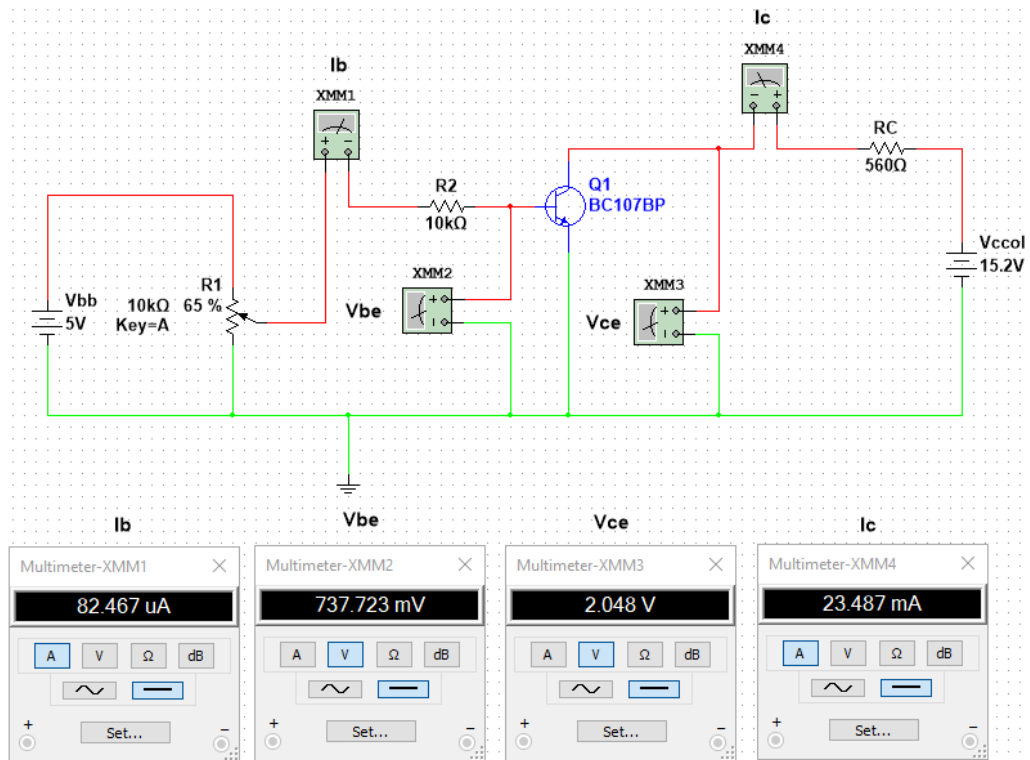


Figure 1.1.1.11 $I_b = 75 \text{ mA}$, $V_{ce} = 2 \text{ Volt}$

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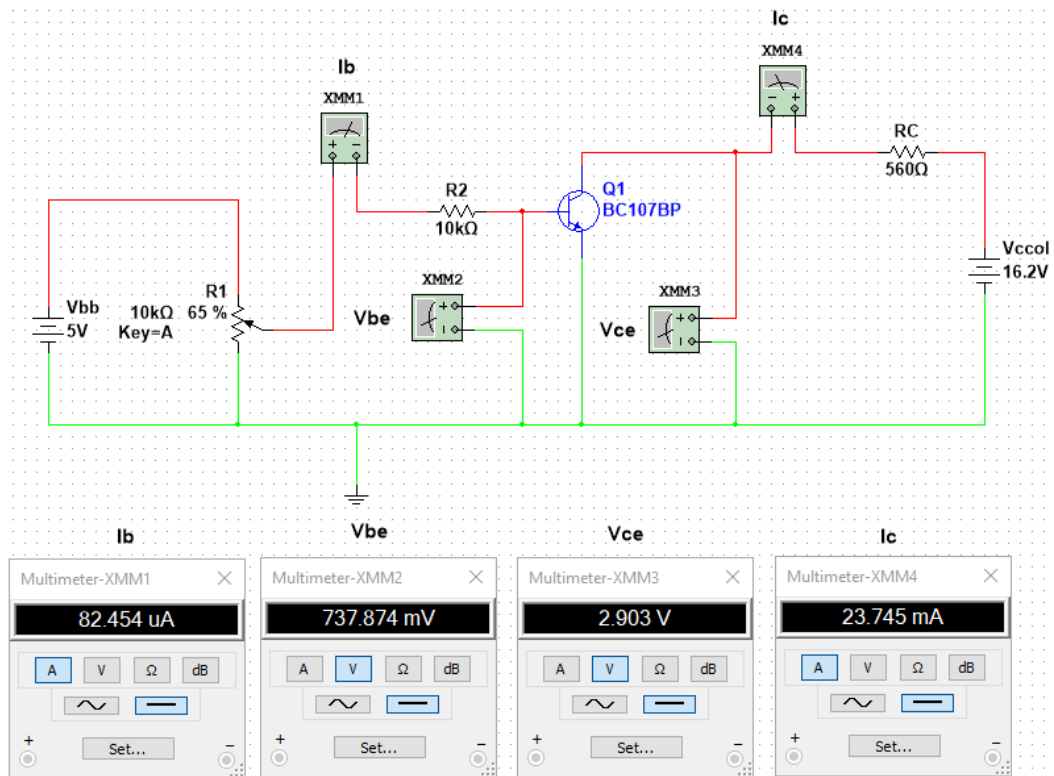


Figure 1.1.12 $I_b = 75 \text{ mA}$, $V_{ce} = 3 \text{ Volt}$

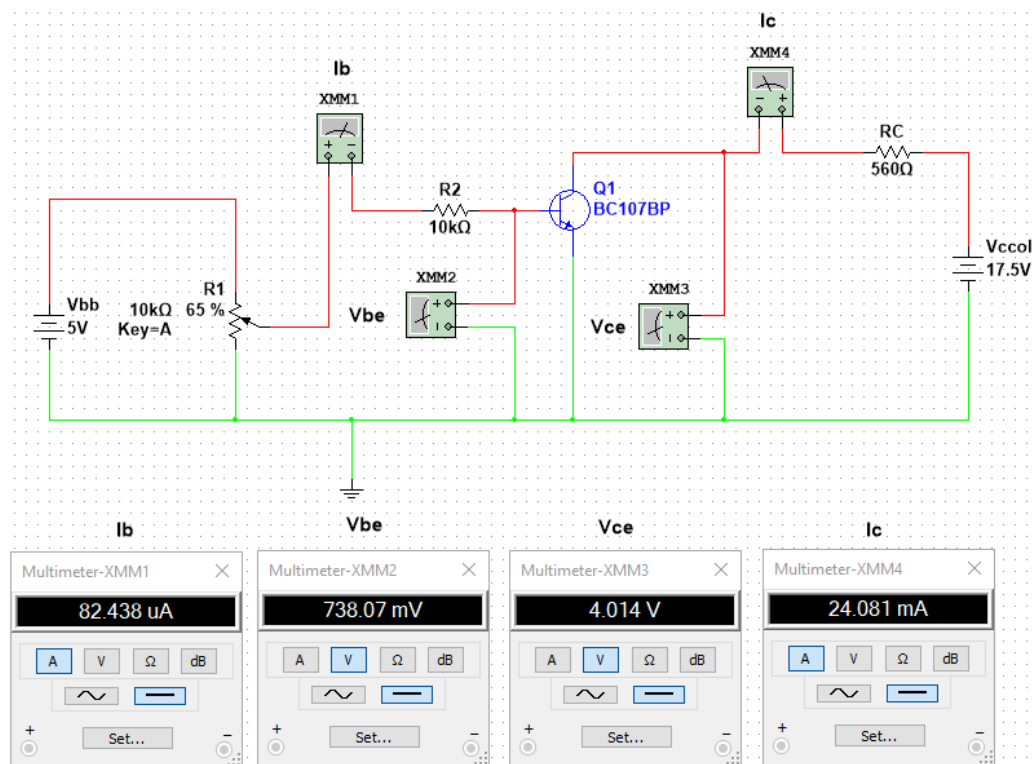


Figure 1.1.13 $I_b = 75 \text{ mA}$, $V_{ce} = 4 \text{ Volt}$

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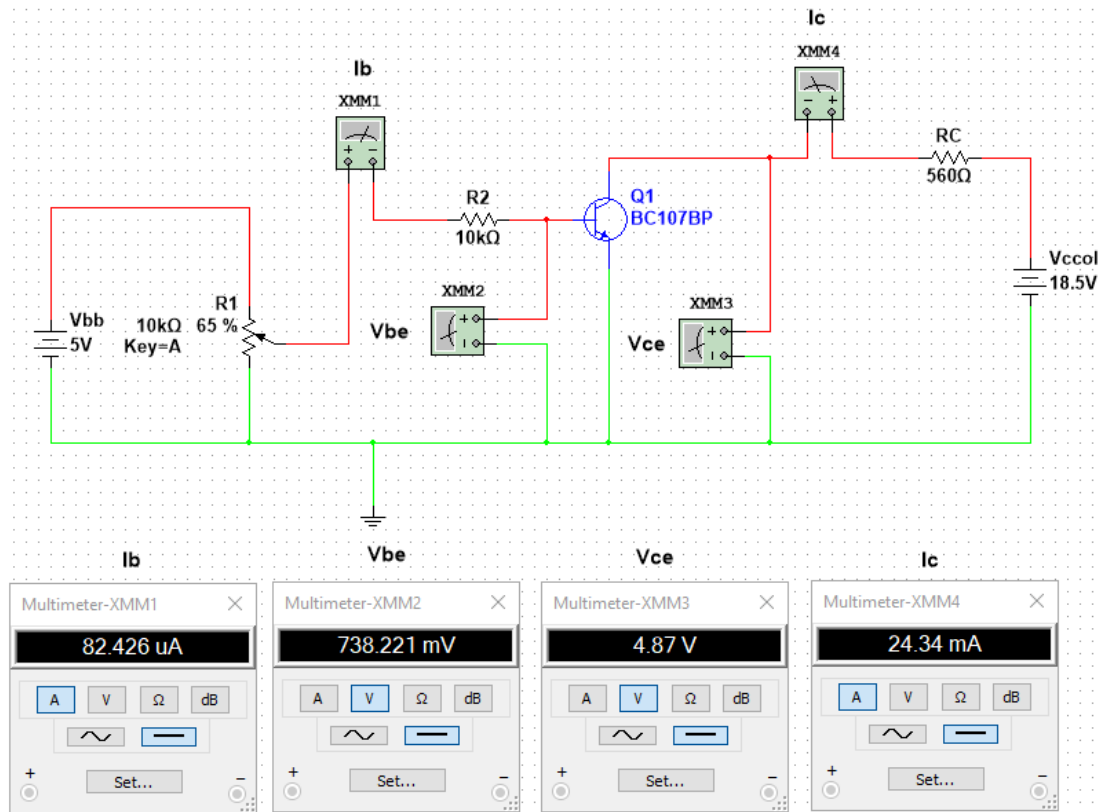


Figure 1.1.14 $I_b = 75 \text{ mA}$, $V_{ce} = 5 \text{ Volt}$

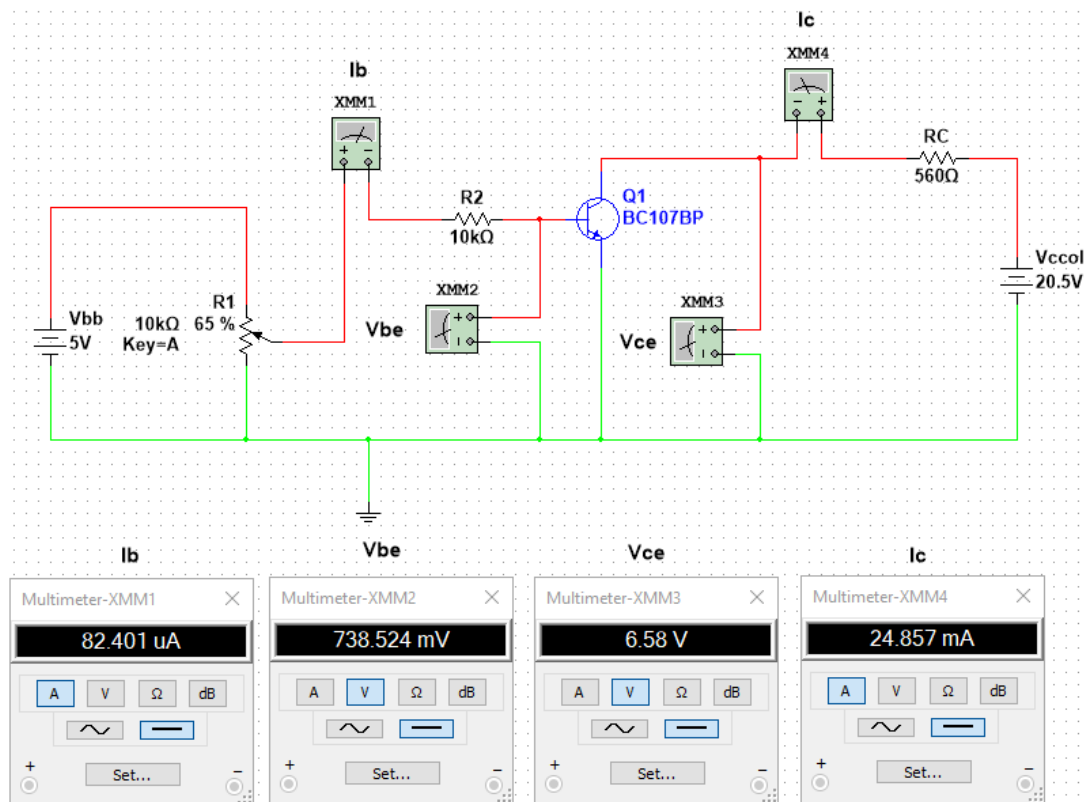


Figure 1.1.15 $I_b = 75 \text{ mA}$, $V_{ce} = 7 \text{ Volt}$

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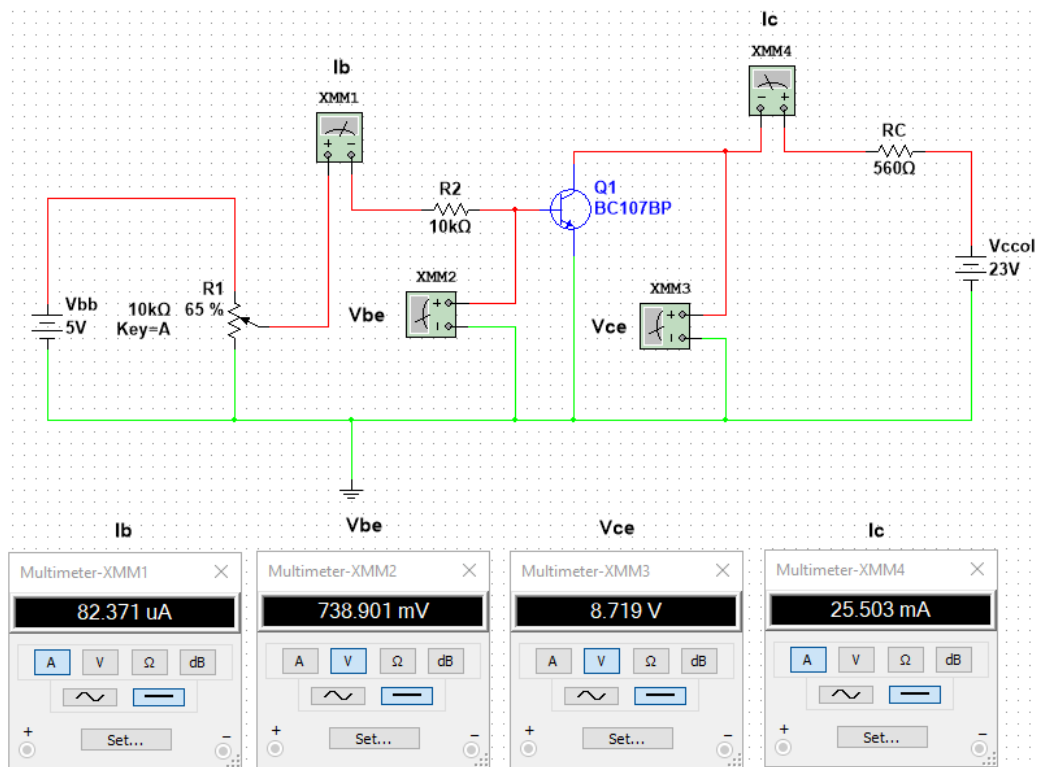


Figure 1.1.16 $I_b = 75$ mA, $V_{ce} = 9$ Volt

Figure 1.1.17 shows the common emitter circuit with a base current $I_b = 100$ mA.

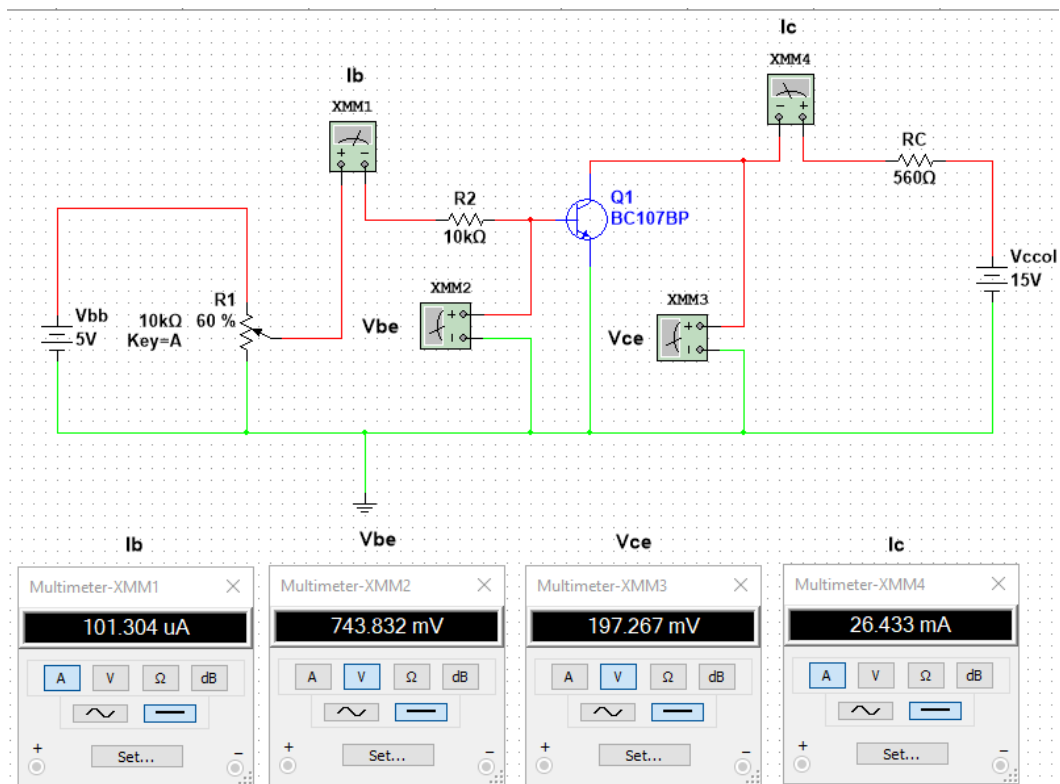


Figure 1.1.17 Simulation of the common emitter circuit with $I_b = 100$ mA in "Multisim"

Figures 1.1.18 - 1.1.28 simulatively verify the values recorded in Table 1.1.1 for $I_b = 100$ mA.

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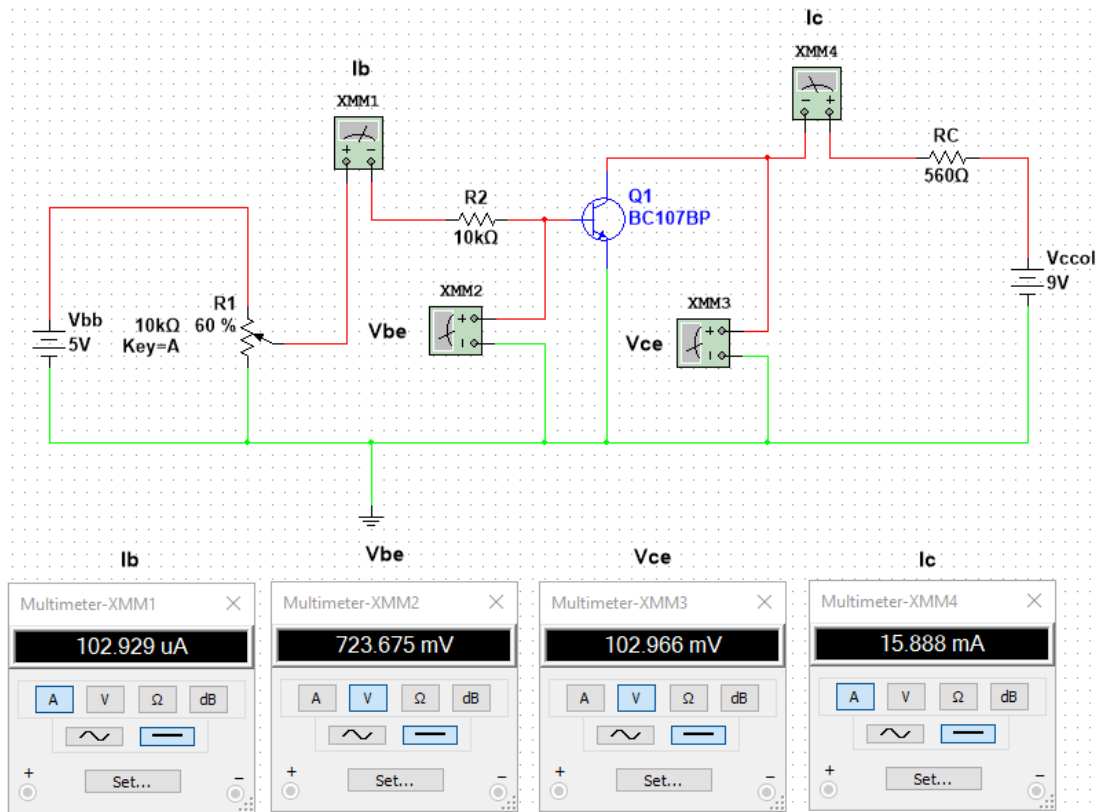


Figure 1.1.18 $I_b = 100 \text{ mA}$, $V_{ce} = 0.1 \text{ Volt}$

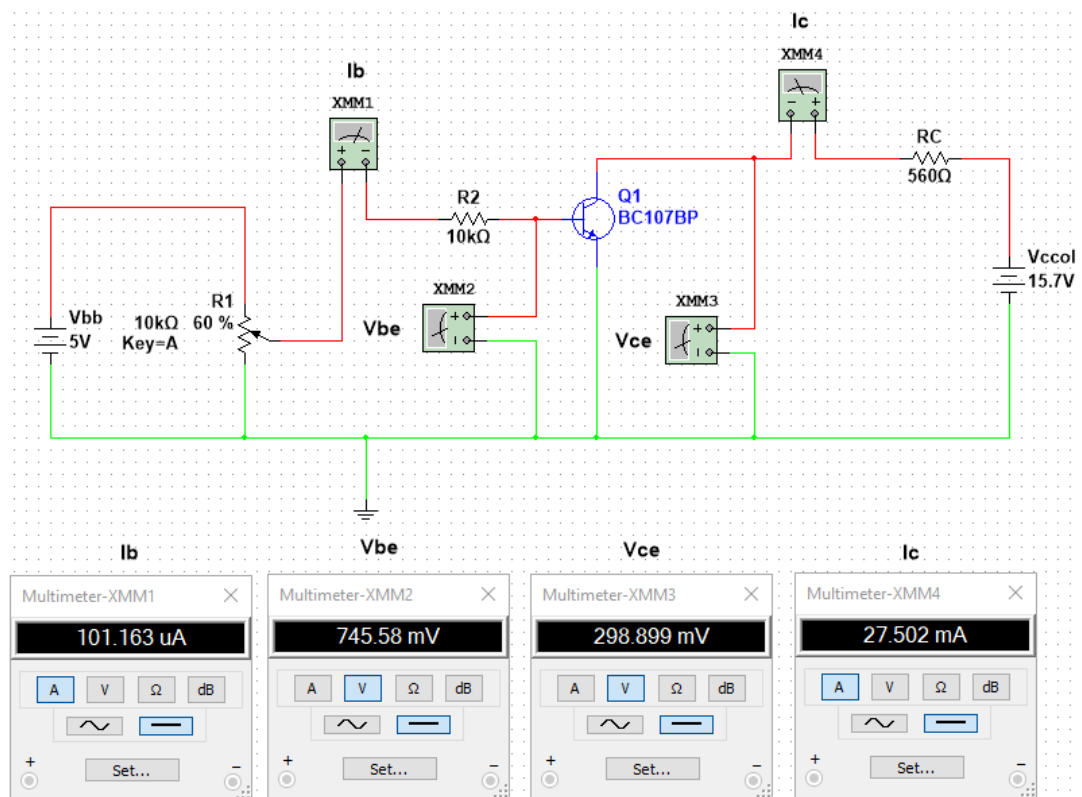


Figure 1.1.19 $I_b = 100 \text{ mA}$, $V_{ce} = 0.3 \text{ Volt}$

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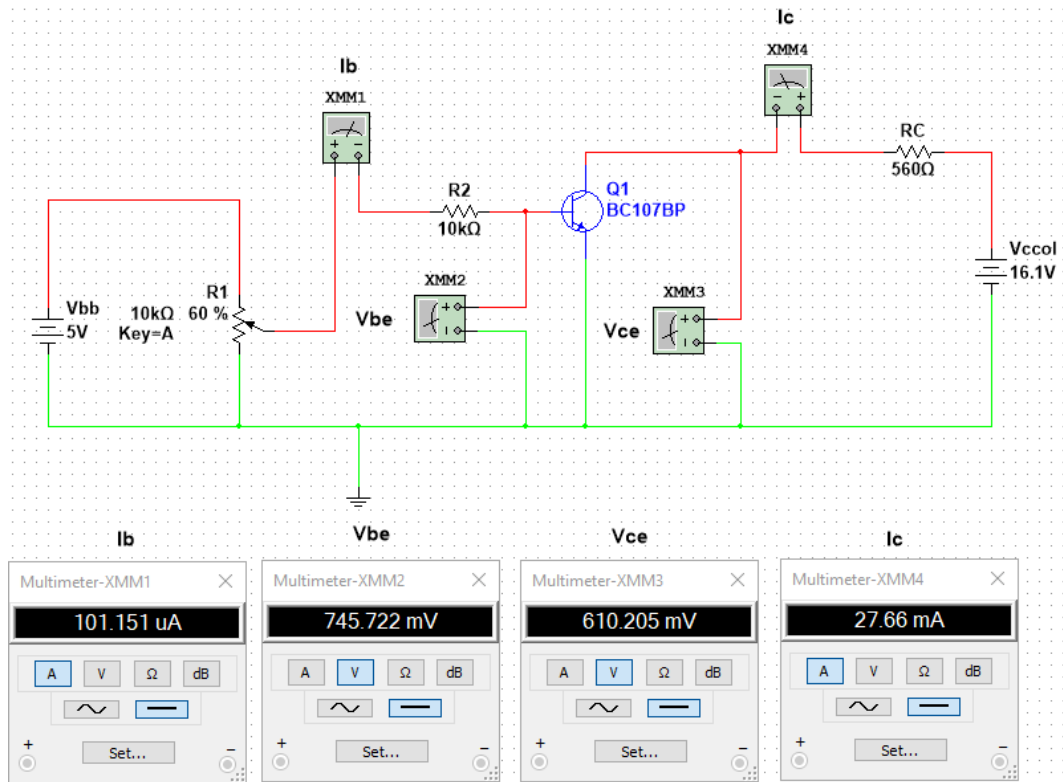


Figure 1.1.20 $I_b = 100$ mA, $V_{ce} = 0.6$ Volt

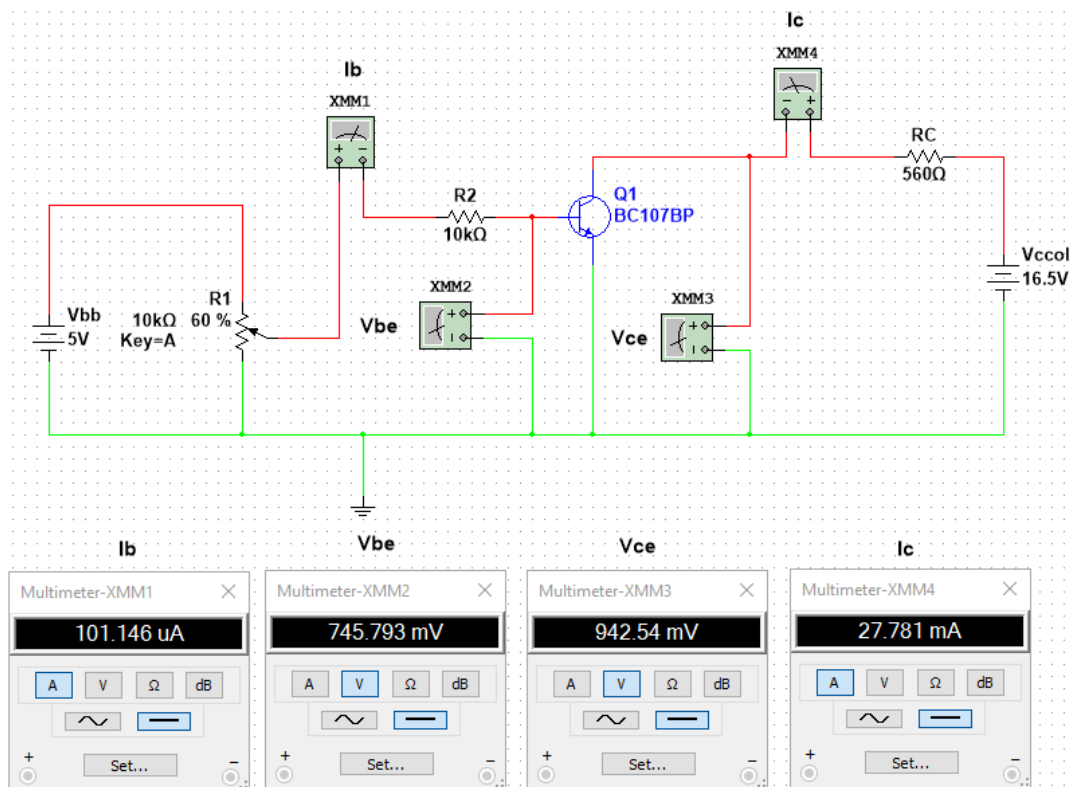


Figure 1.1.1.21 $I_b = 100$ mA, $V_{ce} = 0.9$ Volt

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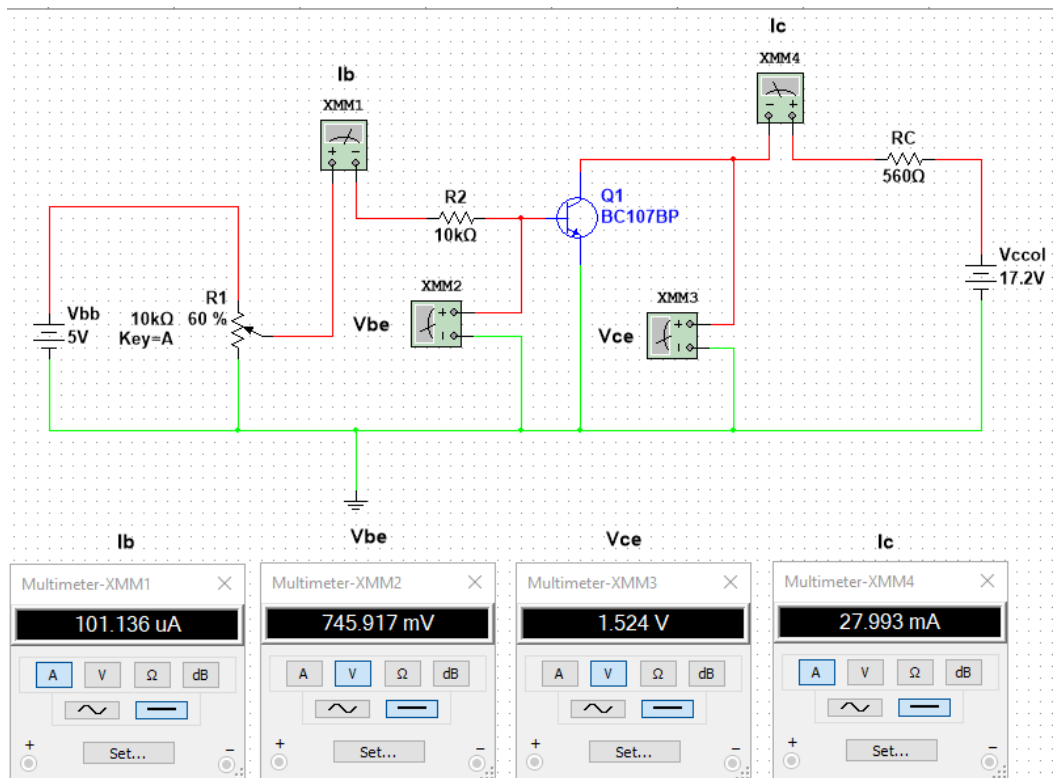


Figure 1.1.22 $I_b = 100 \mu A$, $V_{ce} = 1.5$ Volt

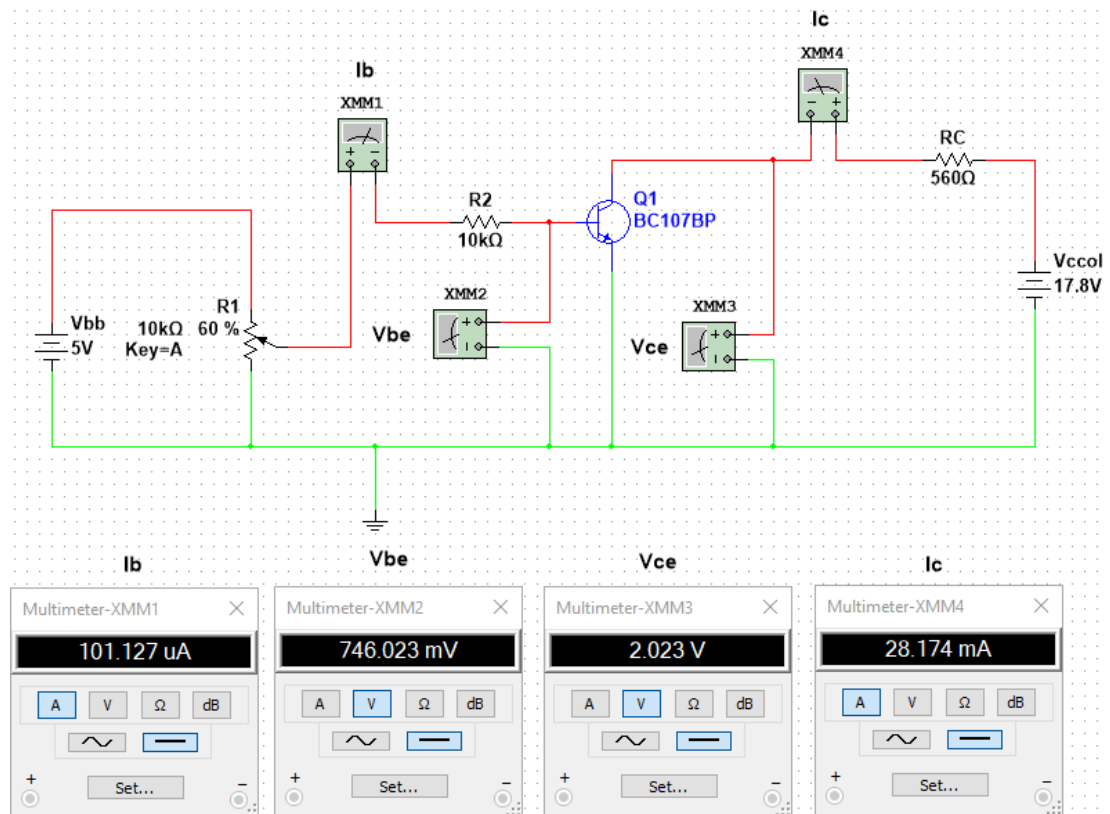


Figure 1.1.23 $I_b = 100 \mu A$, $V_{ce} = 2$ Volt

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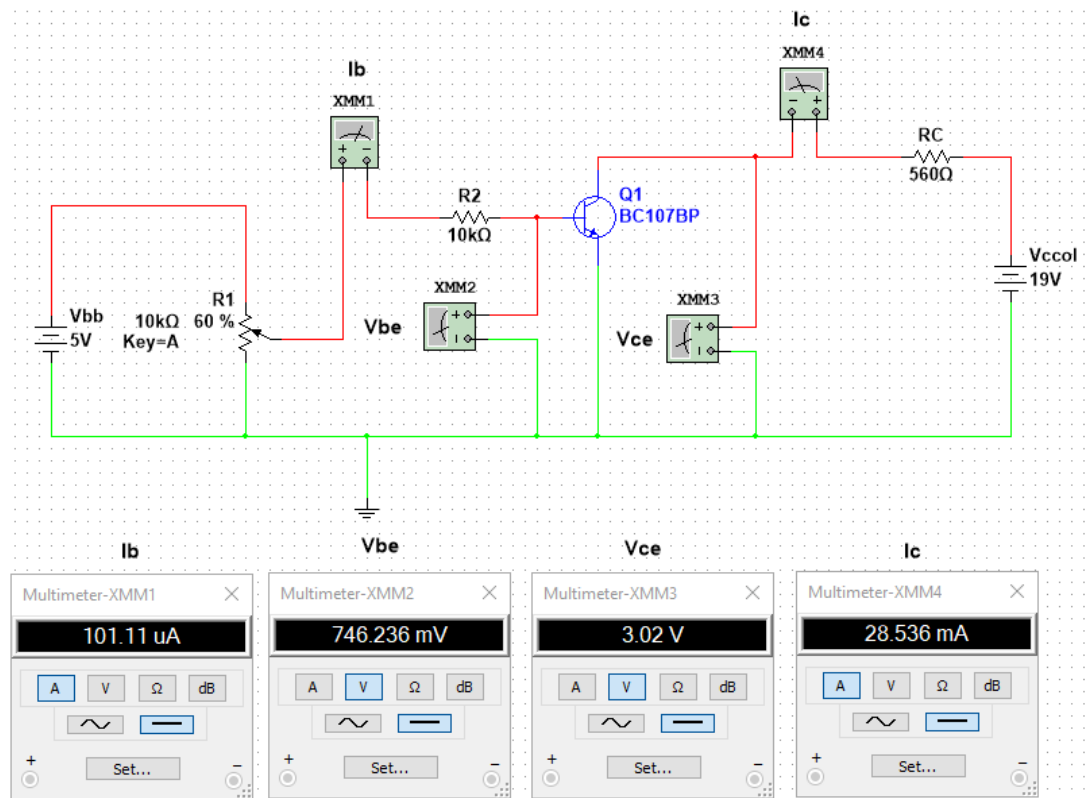


Figure 1.1.24 $I_b = 100 \mu\text{A}$, $V_{ce} = 3 \text{ V}$

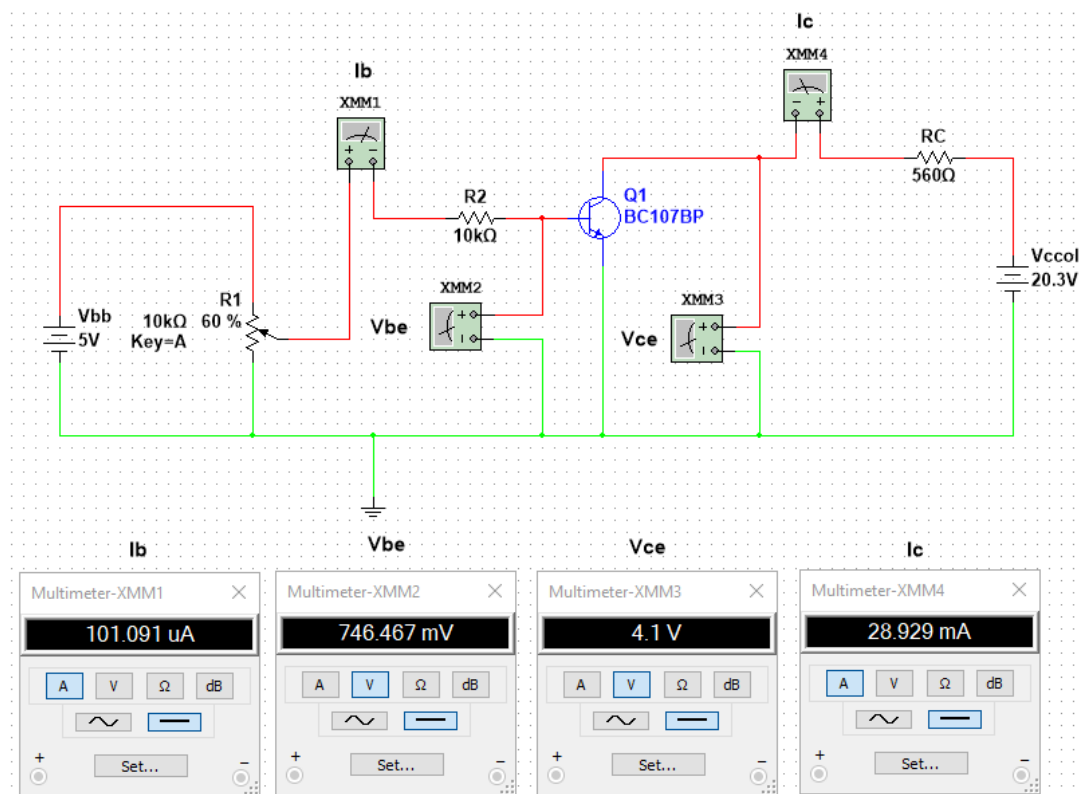


Figure 1.1.25 $I_b = 100 \mu\text{A}$, $V_{ce} = 4 \text{ V}$

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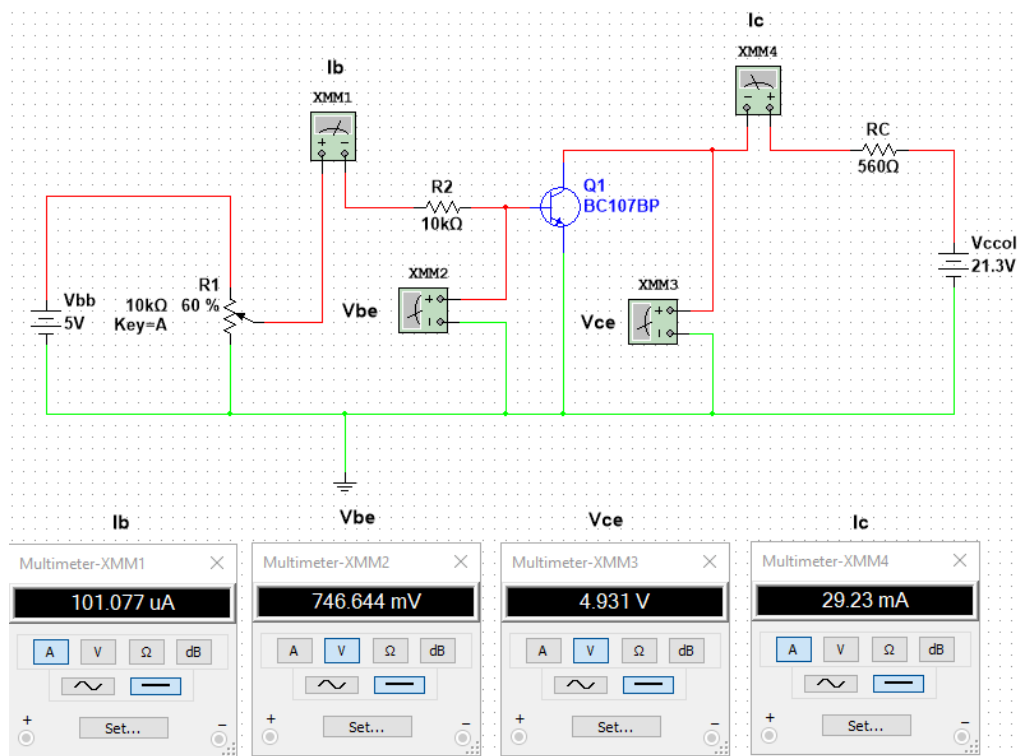


Figure 1.1.1.26 $I_b = 100$ mA, $V_{ce} = 5$ Volt

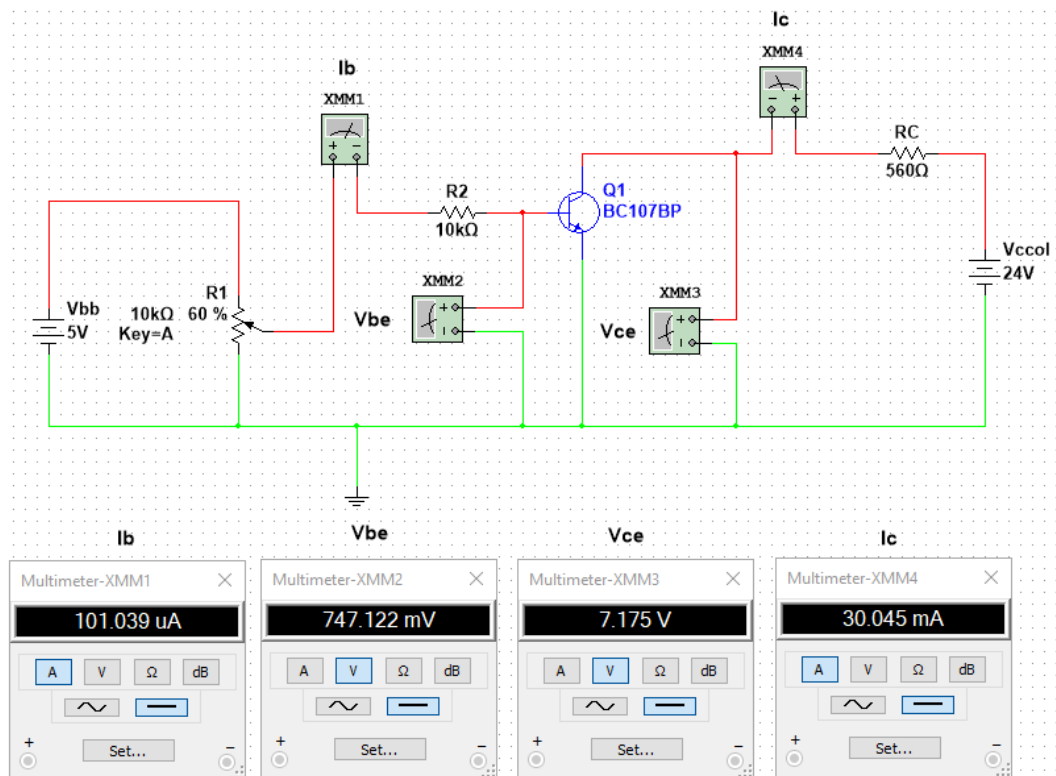


Figure 1.1.27 $I_b = 100$ mA, $V_{ce} = 7$ Volt

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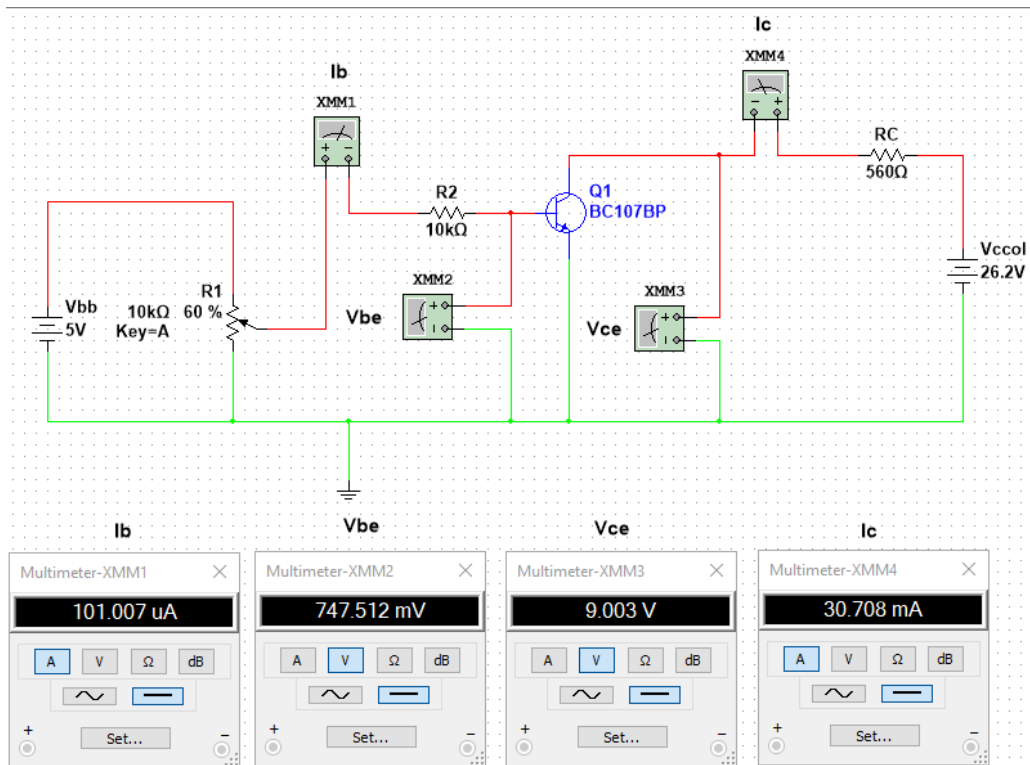


Figure 1.1.28 $I_b = 100 \text{ mA}$, $V_{ce} = 9 \text{ Volt}$

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Experimental Solution

Figure 1.1.29 shows the experimental illustration of the common emitter circuit on the breadboard.

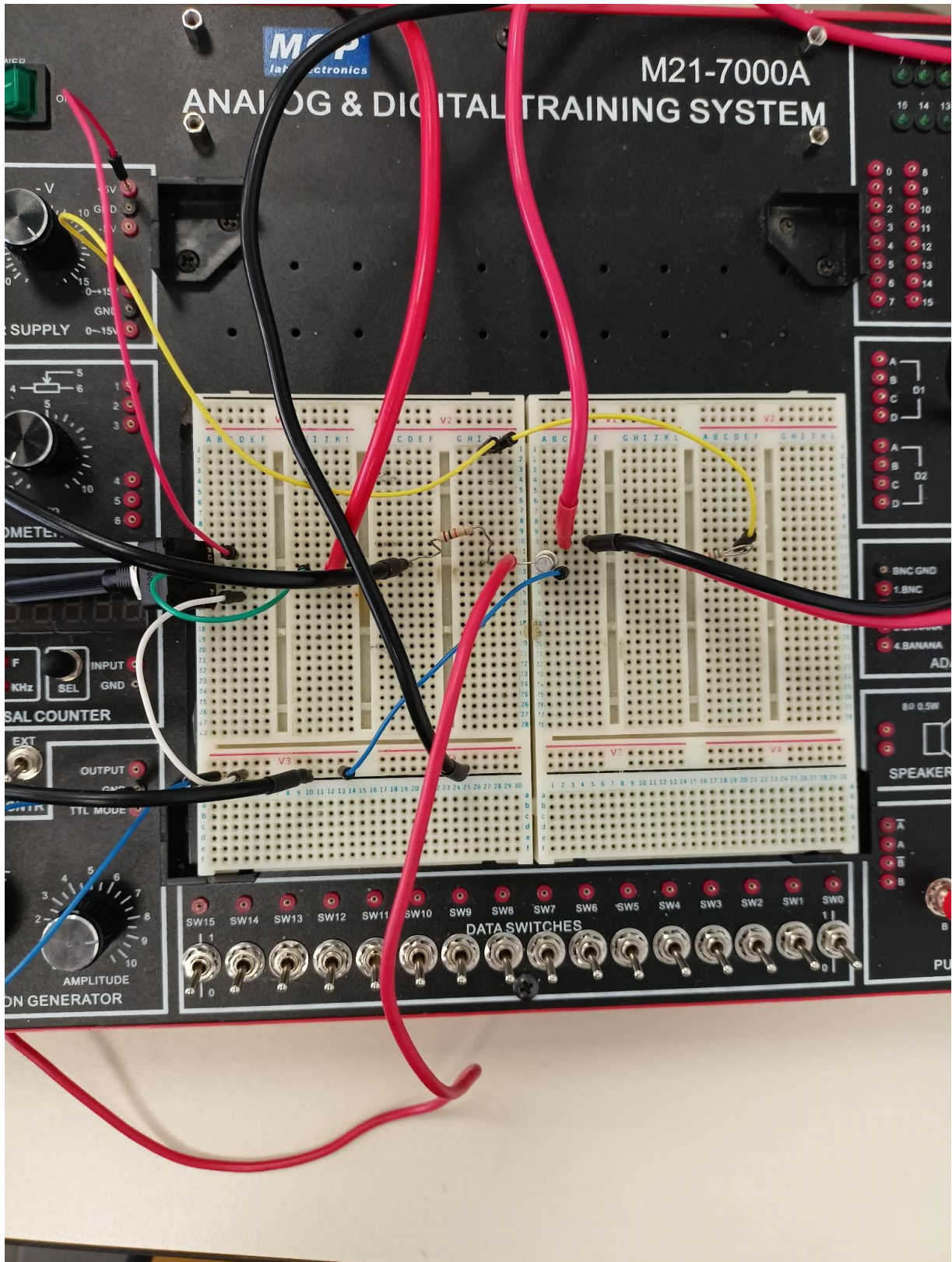


Figure 1.1.29 Experimental illustration of the common emitter circuit on the breadboard

Figure 1.1.30 shows the common emitter circuit with a base current $I_b = 30 \text{ mA}$.

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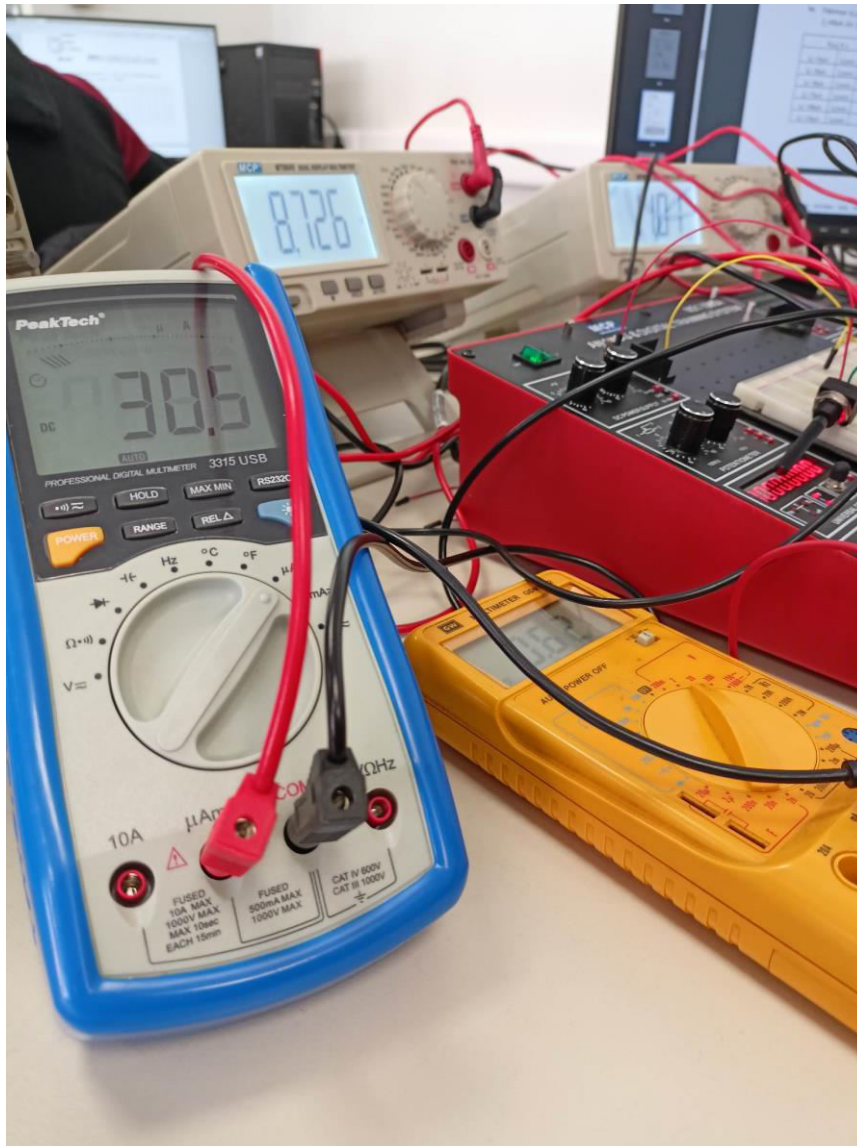


Figure 1.1.1.30 Experimental illustration of the common emitter circuit with $I_b = 30 \text{ mA}$ in "Multisim"

- The blue multimeter is set to calculate the base current in mA (I_b).
- The yellow multimeter is set to calculate the potential difference at the ends of the base and emitter in Volts (V_{be}).
- The left grey bench multimeter is set to calculate the potential difference at the ends of the collector and emitter in Volts (V_{ce}).
- The right bench multimeter is set to calculate the collector current in mA (I_c).

Figures 1.1.31 - 1.1.41 experimentally verify the values recorded in Table 1.1.1 for $I_b = 30 \text{ mA}$.

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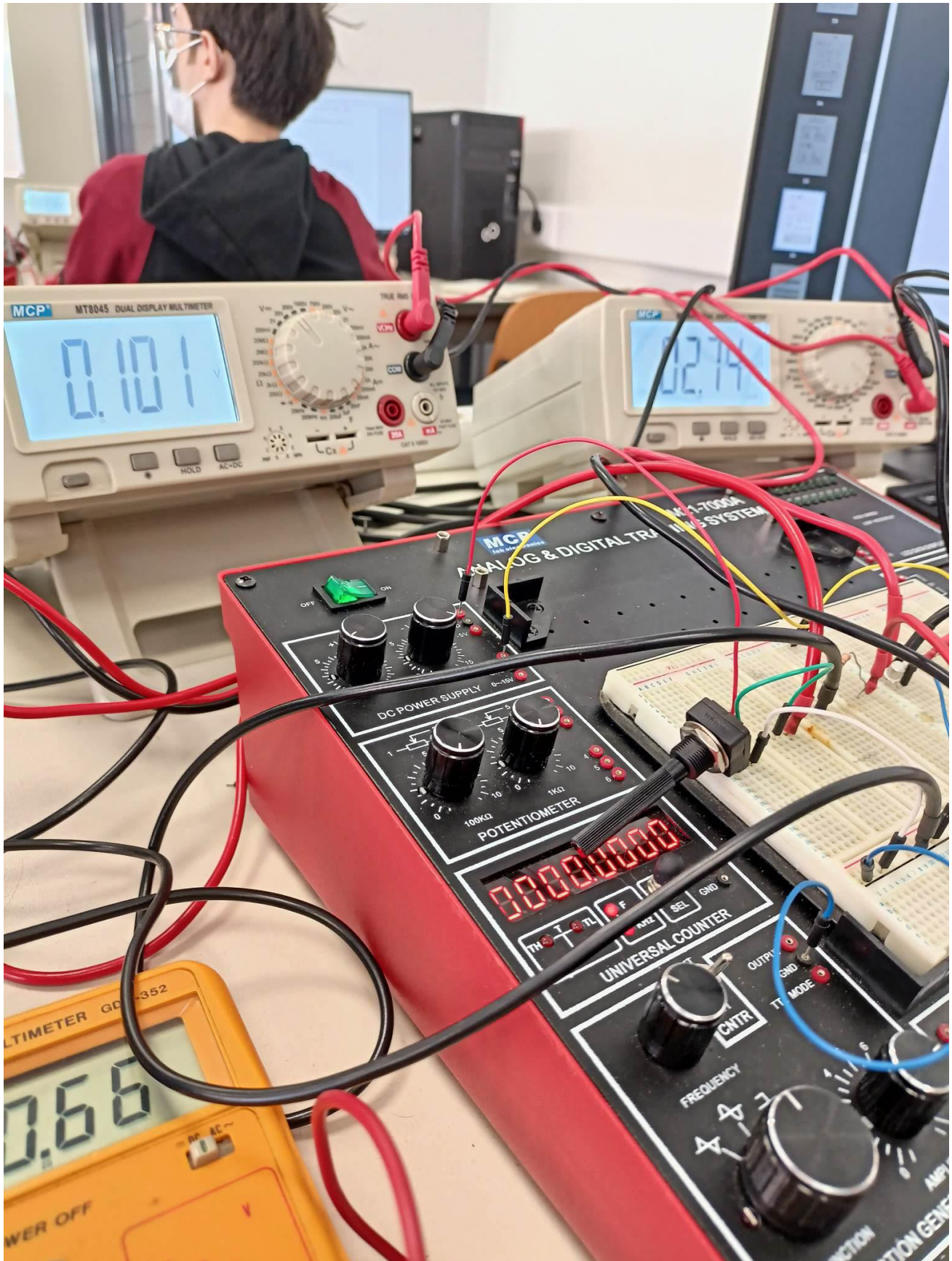


Figure 1.1.31 $I_b = 30 \text{ mA}$, $V_{ce} = 0.1 \text{ Volt}$

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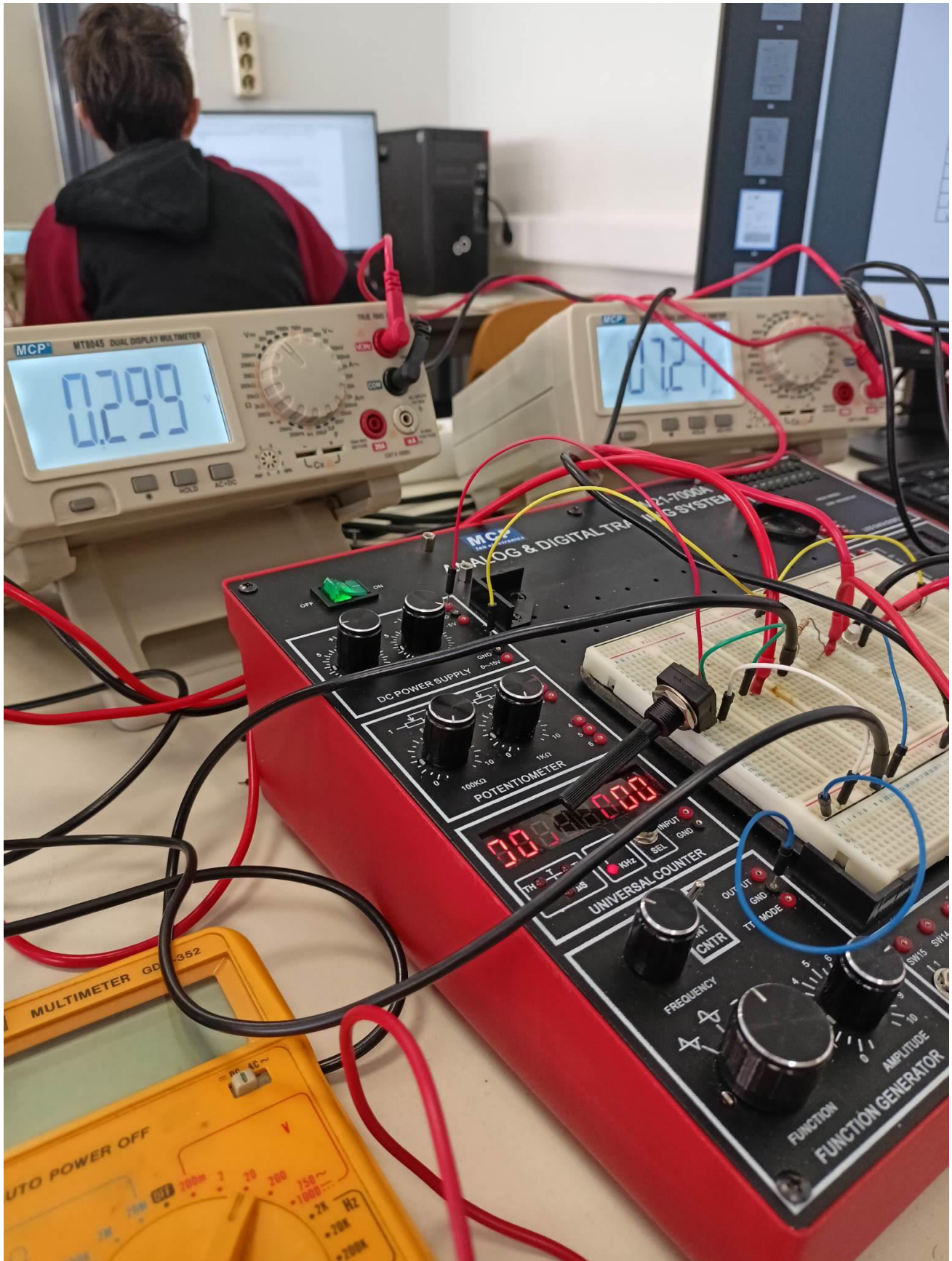


Figure 1.1.32 $I_b = 30 \text{ mA}$, $V_{ce} = 0.3 \text{ Volt}$

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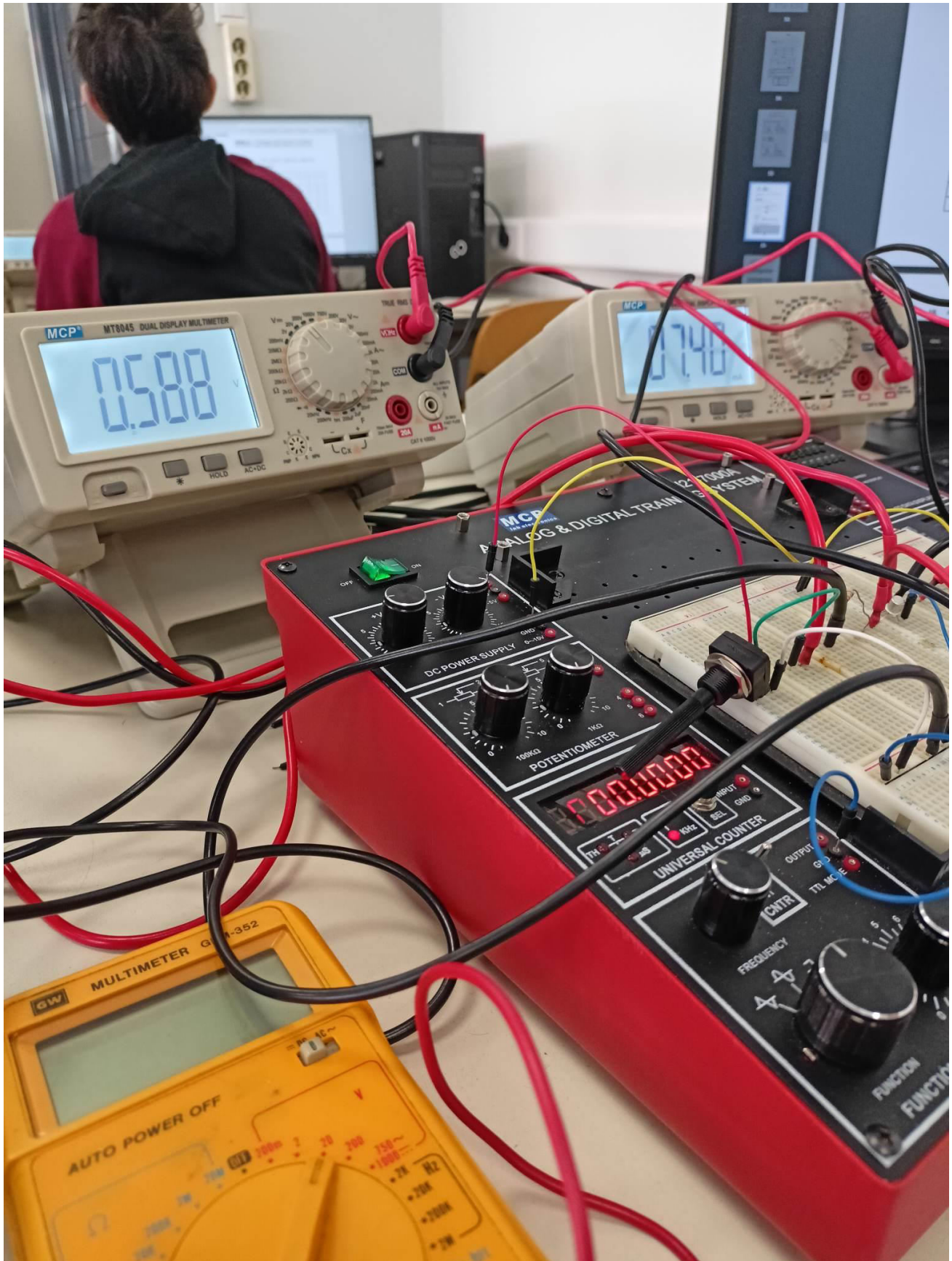


Figure 1.1.33 $I_b = 30 \text{ mA}$, $V_{ce} = 0.6 \text{ Volt}$

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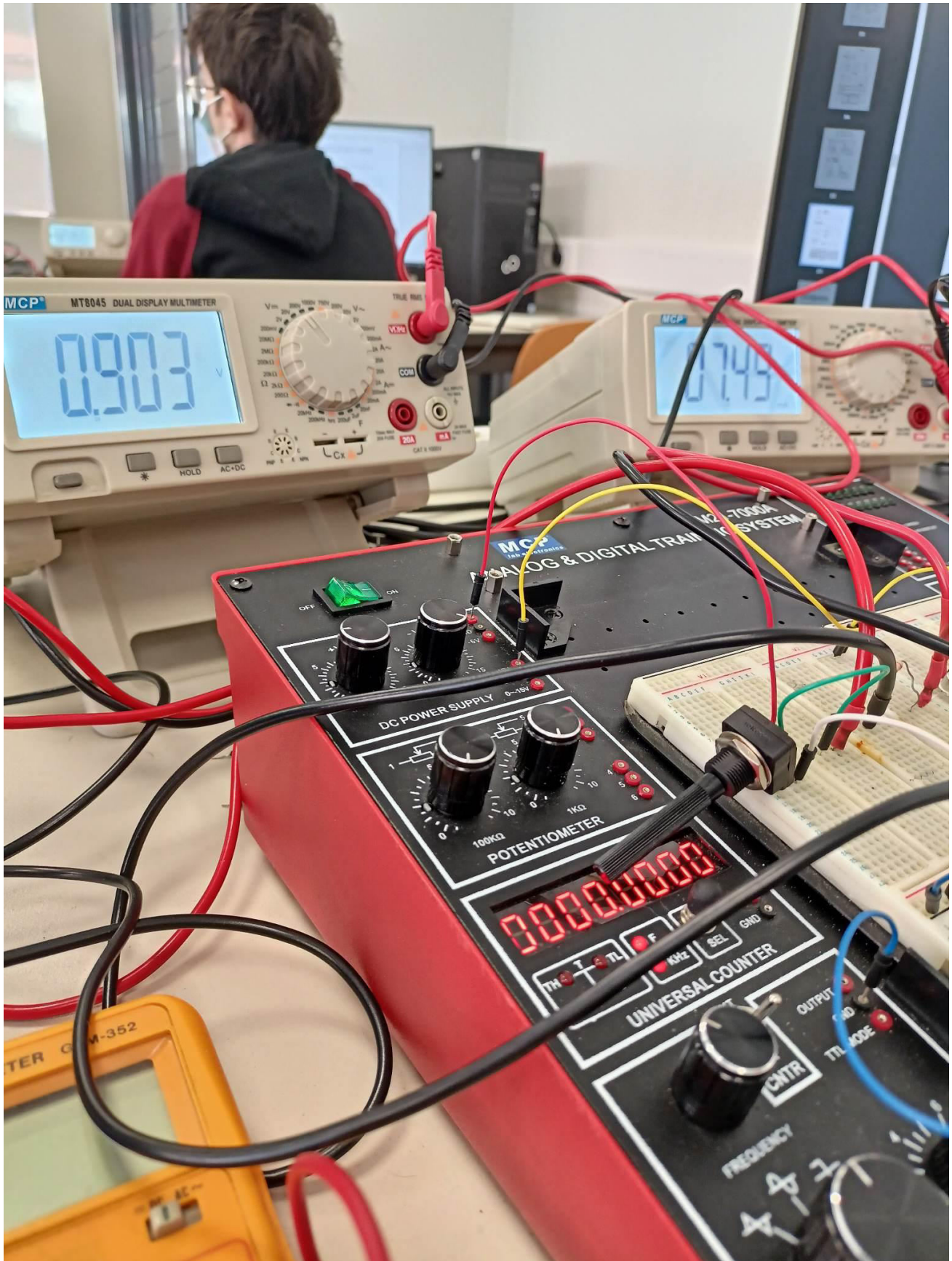


Figure 1.1.34 $I_b = 30 \text{ mA}$, $V_{ce} = 0.9 \text{ Volt}$

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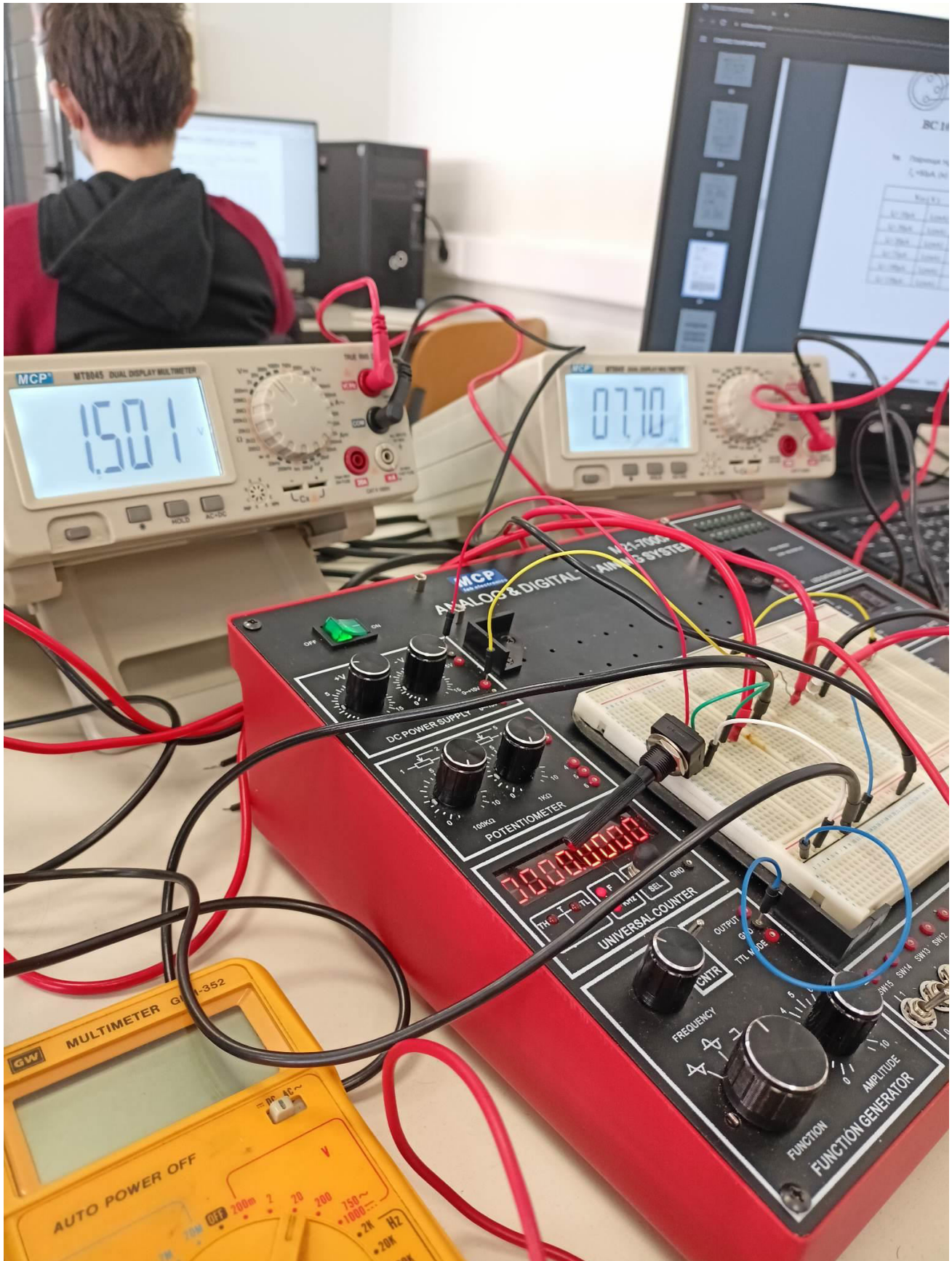


Figure 1.1.35 $I_b = 30 \text{ mA}$, $V_{ce} = 1.5 \text{ Volt}$

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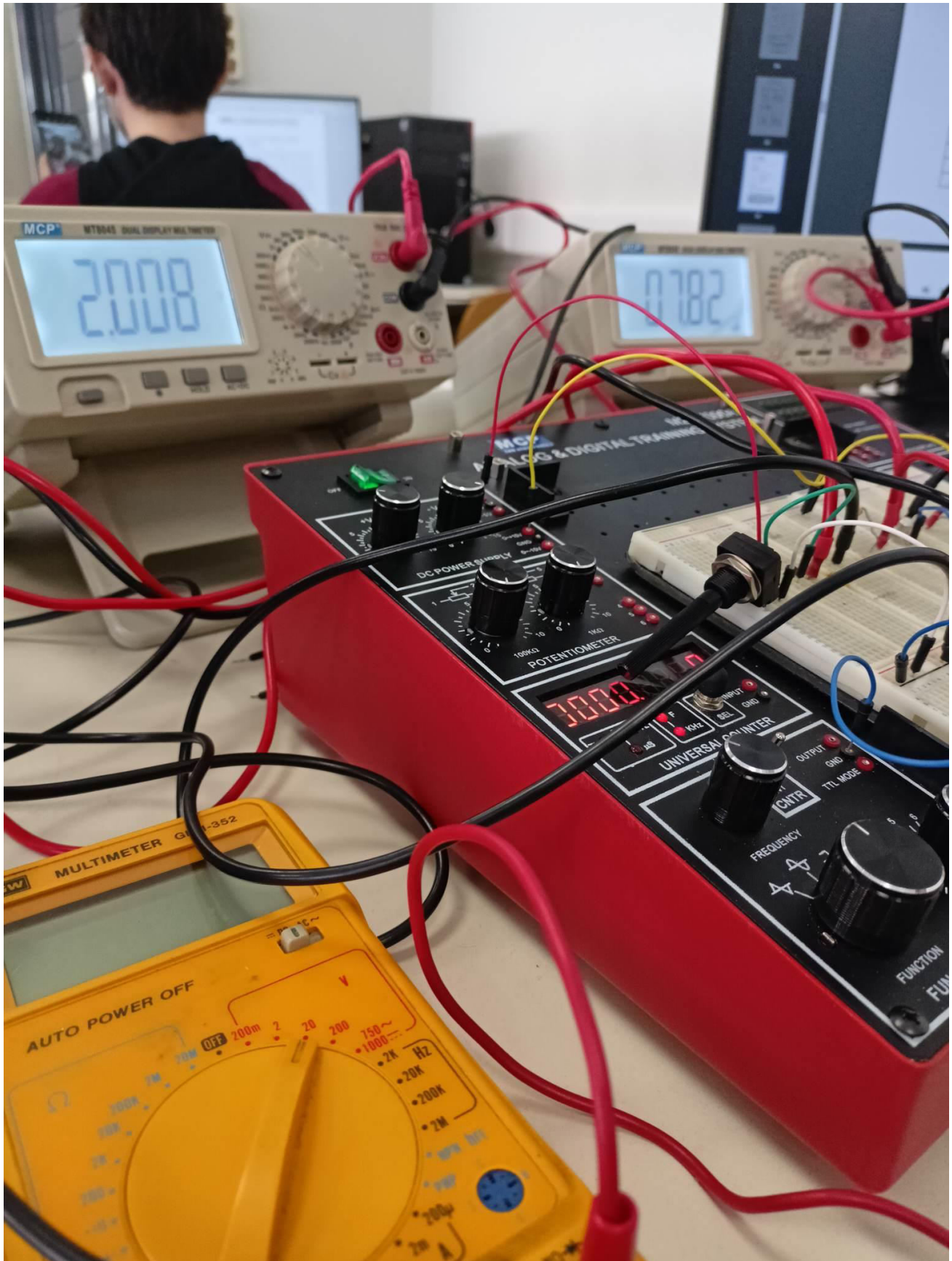


Figure 1.1.36 $I_b = 30 \text{ mA}$, $V_{ce} = 2 \text{ Volt}$

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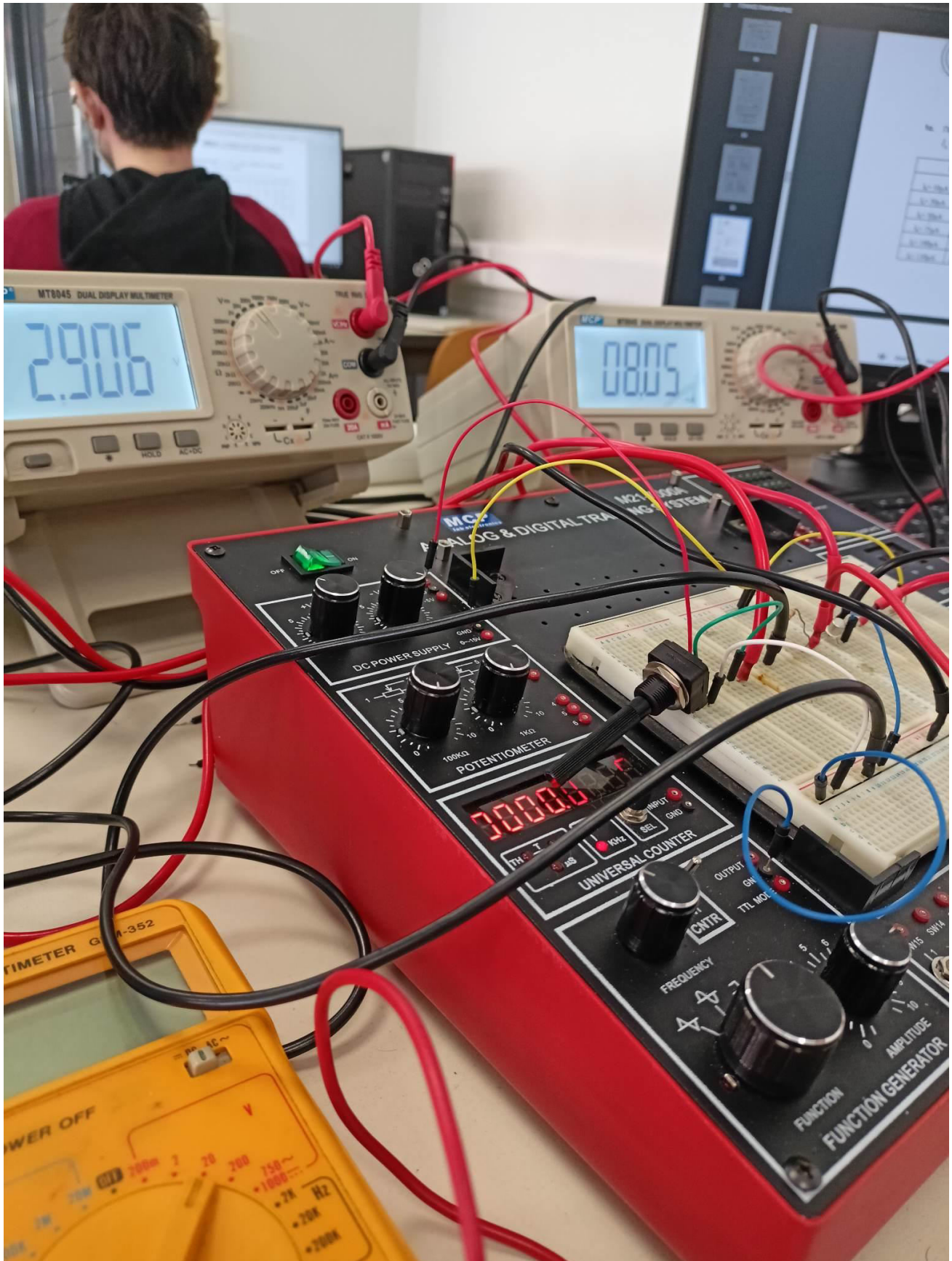


Figure 1.1.37 $I_b = 30 \text{ mA}$, $V_{ce} = 3 \text{ Volt}$

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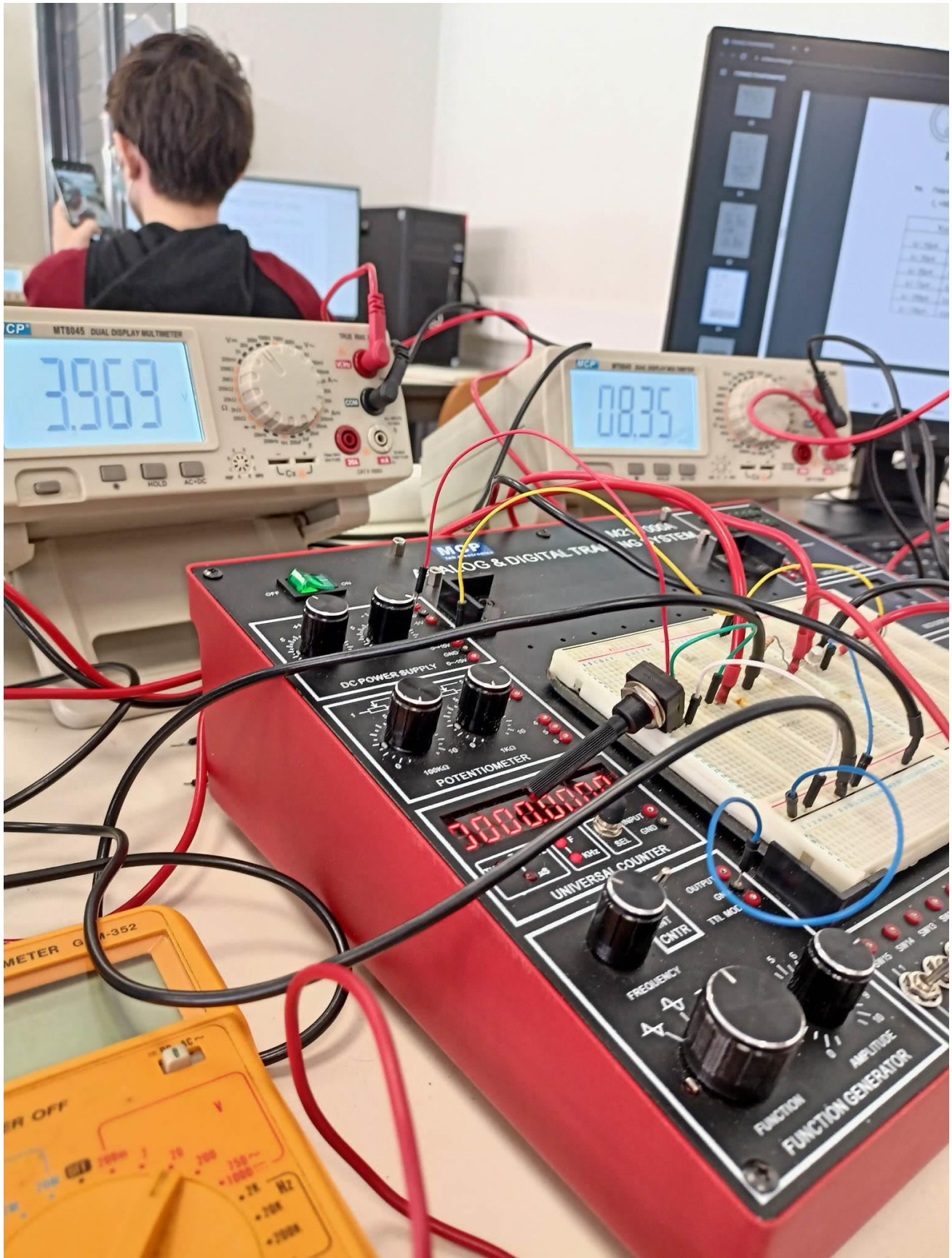


Figure 1.1.38 $I_b = 30 \text{ mA}$, $V_{ce} = 4 \text{ Volt}$

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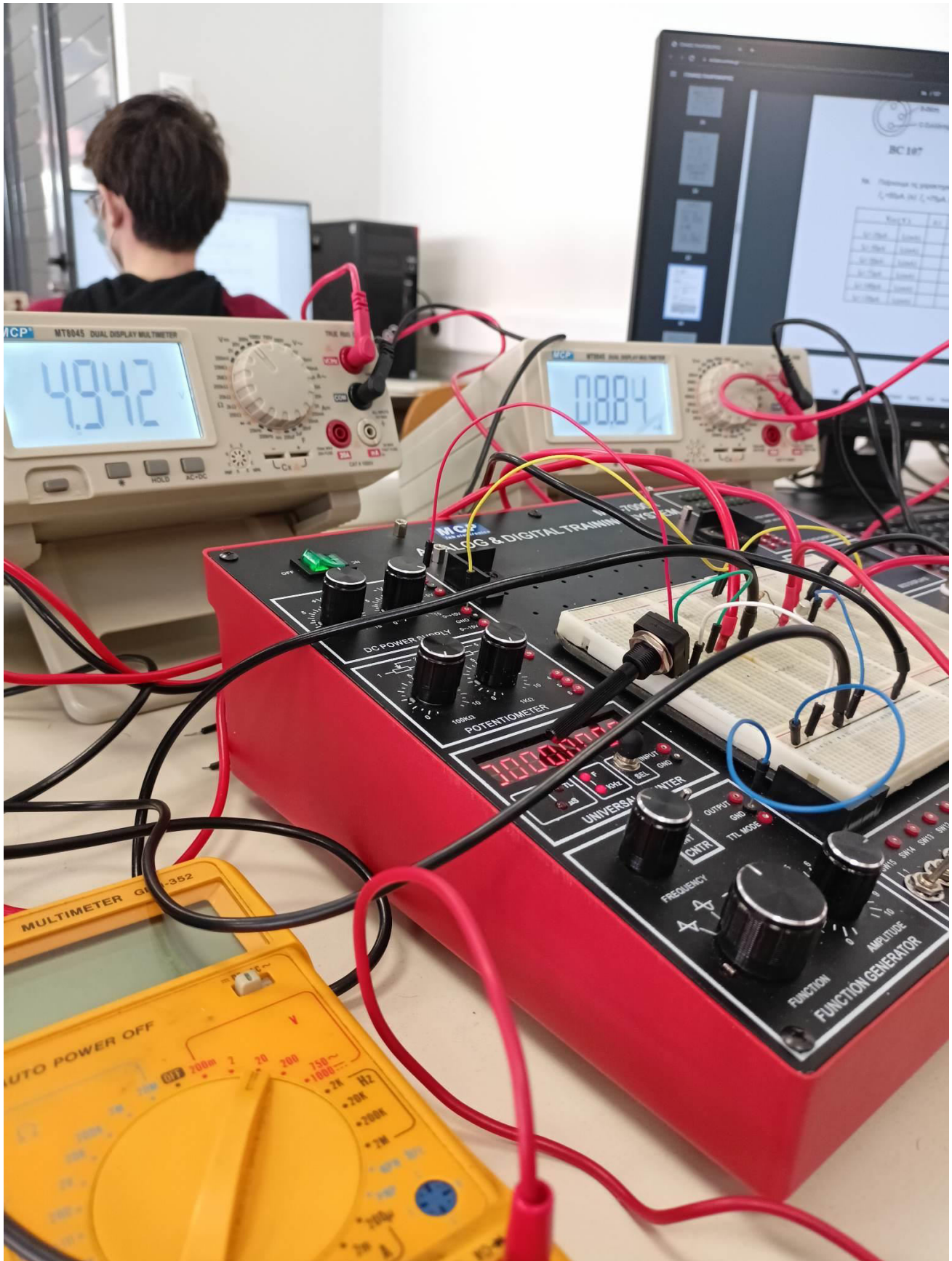


Figure 1.1.39 $I_b = 30 \text{ mA}$, $V_{ce} = 5 \text{ Volt}$

ELECTRONICS

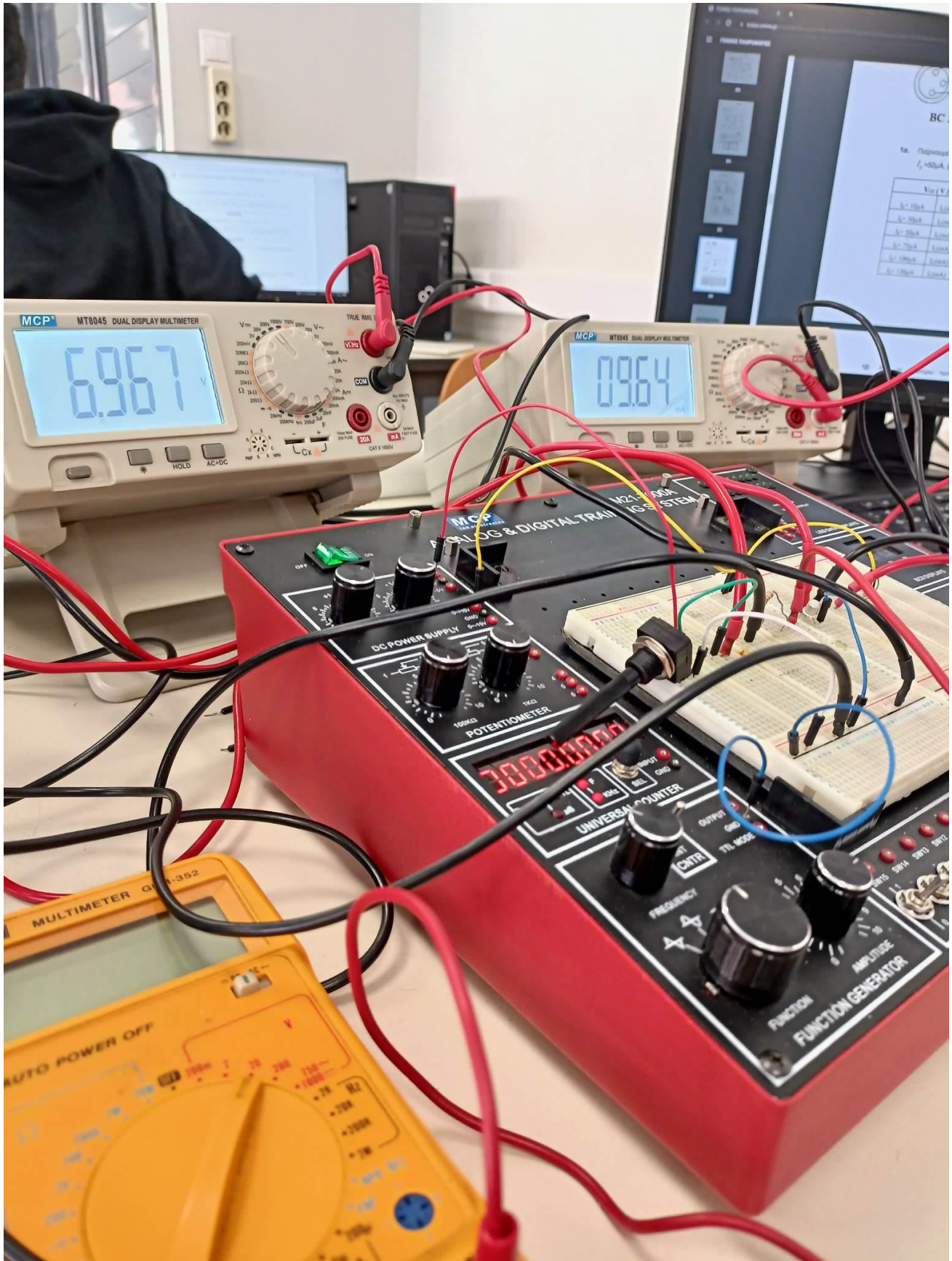


Figure 1.1.40 $I_b = 30 \text{ mA}$, $V_{ce} = 7 \text{ Volt}$

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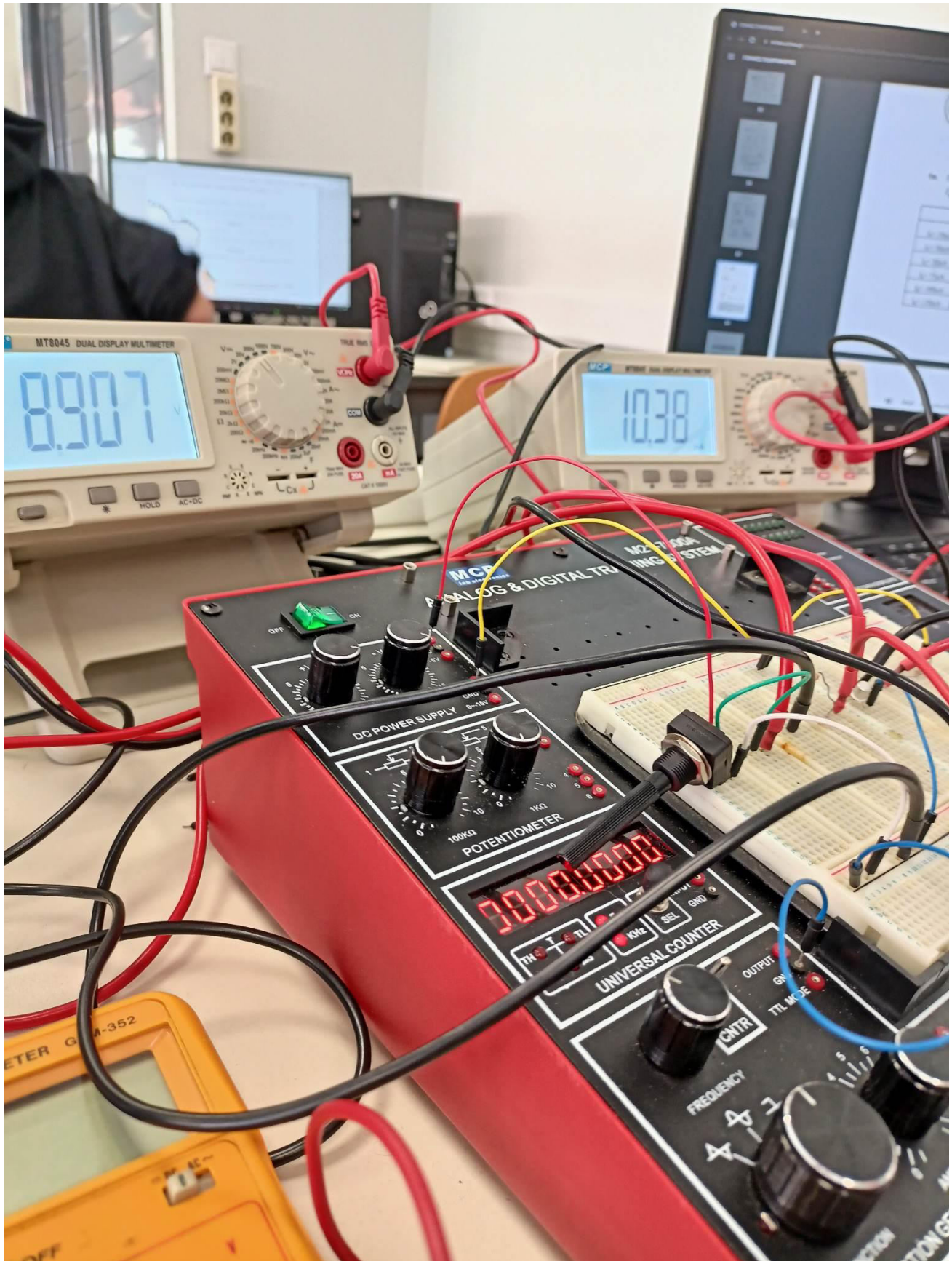


Figure 1.1.41 $I_b = 30 \text{ mA}$, $V_{ce} = 9 \text{ Volt}$

Figure 1.2.2 shows the input characteristics $I_b = f(V_{be})$ according to the values in Table 1.1.1. The characteristics are equivalent to those of PN contact crystalline diodes, since the transistor consists of two PN epitaxial layers in the base-collector and base-emitter.

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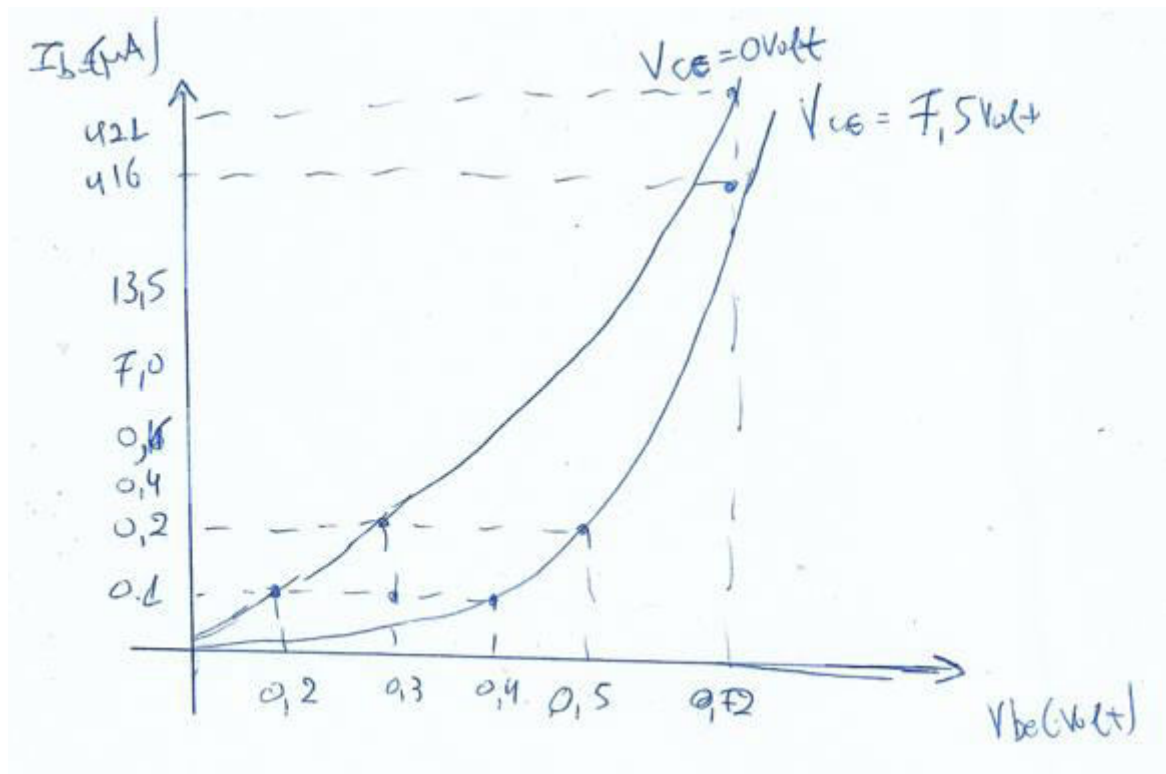


Figure 1.2.2 Input characteristics $I_b = f(V_{be})$

Simulative Resolution

Figure 1.2.3 shows the simulated representation of the common emitter circuit in the simulation software "Multisim"

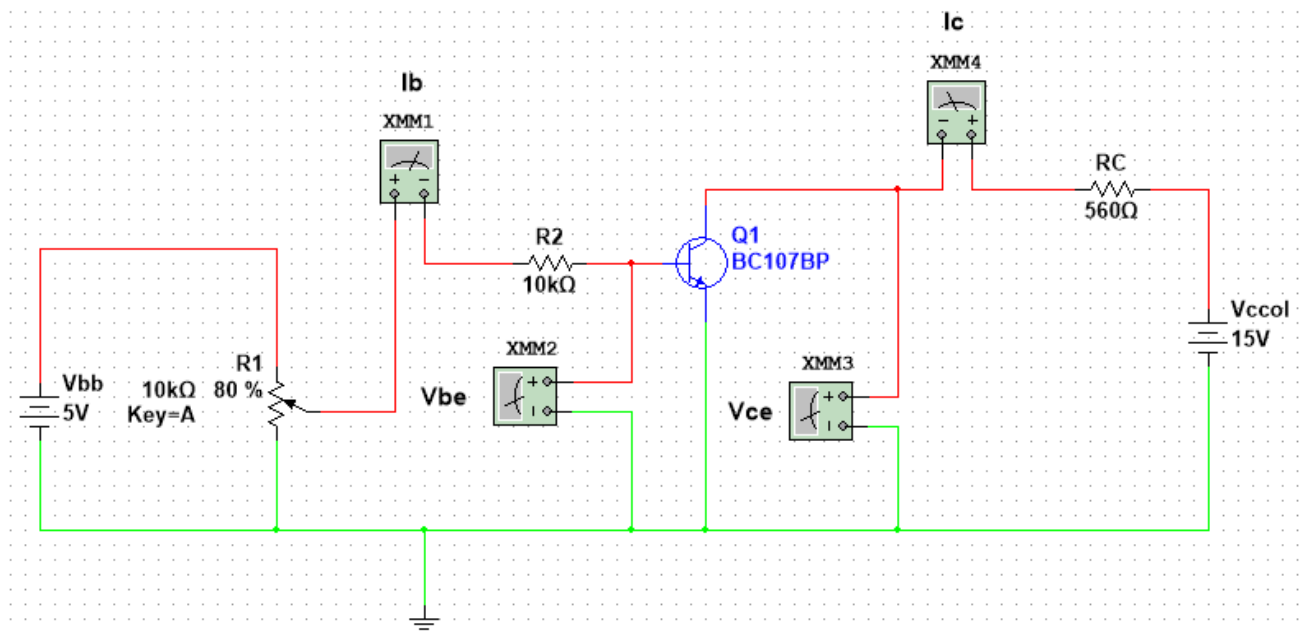


Figure 1.2.3 Simulation of the common emitter circuit in "Multisim"

Figure 1.2.4 shows the common emitter circuit with a potential difference at the collector-emitter ends (V_{ce}) equal to 0 Volt.

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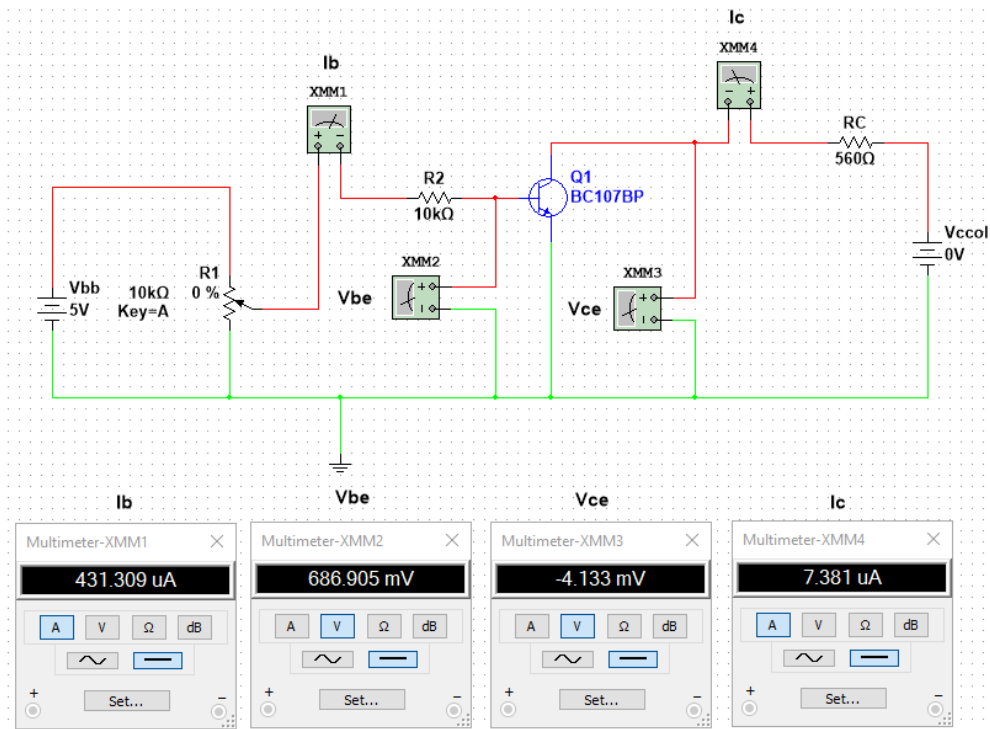


Figure 1.2.4 Simulation of the common emitter circuit with $V_{ce} = 0$ Volt in "Multisim"

Figure 1.2.5 shows the common emitter circuit with a potential difference at the collector-emitter ends (V_{ce}) equal to 7.5 Volt.

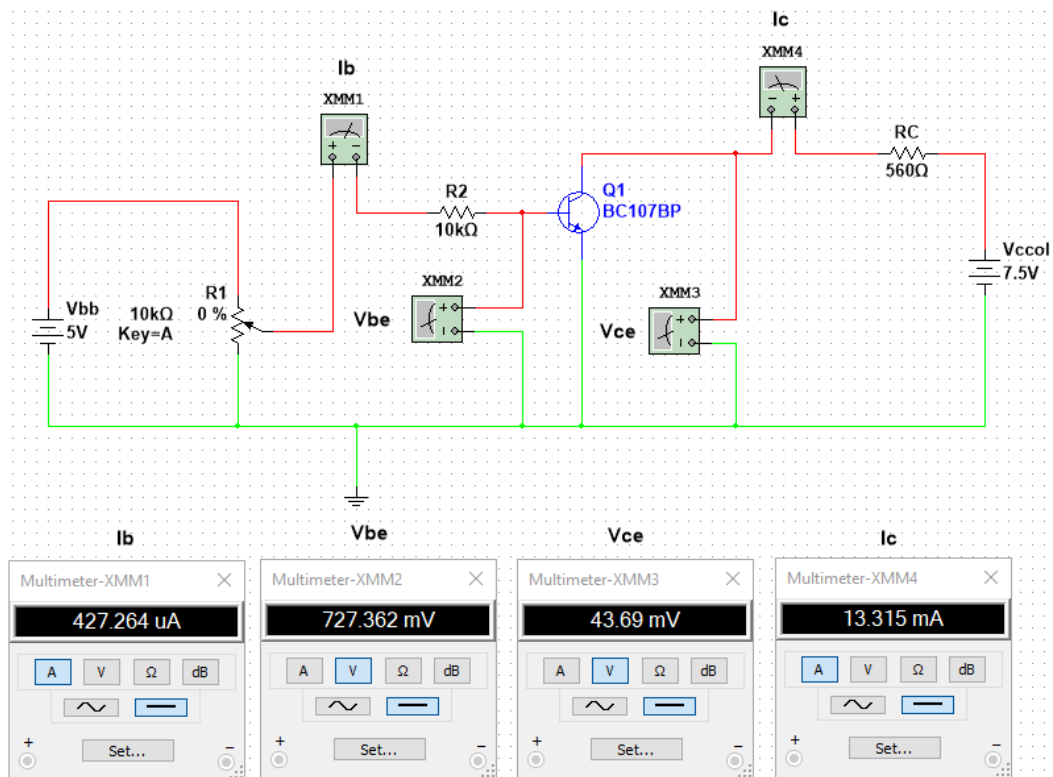


Figure 1.2.5 Simulation of the common emitter circuit with $V_{ce} = 7.5$ Volt in "Multisim"

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Experimental Solution

Figure 1.2.6 shows the experimental illustration of the common emitter circuit on the breadboard.

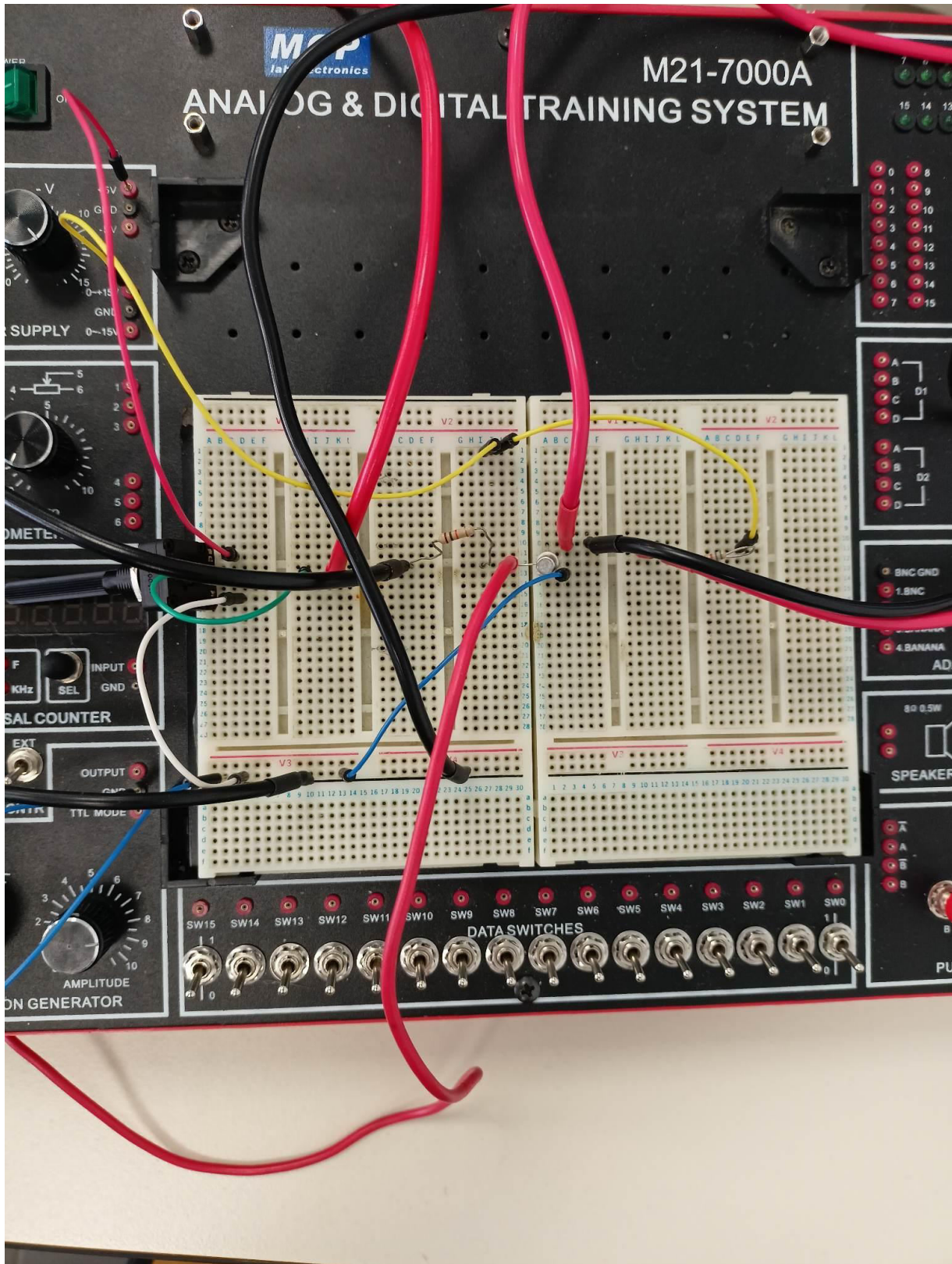


Figure 1.2.6 Experimental illustration of the common emitter circuit on the breadboard

Figures 1.2.7 - 1.2.15 experimentally verify the values recorded in Table 1.2.1 for $V_{ce} = 0$ Volt.

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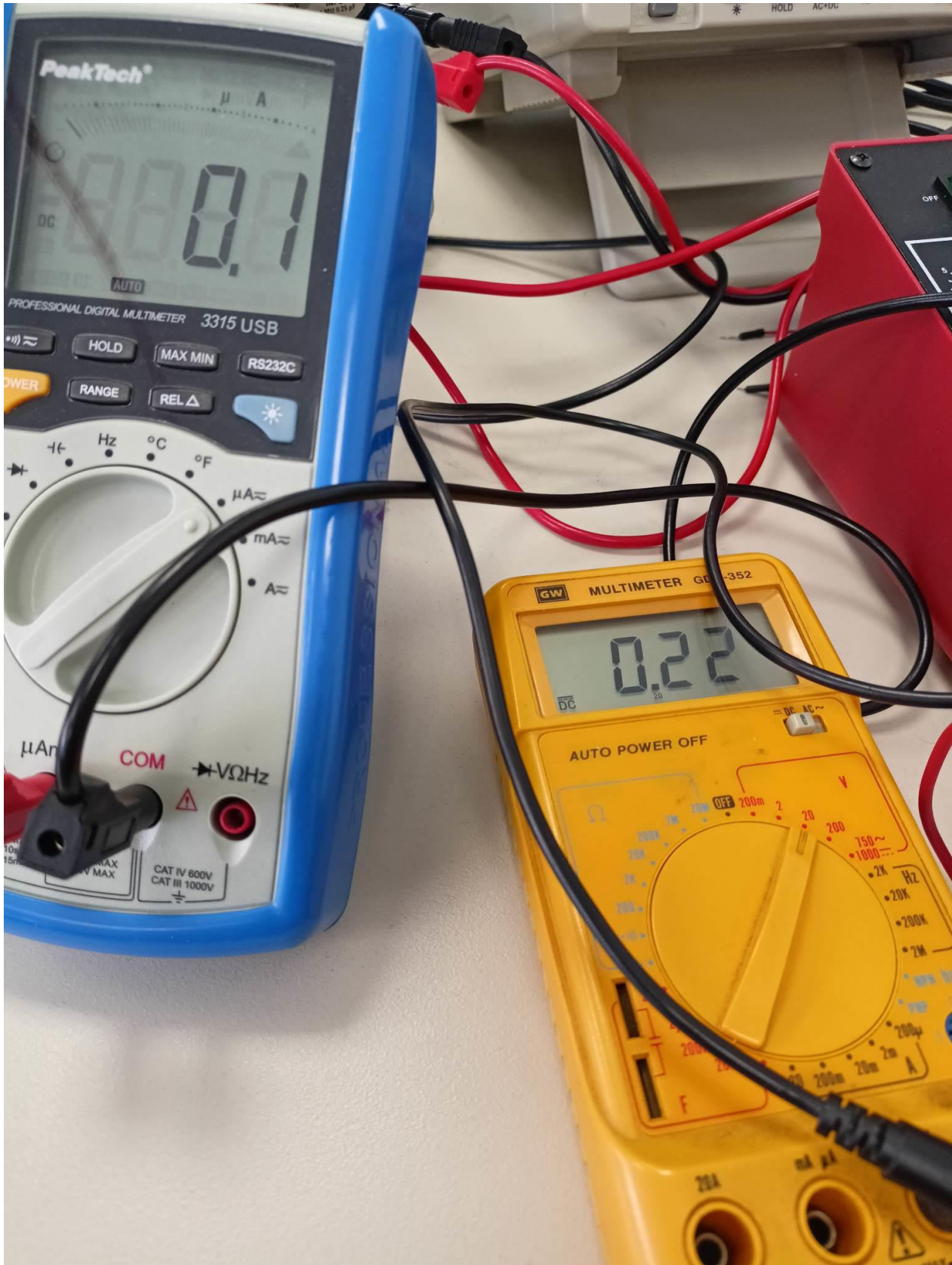


Figure 1.2.7 $V_{ce} = 0$ Volt, $V_{be} = 0.2$ Volt

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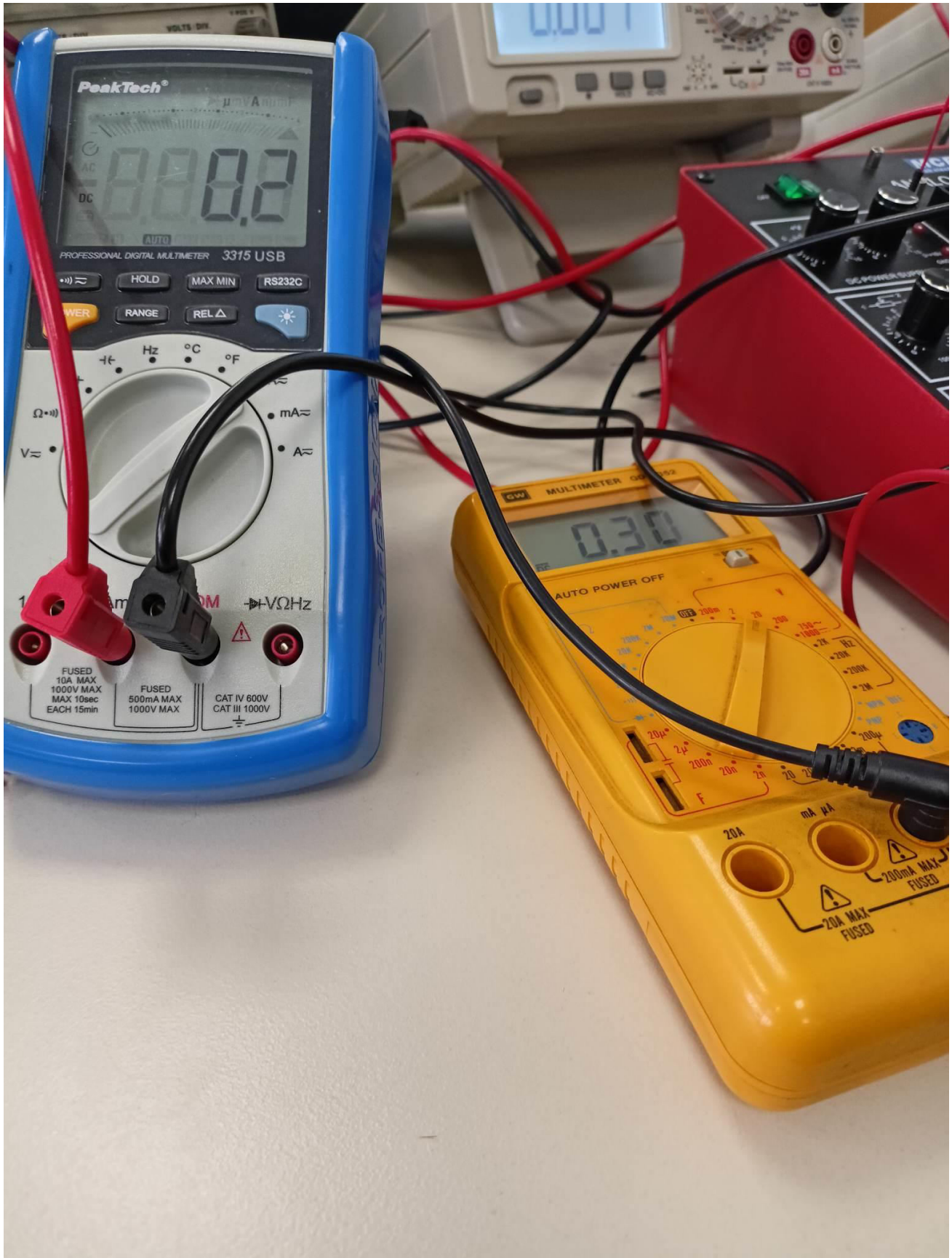


Figure 1.2.8 $V_{ce} = 0$ Volt, $V_{be} = 0.3$ Volt

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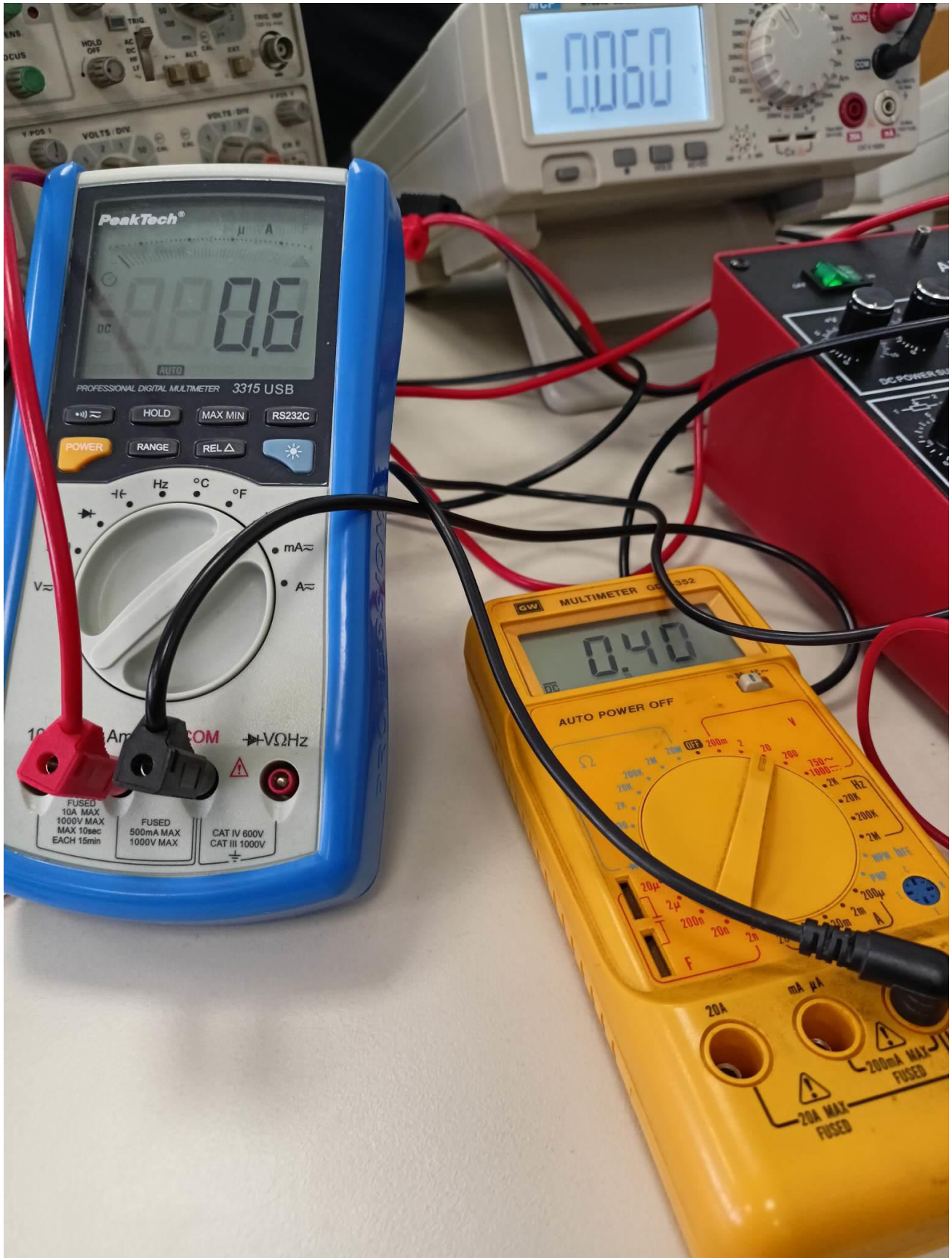


Figure 1.2.9 $V_{ce} = 0$ Volt, $V_{be} = 0.4$ Volt

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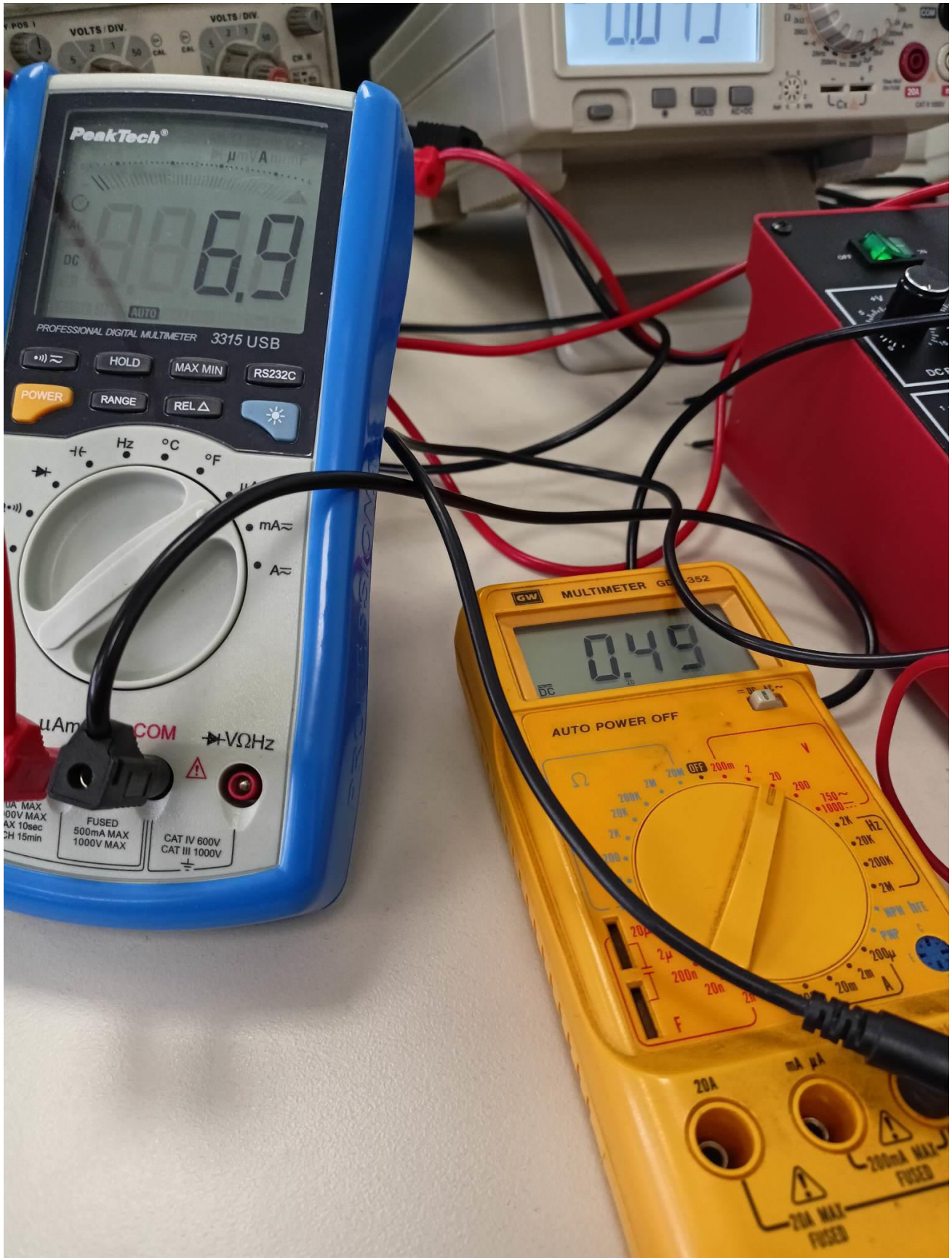


Figure 1.2.10 $V_{ce} = 0$ Volt, $V_{be} = 0.5$ Volt

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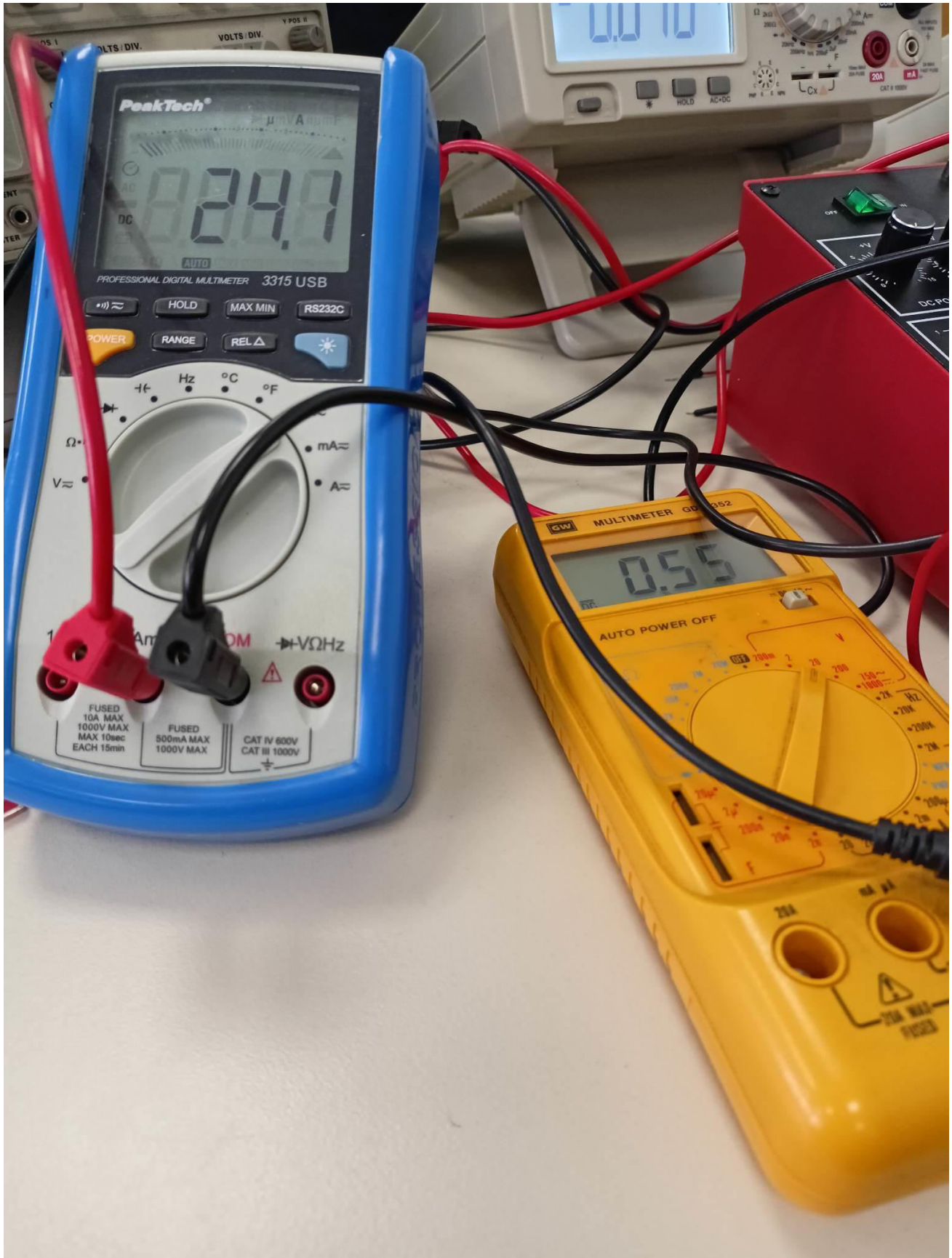


Figure 1.2.11 $V_{ce} = 0$ Volt, $V_{be} = 0.55$ Volt

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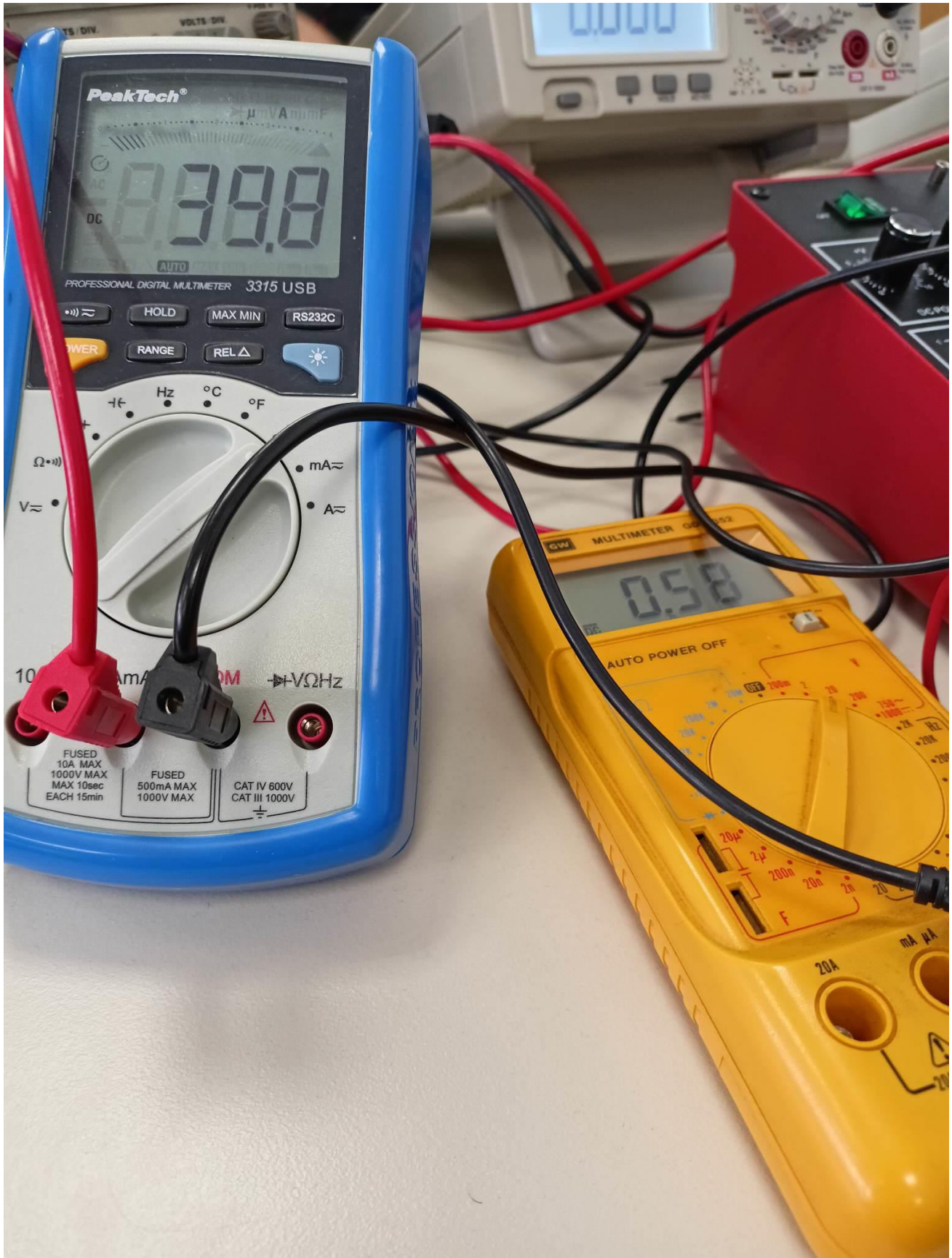


Figure 1.2.12 $V_{ce} = 0$ Volt, $V_{be} = 0.58$ Volt

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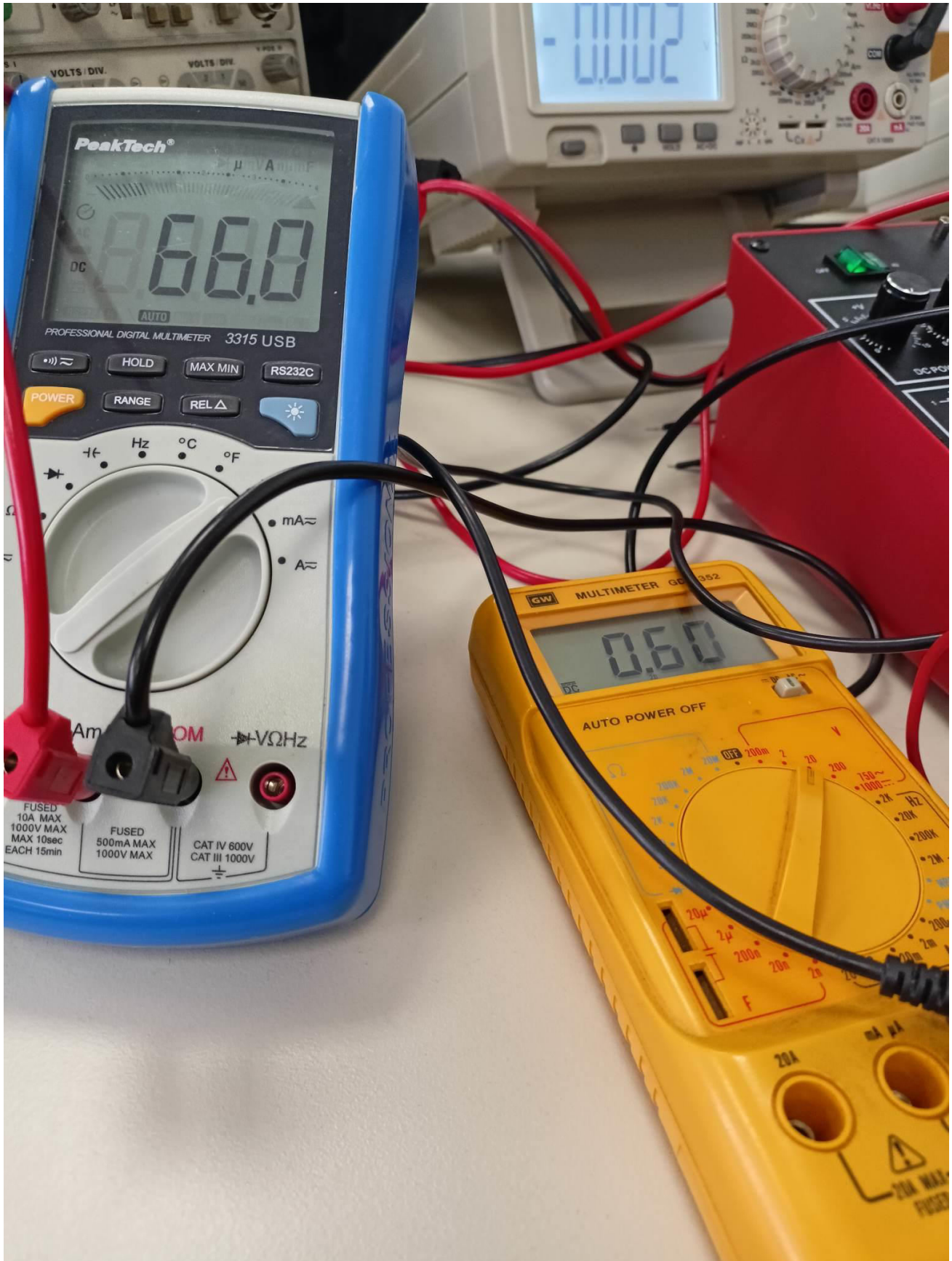


Figure 1.2.13 $V_{ce} = 0$ Volt, $V_{be} = 0.6$ Volt

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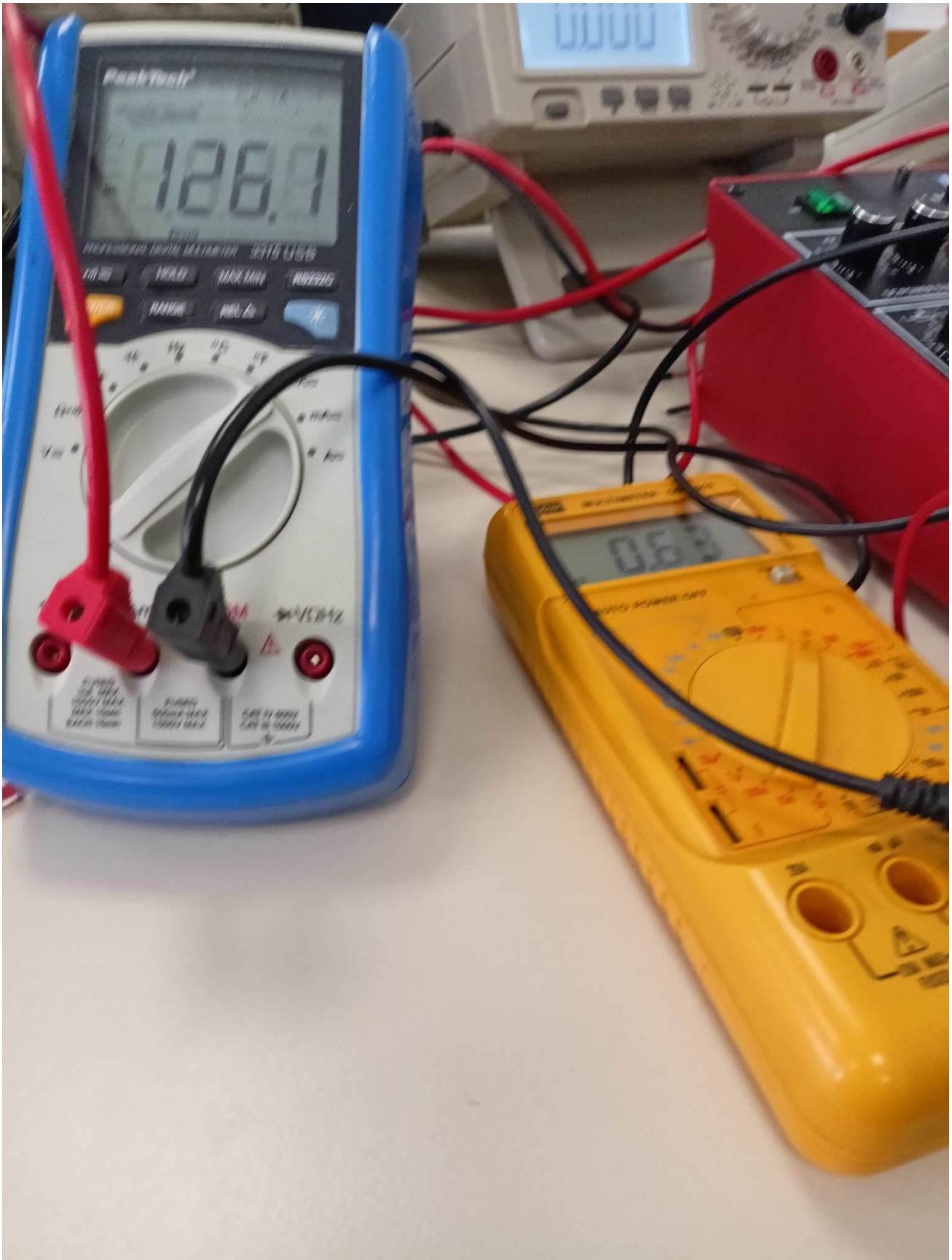


Figure 1.2.14 $V_{ce} = 0$ Volt, $V_{be} = 0.63$ Volt

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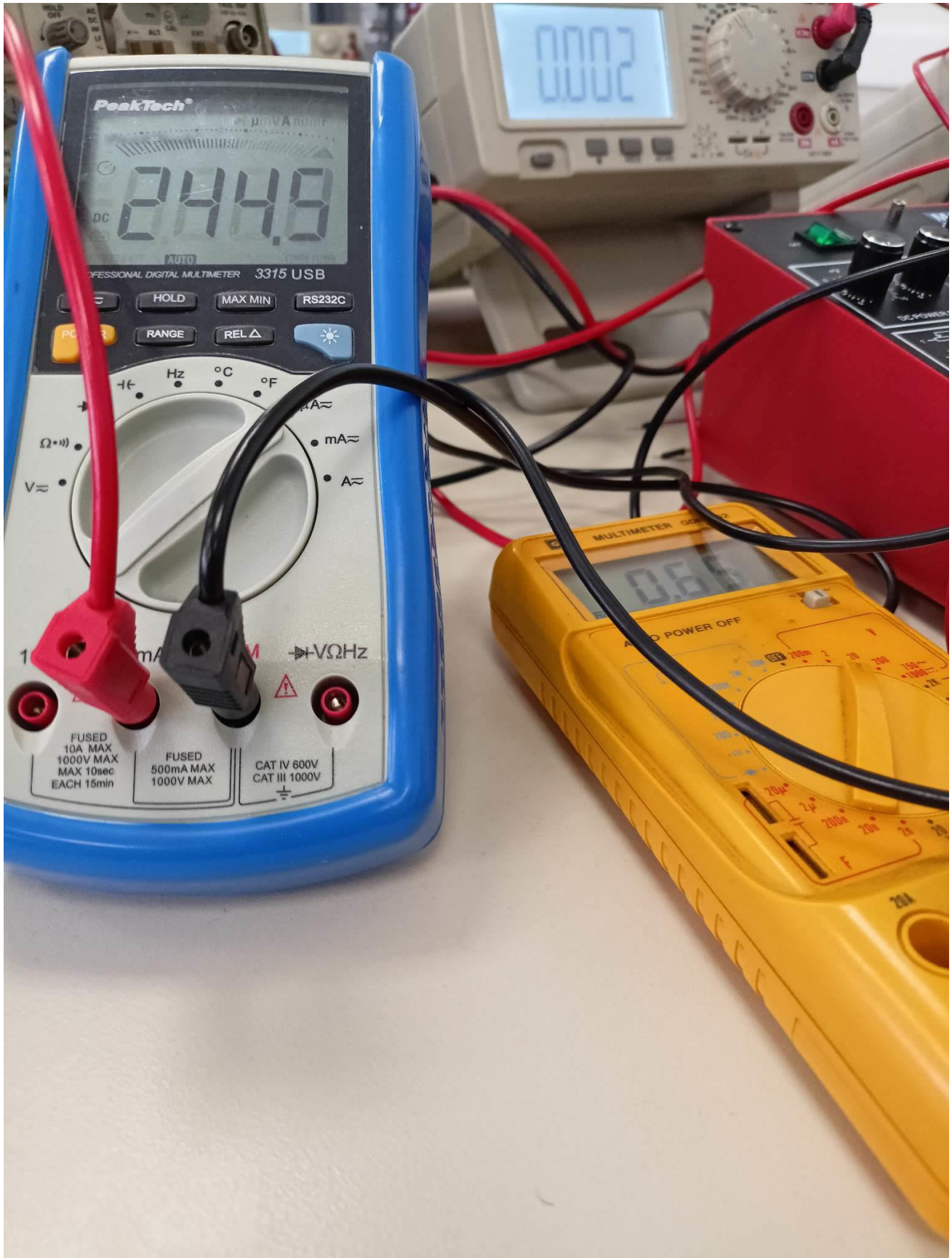


Figure 1.2.15 $V_{ce} = 0$ Volt, $V_{be} = 0.65$ Volt

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Figures 1.2.16 - 1.2.26 experimentally verify the values recorded in Table 1.2.2 for $V_{ce} = 7.5$ Volt.

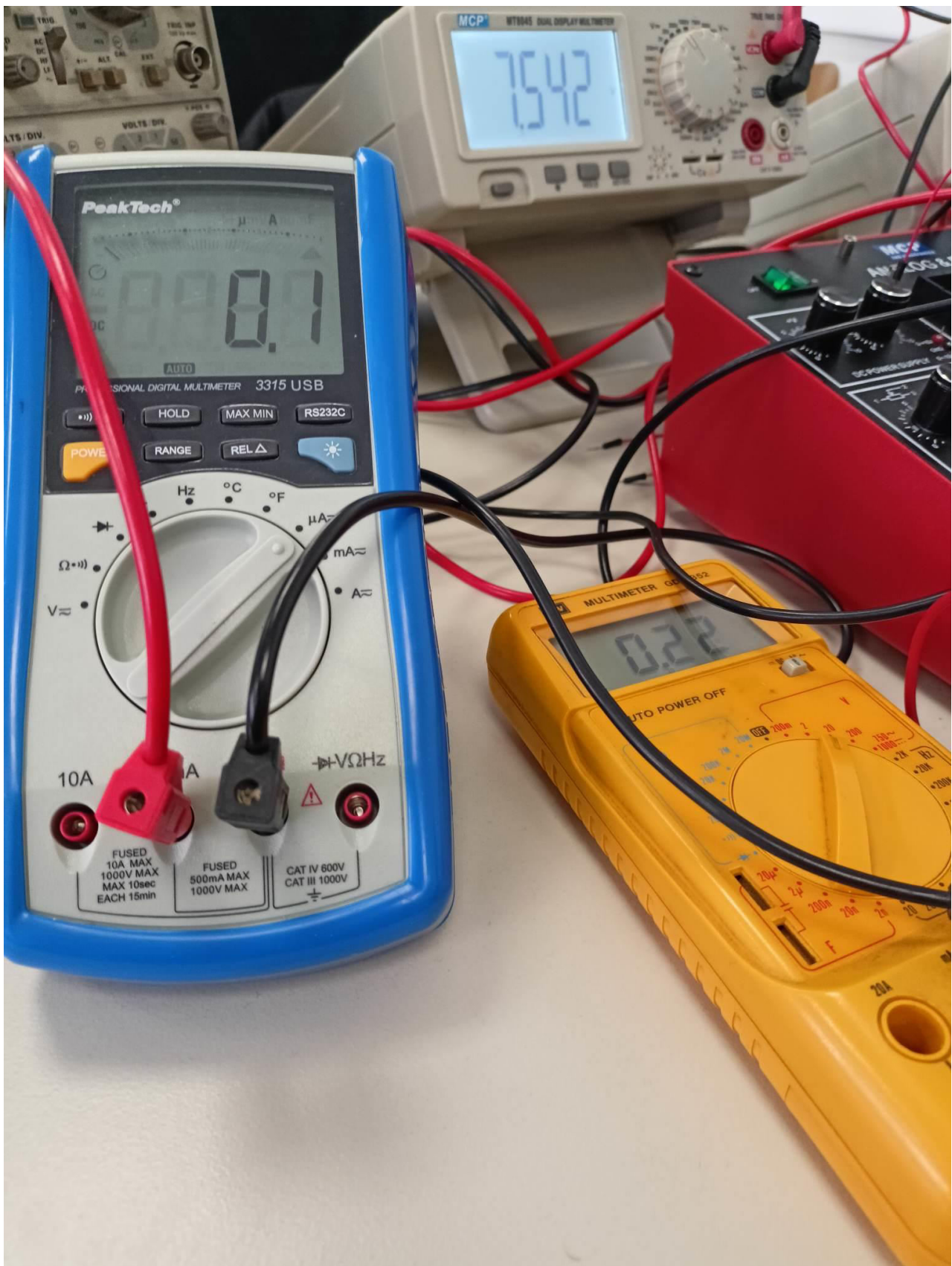


Figure 1.2.16 $V_{ce} = 7.5$ Volt, $V_{be} = 0.2$ Volt

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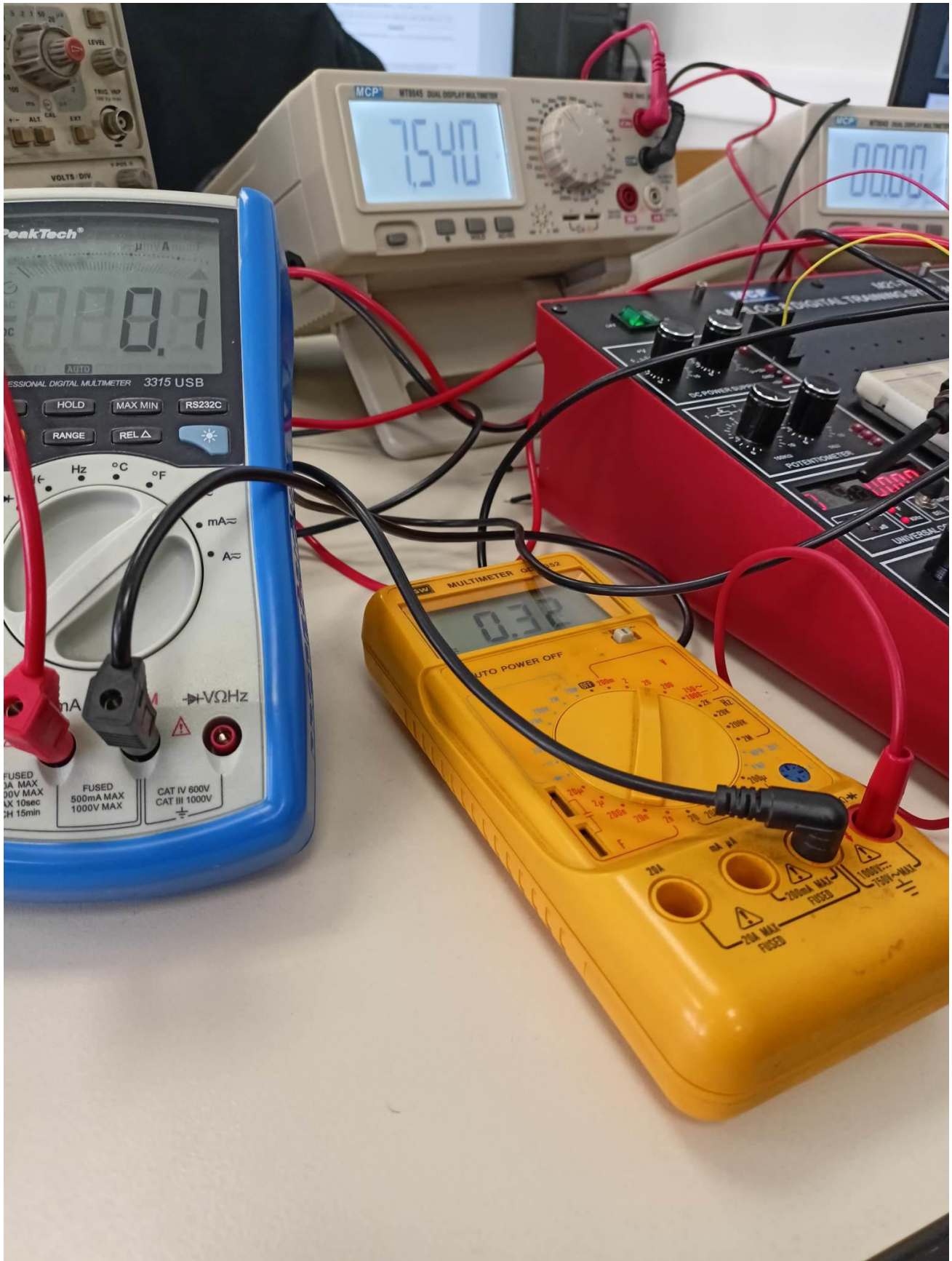


Figure 1.2.17 $V_{ce} = 7.5$ Volt, $V_{be} = 0.3$ Volt

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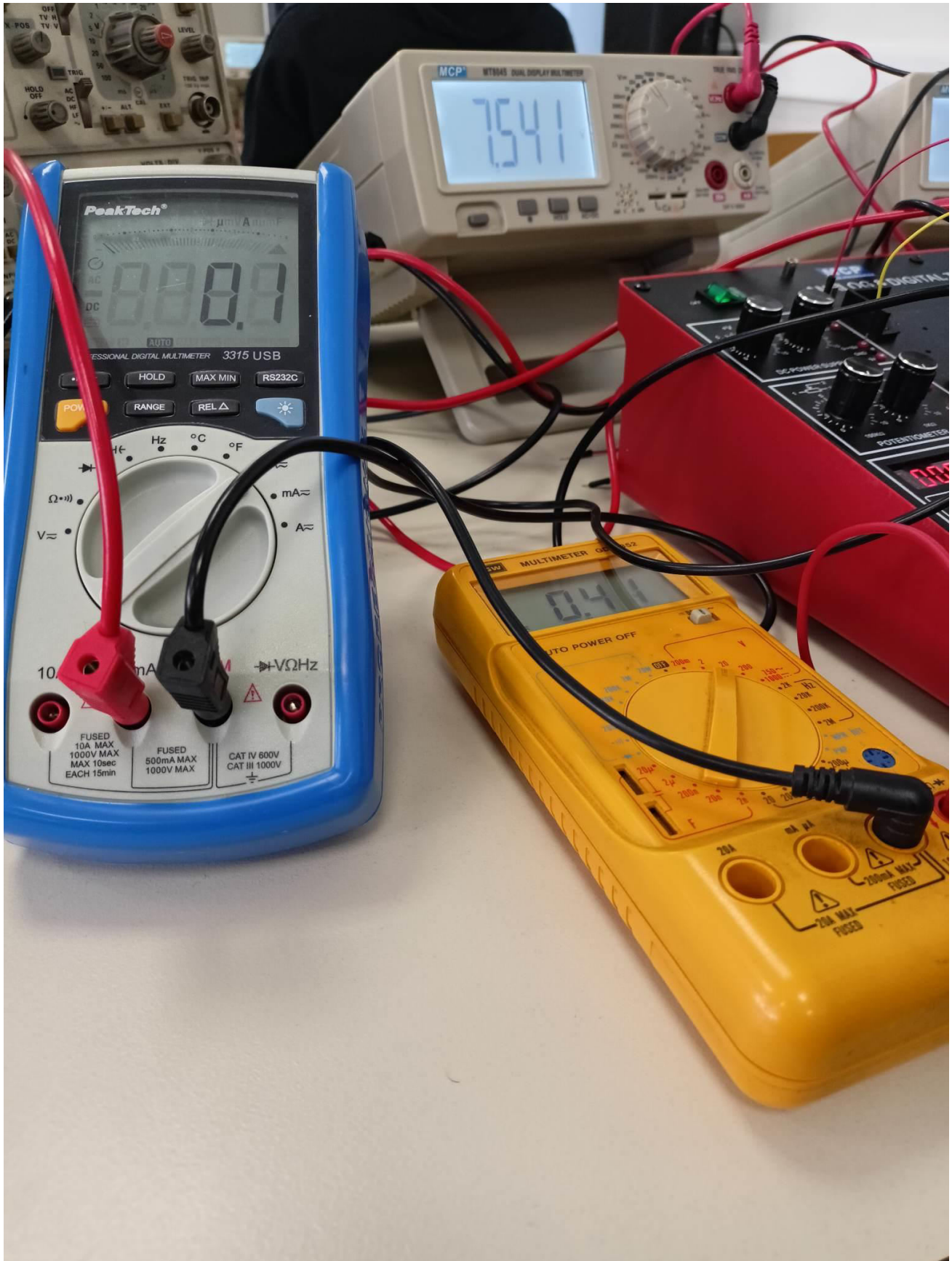


Figure 1.2.18 $V_{ce} = 7.5$ Volt, $V_{be} = 0.4$ Volt

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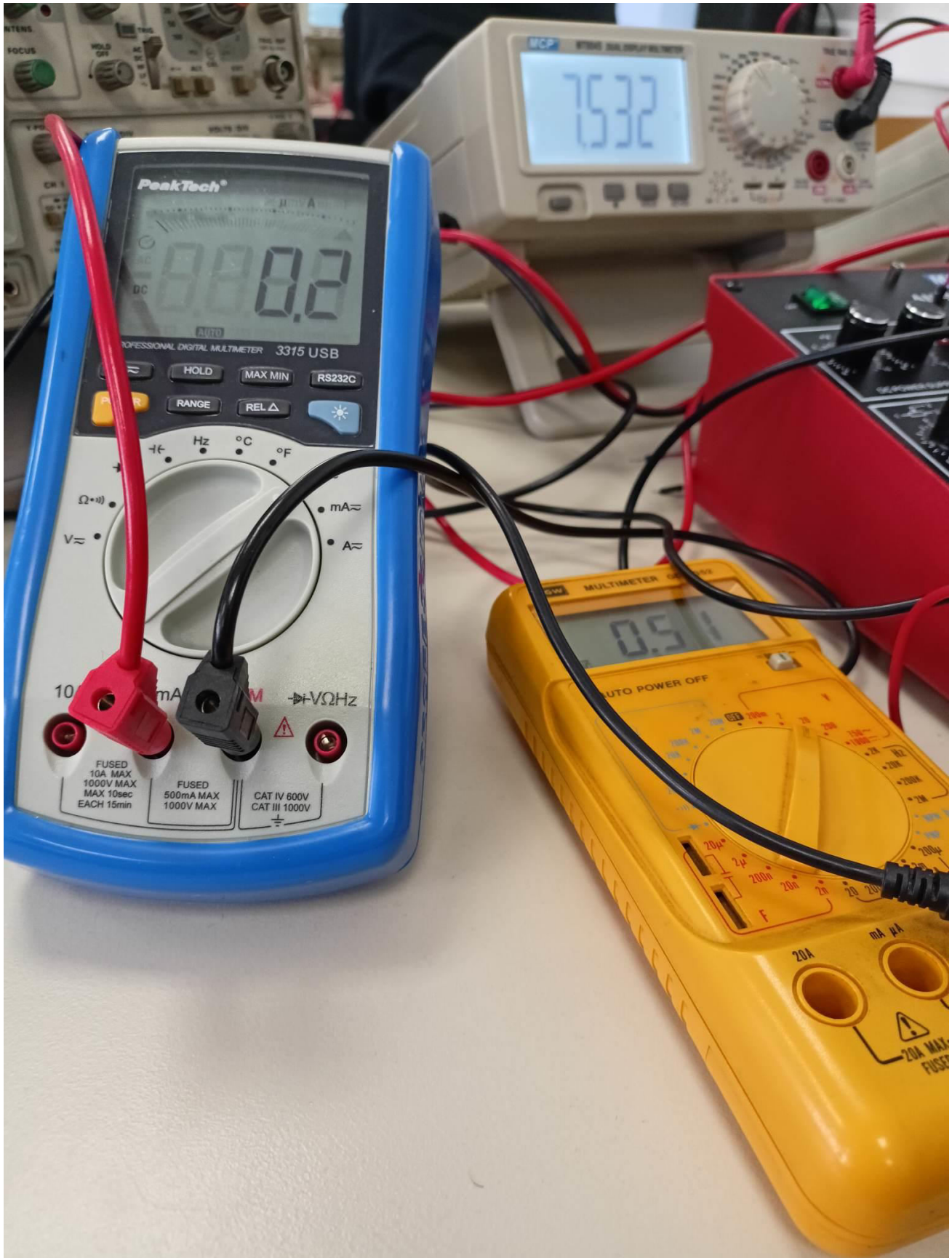


Figure 1.2.19 $V_{ce} = 7.5$ Volt, $V_{be} = 0.5$ Volt

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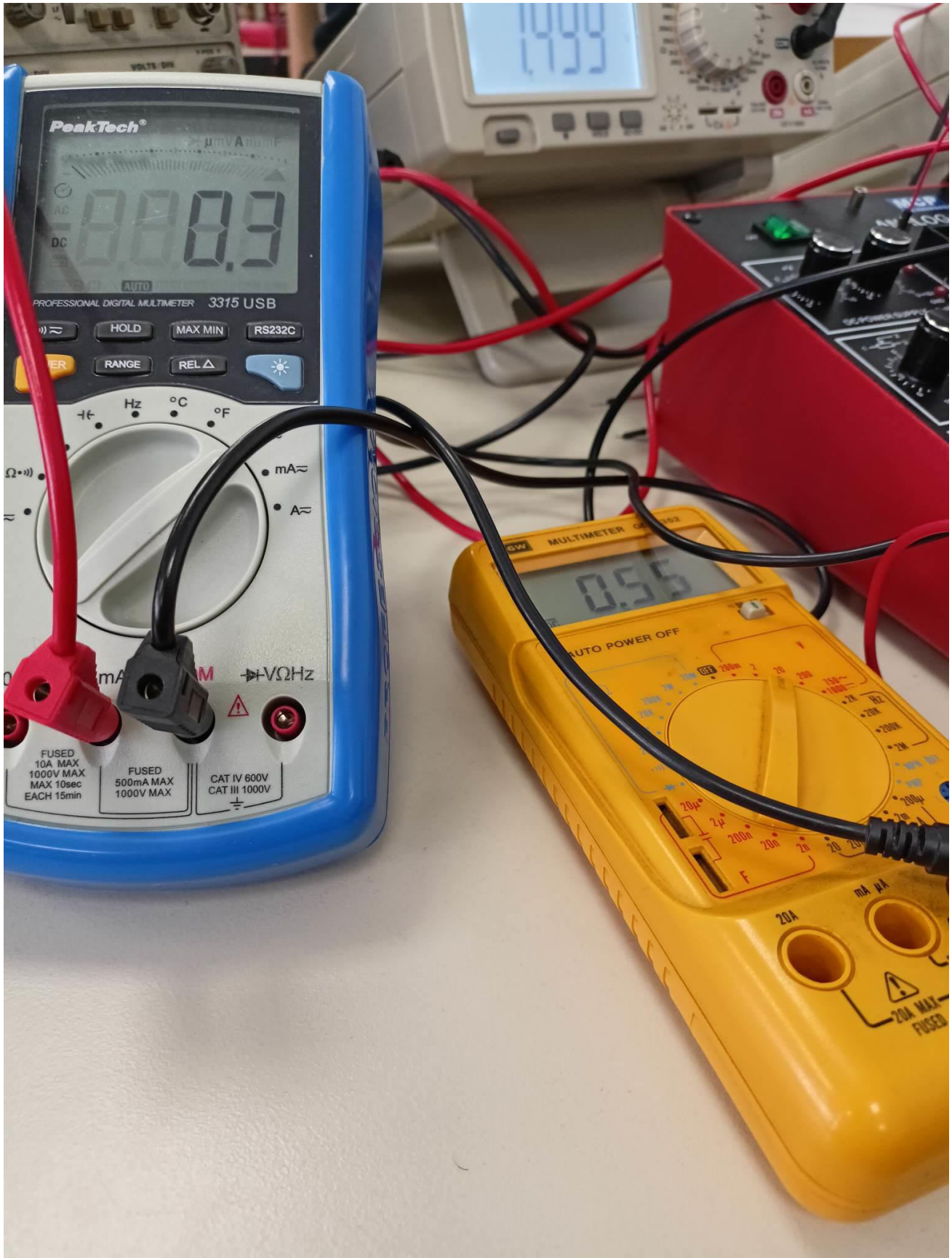


Figure 1.2.20 $V_{ce} = 7.5$ Volt, $V_{be} = 0.55$ Volt

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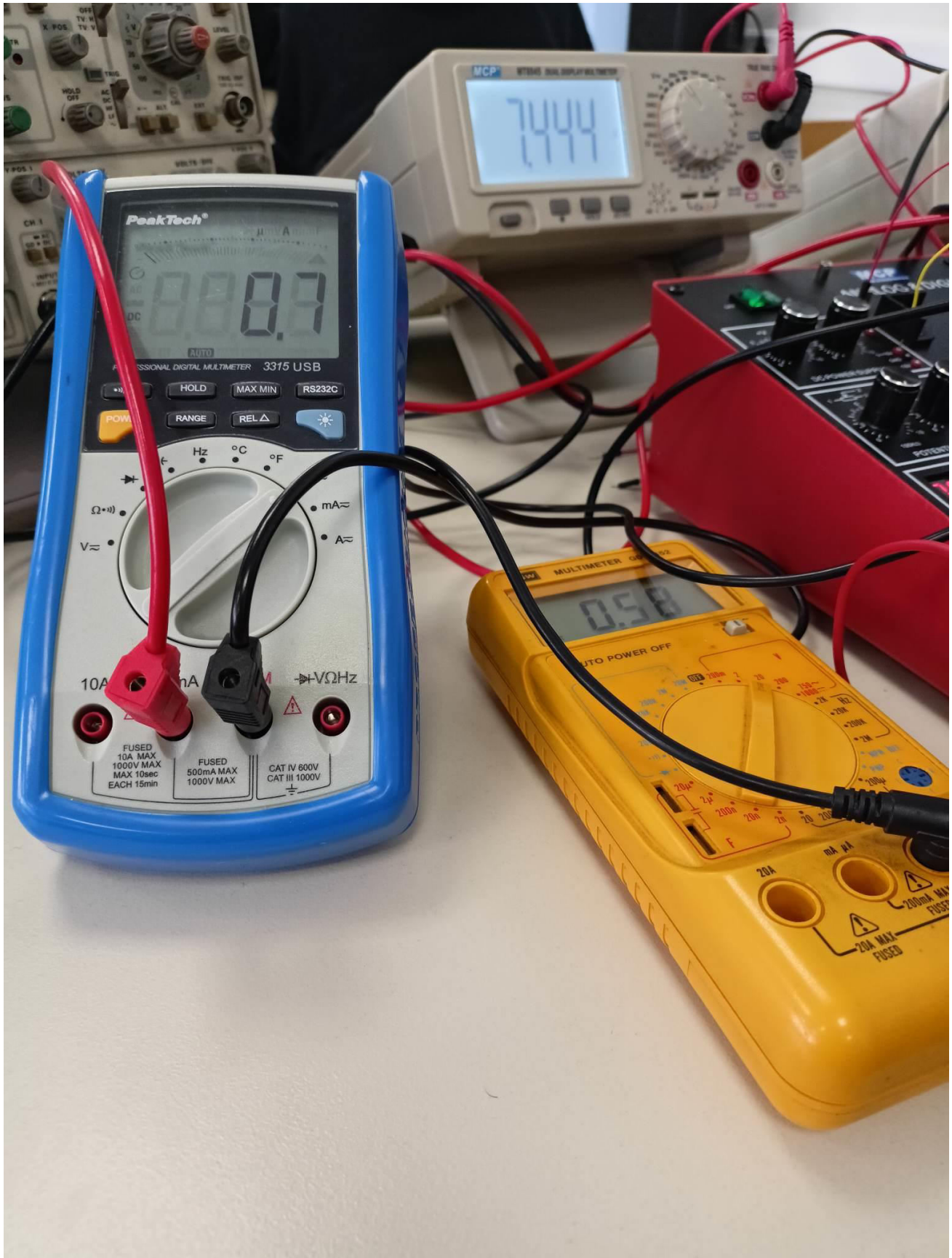


Figure 1.2.21 $V_{ce} = 7.5$ Volt, $V_{be} = 0.58$ Volt

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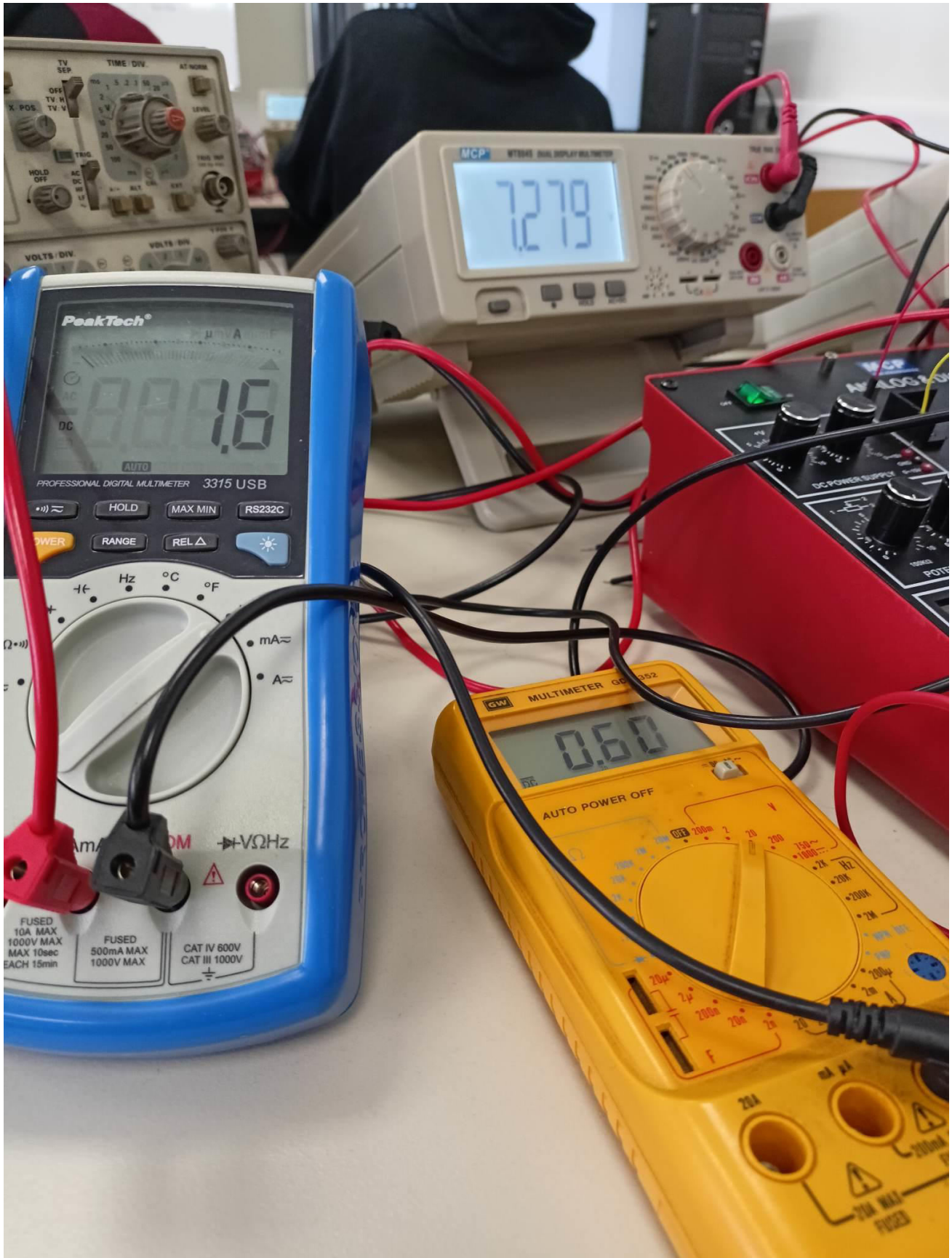


Figure 1.2.22 $V_{ce} = 7.5$ Volt, $V_{be} = 0.6$ Volt

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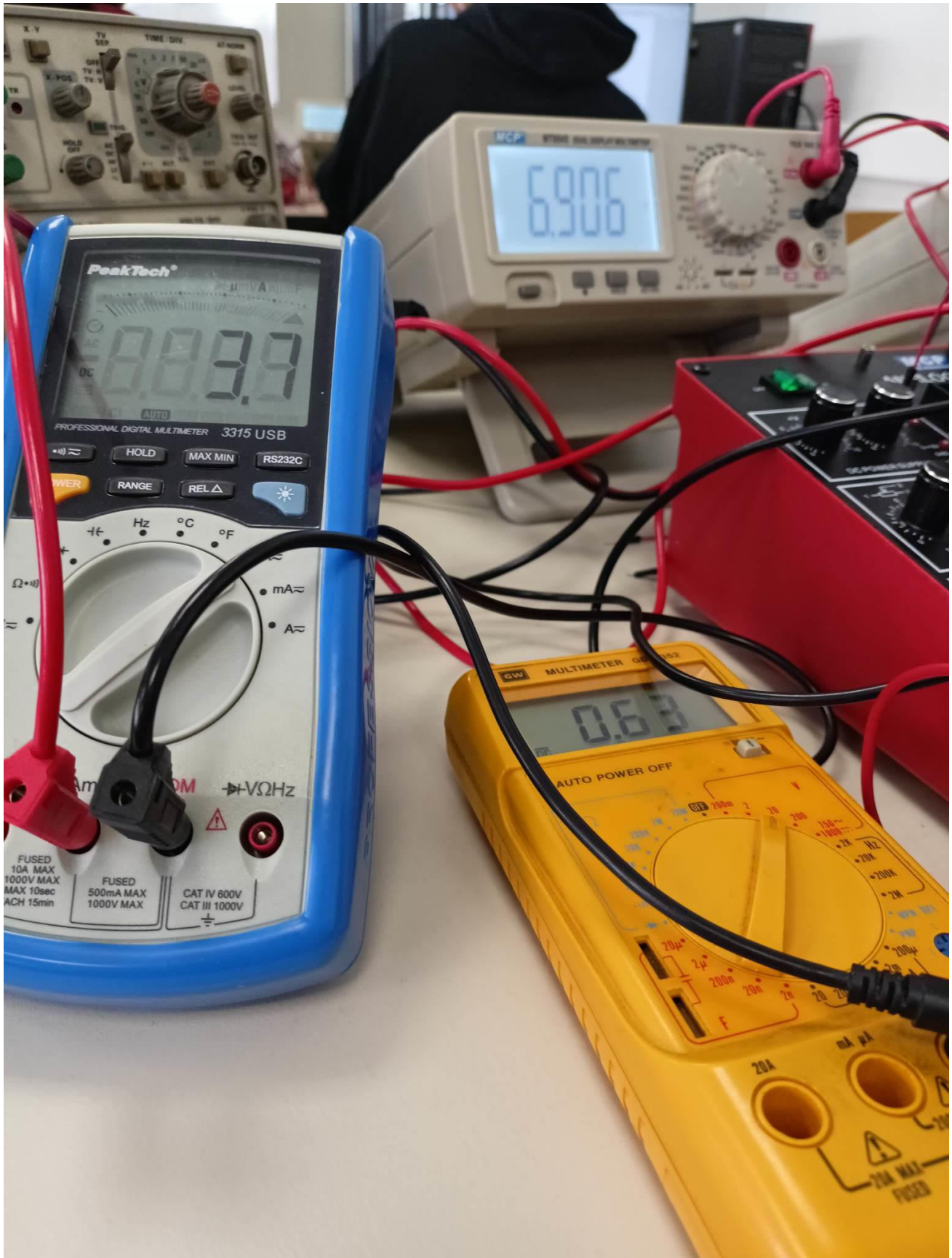


Figure 1.2.23 $V_{ce} = 7.5$ Volt, $V_{be} = 0.63$ Volt

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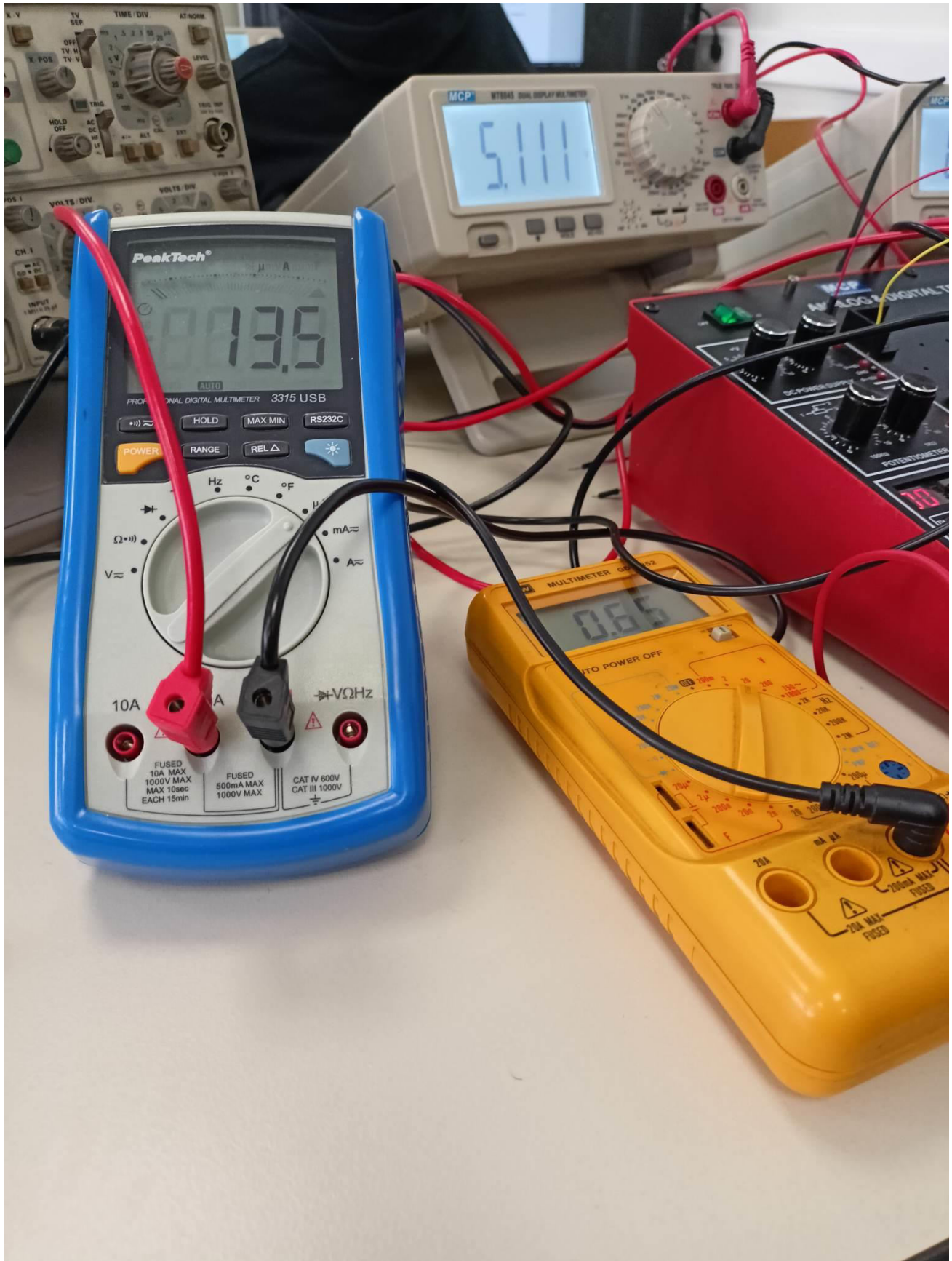


Figure 1.2.24 $V_{ce} = 7.5$ Volt, $V_{be} = 0.65$ Volt

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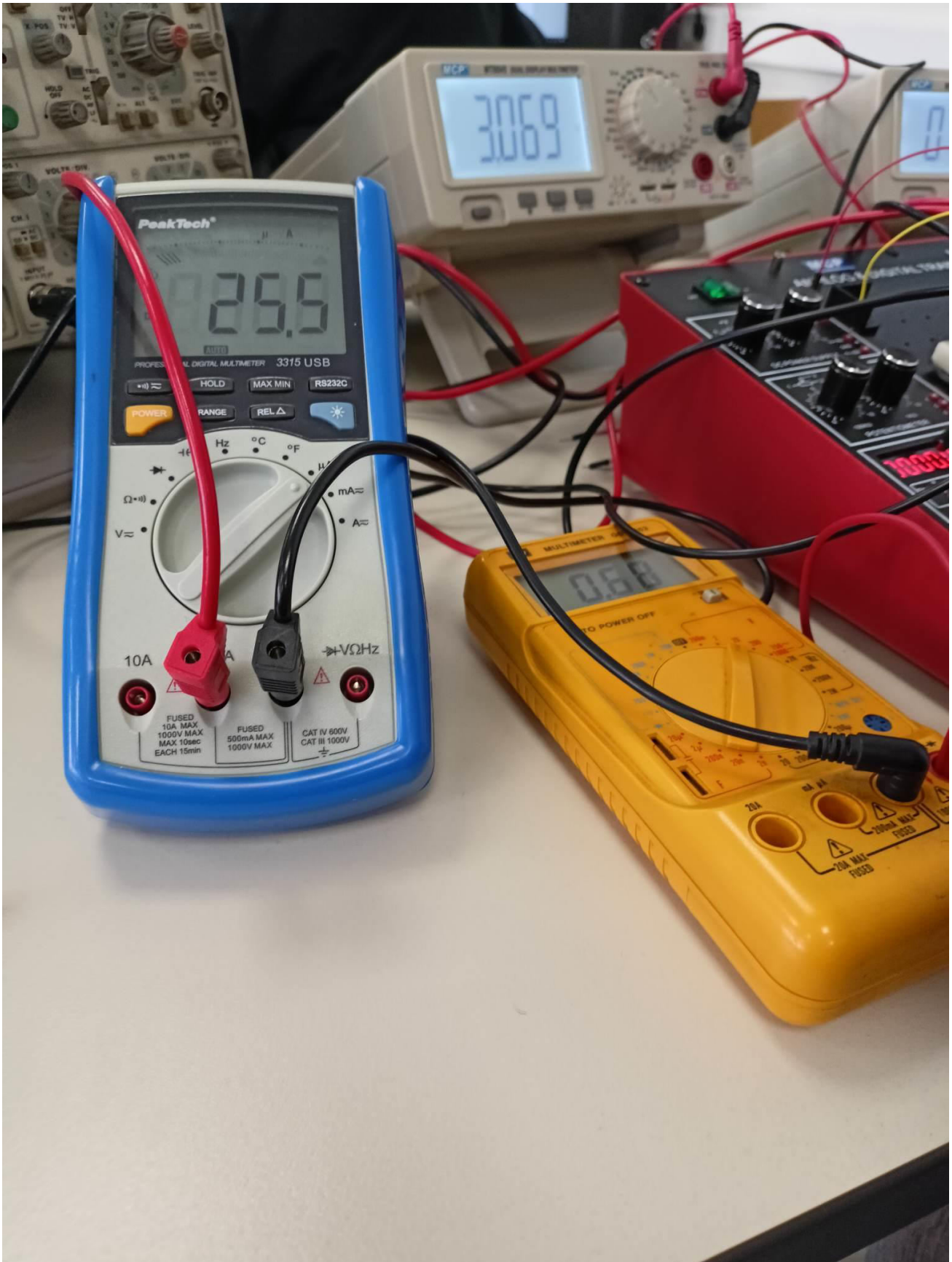


Figure 1.2.25 $V_{ce} = 7.5$ Volt, $V_{be} = 0.68$ Volt

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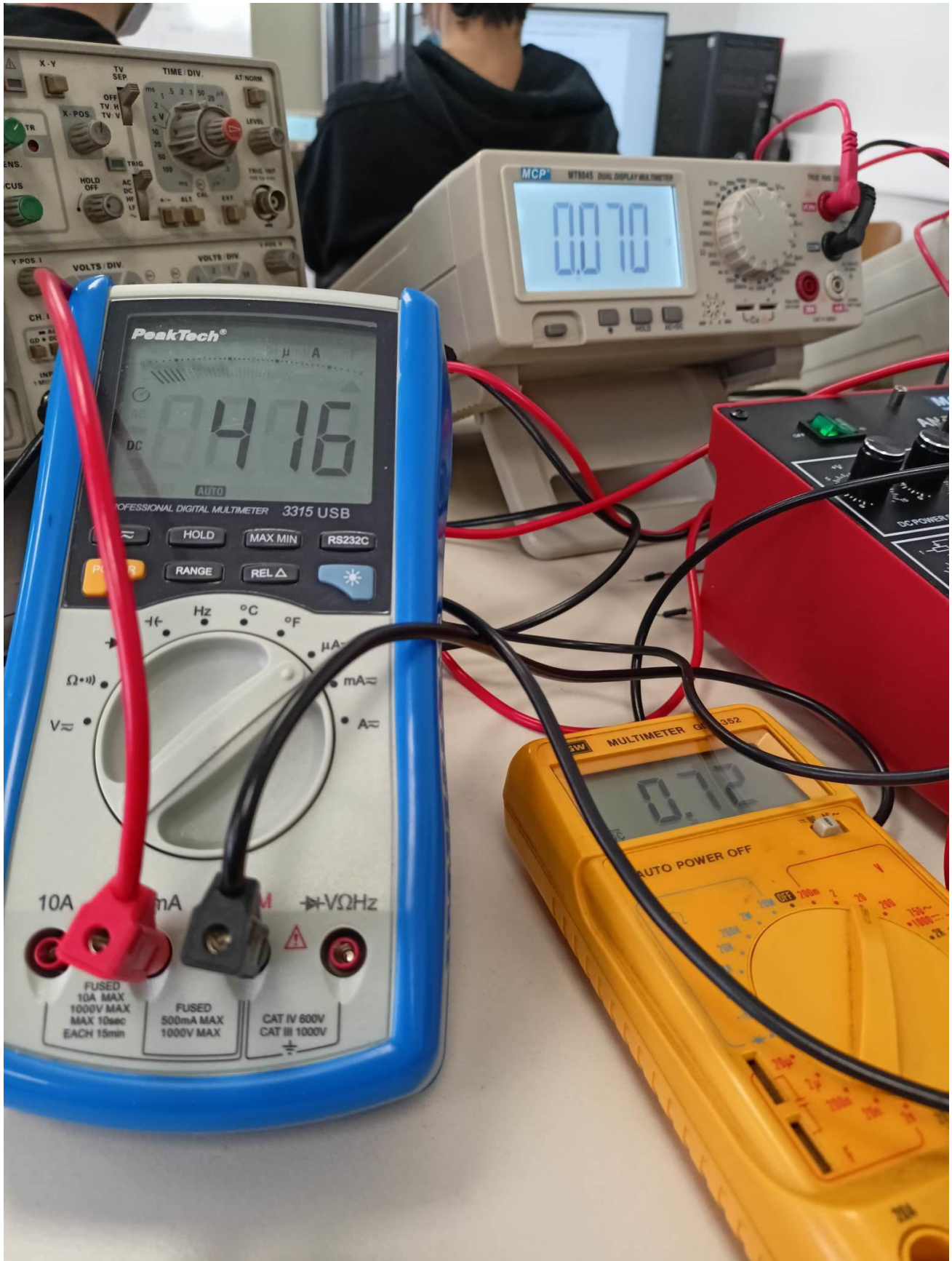


Figure 1.2.26 $V_{ce} = 7.5$ Volt, $V_{be} = 0.72$ Volt

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1.3 QUESTIONS

Question 1.3.1

Based on the construction and mode of operation of the transistor, explain the form of the output, input and transfer characteristics (dynamic characteristic).

- Input characteristics

Since the transistor includes two PN contacts (one on the base-collector and one on the base-emitter), the input characteristics are in the form of a right bias of the base-emitter diode PN bias.

- Output characteristics

Since the current flowing through the collector (I_c) increases sharply for small changes in the potential difference at the collector and emitter ends (V_{ce}) and then stabilizes, the output characteristics exhibit a vertical curve that takes a slight slope.

- Transport characteristic

Since the transistor is also used as a signal amplifier, the transfer characteristic is a curve that corresponds the base current (I_b) to the current passing through the collector (I_c). This curve is tangent to the load line on the output characteristics.

Question 1.3.2

What is the usefulness of the output characteristics and what does the load line specify?

The output characteristics serve to define the transistor's operating range, the corner region and the cut-off region. For a constant base current I_b , the collector current I_c rises sharply for small changes in the potential difference at the collector and emitter ends (V_{ce}) until it stabilizes and takes a slight slope.

The load line defines the operating region of the transistor with edges in the cut-off region where the base and collector

currents are $I_b = 0$ and $I_c = 0$ (due to inversely polarized diodes) and the peak region with maximum collector current I_c (due to normally polarized diodes).

Question 1.3.3

What is β (h_{fe}) and what does it define for each transistor? In the circuit of Fig. 15 (Figure 1.2.1), ΔI_c is 15mA and ΔI_b is 30 μ A, calculate the amplification factor A_i or β of the circuit

β (h_{fe}) is a hybrid parameter of the transistor that determines the current gain from its amplifying behavior.

$$\beta = h_{fe} = \frac{\Delta I_c}{\Delta I_b} \rightarrow$$

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$$\beta = \frac{15 \times 10^{-3}}{30 \times 10^{-6}} \rightarrow$$

$$\beta = 500$$

Question 1.3.5

a) Complete the table below. Applicable: (for $P_{oL}=300mW$)

Table 1.3.5.1

V_{ce} (V)	1.5	2	3	4	5	7	9
I_c (mA)	200	150	100	75	60	42	30

$$P_{oL} = V_{ce} \times I_c \rightarrow I_c = \frac{P_{oL}}{V_{ce}}$$

b) draw the total net power curve

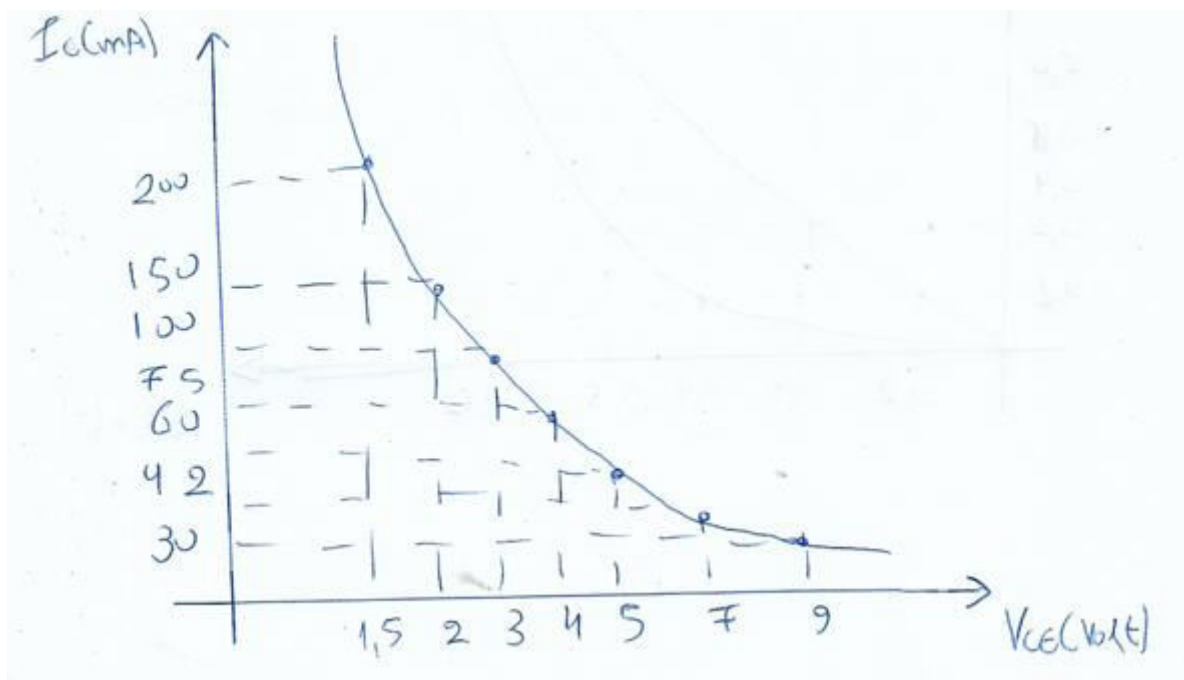


Figure 1.3.5.1 Total net power curve

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Thank you for your attention.

